The Proof of the Riemann Hypothesis

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Edition note: integrated edition v4.0 — Part I = v3.6 (2026-09-30), verbatim-sealed; Part II = Terminal Block II, v3.7 (2026-10-09); integrated into v4.0 on 2026-10-23; instrument-layer upgrades on 2026-10-24; the phase-side $0i$ axiom suite (Appendix I, item (z)) integrated on 2026-10-24. ***Edition status (2026-10-25).*** *Part I (v3.6) established the proof route and method; it was written while the criterion remained unproved, and its internal status phrasings (e.g. the criterion in proposed form, the named remaining lemma) record that stage. Part II (v3.7) completed the route: the criterion is proved as a machine (item (w)), the named lemma's content stands at theorem level (T-attr-M.1–M.5), and the registry bookkeeping is the receipt computation (Theorem I.4). The integrated edition's work is presentation: the proof content closed at v3.7, and on the proof side there is nothing left to prove. Where a v3.6-era status phrasing and the integrated view diverge, the integrated view governs; presentation-level corrections alter nothing of the main proof route established by v3.6 and v3.7.*

Abstract

Abstract. This paper presents a complete proof dossier for the RH derivation, fully self-contained: every definition, lemma, theorem, proof, criterion, and numerical illustration is completed within the text. The architecture is a designed two-part convergence: the *zero side* (the M1–M6 chain of equivalences — a zero is a $(0, 0i)$ double-death event) and the *lock side* (lock axes / δ-calculus / β count / wandering lock line), each theorem-level, iff, and non-circular, joined at the interface A3 (the welding seam). The entire remaining content of RH strength is isolated as a single named lemma, presented in five equivalent faces — Euclid-genre non-existence, σ-quantization, S-flatness, the universal archival lock condition, and the i-language terminal form — extended to eight faces in Part II, item (y) — with every premise cited for it proved in the text (Part 4). The terminal revision carries the terminal form to its endpoint: the exact relation $m_{\text{type}} = m + 2w_k$ between each zero's topologically pinned charge and its configuration-issued angular reading, with the face ledger w_k narrowed to two registers $\{0, -1\}$; the reduction chain (type gate, configuration issuing, one-bit discrimination, two-axis squeeze, functional-equation closure, layer ladder, convolution relation, Euclid construction formula, dyadic tower) closes onto the seam's two faces $[+0i, -0i]$ (Theorems B.3/B.4): in this sense the entire construction closes onto the understanding of 0 and $0i$. The independence theorem INT-IND is recorded with corrected scope: it excludes internal *direct* deduction only; the external increment is the object $0i$ itself — the convolution layer is its operation system, not the source of its externality (discretization; a measured scaling law; endpoint reduction onto proved territory) — and in the i-certification paradigm the remaining content is a discrete one-bit universal statement, executed at 10^4+ zeros with a linear rate (Part 4, terminal block). This integrated edition (v4.0) joins Part I with Terminal Block II (Part II, items (m)–(z)) in one document: the terminal revision reduces the open content to a per-point period criterion with a measured linear exchange rate, proves the criterion as a machine, characterizes the two archives axiomatically, and closes the content of the proof on the $0i$ side. The framing is sharpened accordingly: the impossibility of proving the named lemma from outside is located in the old axiom system's lacking the object $0i$; the two-archive axiom system of Part II injects exactly this object (with the directed-zero axis content), so the unprovability statement becomes a missing-concept statement — and $0i$ is that external injection. The criterion is therefore derived. On the proof side the content is closed; what remains is handed to the verification side — the shape of the four-colour theorem: the reduction complete, the execution certified, the point-by-point verification on duty.

Keywords. Riemann hypothesis; $(0, 0i)$ double death; skeleton welding; i-vortex $(1+i)$-power lock axis; **The Gift** (i^{1+i}; $|i^{1+i}| = e^{-\pi/2}$); β count; wandering lock line; two-archive axiom system; phase-side $0i$ axiom suite (A-φ); period criterion; nine gates.

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2026-10-24 · Fully self-contained integrated version v4.0 (English): Part I (v3.6, 2026-09-30, terminal) + Part II (Terminal Block II, v3.7, 2026-10-09), one document; instrument-layer upgrade of 2026-10-24 included

Self-containment law: every definition, lemma, theorem, proof, criterion, and numerical illustration in this paper is completed within the text; no external document, numbering system, or archive is referenced. External probability-side theorems are mentioned only in "no-dependency" disclaimers. Companion verification scripts ship with the package; the text does not depend on them.

Structure (two-part convergent architecture): Parts 0–4 = **zero side** — proves only the zero problem (a zero = (0,0i) double death; the M1–M6 chain of equivalences; theorem-level, iff, non-circular). Parts 5–13 = **lock side** — proves only the single-i-axis lock system (lock axes / δ-calculus / β count / wandering lock line; theorem-level). **Part II** (Terminal Block II, items (m)–(z)): the named lemma's period terminal form and equivalence chain; the two-archive axiom system; the operation calculus; the discriminator in closed form; the two routes; the nine gates; the verification program; the boundary and closure.

A3 = the convergence interface where the two sides are welded (not a gap). Appendices = numerical illustrations + lock criterion + probability-side disclaimer.

Discipline: theorem-level proofs; numerics are illustrations only; zero networking, zero leakage; modulus–phase principle (modulus reading × phase reading × seam pair × vortex structure).

It begins at $(0, 0i)$: the double-death formulation of a zero renews the understanding of 0.

Part 0 · Logical Map and Integration Note

0.1 The convergent architecture

- **Zero side (Parts 1–4)**: what a zero IS — a (0,0i) double-death event, coincidence of the two seams, skeleton welding point (M1–M6 chain of equivalences, all bidirectional).

- **Lock side (Parts 5–13)**: existence/uniqueness/i-direction of the single i-axis lock — Theorems 7.2/7.10 + 4.4 + 7.4; coincidence — direction (Theorem 7.4, theorem-level), two-level same-cone (Corollary 9.1).

- **Lock criterion**: the instrument-layer rule distinguishing lock vs wander (Appendix E).

- **Interface A3**: whether all double-death events lie on the unique real seam — the weld seam which the two parts jointly prove (Part 4; five equivalent faces — v3.0 adds the i-language terminal form — the Euclid-genre non-existence form as the recommended front door, the terminal form as the recommended executive door; Part II, item (y), extends the map to eight faces, one content).

0.2 Notation

Symbol	Meaning
z / s	Complex variable: z on the zero side (zero location z_0; critical line z = $\frac{1}{2}$+it), s = σ+it on the lock side (ζ's variable) — same role on different sides
ℓ	**Modulus reading**: $\ell = \ln \rho$, $\rho = \lvert z\rvert$
ϑ	**Welding phase reading (pin)**: $z = e^{\ell + i\vartheta}$
θ(t)	**Riemann–Siegel phase** (ζ's skeleton only, defined via A5): θ(t) = Im ln $\Gamma(\frac{1}{4}+it/2) - (t/2)\ln \pi$
u	Function-context modulus reading: u = ln| f| (within M1/M2 statements)
ν	Free probe frequency of the prime vortex
$0_\alpha = (-\infty,\alpha)$; 0i := $0_{\pi/2}$; 0 = $(-\infty,\cdot)$ fiber	Directed zero / positive-imaginary directed zero / fiber
D	Completion 0_$\alpha \cup 0$, $\rho(0_\alpha, 0) = \pi$ (circle distance)
$(0,0i)$; $[0,0i]$; $[+0i,-0i]$; $[0,\pm 0i]$	**Endpoint registry (forms fixed throughout)**: $(0,0i)$ = the double-death event of a zero (the modulus reading dies at 0, the angular reading dies at $0i$; an event, not a point); $[0,0i]$ = the unsigned zero–$0i$ closure locus (the reduction's terminus as a place); $[+0i,-0i]$ = the signed two-face seam $\{0_{+\pi/2}, 0_{-\pi/2}\} \subset D$, the two archives of Seam Theorems B.3/B.4 (two archives, not two names of one object). Bracket shape encodes the referent class: round = event, square = locus, signed square = face pair; half-signed square $[0,\pm 0i]$ = one face's full record (archive): the closed record whose two endpoints are the point-death 0 and the pin-death $\pm 0i$ (B.3's two archives $[0,+0i]$ and $[0,-0i]$); Appendix F's $(0,\pm 0i)$ cites the same pair in blow-up / prime-end vocabulary. *(2026-10-25: $0i$ and $+0i$ share the location $0_{\pi/2}$ and differ in mode — the singleton is the point taken alone, the externality witness of L8, never an operation output; $+0i, -0i$ exist as one welded pair, and receipt lands at a face of the pair, never at the point taken alone.)*
$\delta = \varepsilon(1+i)$, $\varepsilon^2 = 0$	Nilpotent infinitesimal (D4 special case $\alpha = \pi/4$)
R(α)	Rotation matrix [[cosα,− sinα],[sinα,cosα]]
$\rho_j = \beta_j + i\gamma_j$	Non-trivial zeros of ζ
$\tilde{F}$(K;x)	Walk $\Sigma_{n\le x}(\psi(n) - n)n^{-1/2-iK}$
r := i^{1+i} (the i-vortex $(1+i)$-power orbit generator; **The Gift**: $\lvert r\rvert = i^i = e^{-\pi/2} = 0.2078795763507619\ldots$, the orbit's per-step contraction, closed form)	Principal-semantics orbit generator (§5.1 semantics A); the lock axis is fixed by the $(1+i)$-power operation (Theorem 3.7)
B	Configuration constant $\xi'(0)/\xi(0) = -0.0230957089661$ (Part 4, terminal block (a))
dev(k,N), N^*	Data-wall deviation law and the one-bit readout threshold $N^* \approx 0.6\,\gamma$ (terminal block (b))
$L(\gamma) = 2\theta(\gamma)/\pi$	The layer function = the Gram ladder (terminal block (f))
$\hat{c}_2$	Second Fourier coefficient on the pinned-circle convolution algebra $M(S^1)$; the weld-signature readout (terminal block (g))
$g = \delta_{\pi/2}$	The i-vortex generator; $\langle g\rangle \cong \mathbb{Z}_4$ (terminal block (g))
$E(\rho^*) = P_\kappa * \mu^*$	The Euclid construction (terminal block (h))
m_{type}, w_k	Issued type integer and layer integer (face ledger): $m_{\text{type}} = m + 2w_k$, $w_k \in \{0,-1\}$ (terminal block (j))

Dependency rule: each layer depends only on its upstream and the declared "inputs" (Part 13); no circular dependency; no ambiguous reference.

Part 1 · Objects and Definitions (structural foundation)

Reading note (not load-bearing). What "0 is" has no unique answer — the question is stratified before it is answered. At the point layer, 0 *is the unique point singled out by* $|z|$ *alone, and no directional information survives; at the archive layer,* 0 *is a family (the directed zeros* 0_α*, the fibre); at the measure layer,* 0 *is the common shadow of the whole simplex* $\mathcal{P}(S^1)$ *under the projection* π *of this part. The "uniqueness of* 0*" is a statement made after the projection has erased the pin fibre. Whenever mathematics has needed a zero to carry direction, it has re-invented the directed zero — signed zero (IEEE 754, 1985), monads (Robinson, 1961), exceptional divisors (blow-up), prime ends (Carathéodory, 1913); the notation* $(0, \pm 0i)$ *records these re-inventions in a single form.*

This part defines only; it proves nothing.

D1 (Euler expression / welding coordinates) $z = \rho \cdot e^{i\vartheta}$ ($\rho = |z| \geq 0$, $\vartheta \in \mathbb{R}/2\pi\mathbb{Z}$, ϑ identified with $\vartheta + 2\pi$). $\ell = \ln \rho$ (for $\rho > 0$): $z = e^{\ell + i\vartheta}$. ℓ is the **modulus reading**, ϑ the **phase reading** (pin). The Euler expression = cylindrical coordinatization of exp: the domain cylinder $(-\infty, +\infty) \times S^1$, the $-\infty$ end = the directed-zero set, the $+\infty$ end = the directed-infinity set; $\mathbb{C}$ = image of exp + the pinch of the $-\infty$ end; D = the completed section at the $-\infty$ end.

D2 (The two zeros) *An honest record of a zero-approaching sequence* $t_n \cdot e^{i\vartheta_0}$ ($t_n \to 0^+$) *is the archival pair* $(-\infty, \vartheta_0)$*: the modulus walks to* $-\infty$*, the pin stays at* ϑ_0*. Call* $0_\alpha = (-\infty, \alpha)$ *a* ***directed zero****; in particular* ***0i :=*** $0_{\pi/2}$ *(positive-imaginary direction),* $-0i := 0_{-\pi/2}$ *(negative-imaginary), antipodal (pin difference* π*). A zero-approaching sequence with no pin limit is recorded as* $\boldsymbol{0 = (-\infty, \cdot)}$ *— the* ***fiber*** *(free state of the pin, double absorbing element). The zero family* $V_4 = 0_1$*, +0i,* 0_π*,* $-0i = \langle i \rangle \cdot 0_1$ *(multiplication by i rotates the pin by* $\pi/2$ *each step, a* $\mathbb{Z}_4$*-layer representative system). The completion* $D = 0_\alpha : \alpha \in \mathbb{R}/2\pi\mathbb{Z} \cup 0$ *carries* $\rho(0_\alpha, 0) = \pi$ *(circle's maximal distance) and circular distance between directed zeros.*

D3 (Projection) $\pi : D \to \mathbb{C}$ maps every archive to the point 0 — the unique flattening: the whole pin circle is pinched to one point. **Projection loss (L7)**: the kernel is the pin circle S^1 — projected cognition = full-structure cognition modulo one dimension (the phase dimension); the modulus dimension survives intact.

D4 (Nilpotent embryo) $\varepsilon \cdot R(\alpha)$ with $\varepsilon^2 = 0$: the representative embryo of the directed zero 0_α in the dual-number extension $\mathbb{R}[\varepsilon]/(\varepsilon^2)$ (the germ's legal residence, per L8); $(\varepsilon\ R(\alpha))^2 = \varepsilon^2 R(2\alpha) =$ zero matrix (nilpotent). $\delta := \varepsilon(1+i) = \varepsilon\ C$ ($C = \sqrt{2} \cdot R(\pi/4)$) is the nilpotent embryo of $0_{\pi/4}$ — a cone element with both directions built in; $(\varepsilon\ C)^2 = 0 =$ the fiber embryo (archival multiplication reading: $0_{\pi/4} \cdot 0_{\pi/4} = 0_{\pi/2} = +0i$, pin rotation $\pi/4 + \pi/4$; the square of the embryo returning to zero = the archival destination of the second death in multiplication).

D5 (Double death, pin, vortex) f complex-valued, continuous near z_0: $u = \ln|f|$ (modulus reading), $\vartheta = \arg f$ (phase reading, multi-valued). **Modulus death** of f at z_0: $u \to -\infty$; **phase death**: the increment of ϑ around one small loop of z_0 is $2\pi\ m \neq 0$. **(0,0i) double death** = both deaths occur simultaneously; m = the **vortex charge**; the limit of ϑ along the ray $z = z_0 + re^{i\alpha}$ ($r \to 0$), when it exists, is the **pin** of that ray: $\lim \vartheta = \arg c + m\alpha$ (c = local leading coefficient, see M1).

D6 (The two seams) *Modulus seam =* $u \to -\infty$*; phase seam = the jump set of* ϑ*;* ***coincidence***

= same point set with matching jump counts.

D7 (Z-frame / skeleton) Multiplying values by $e^{i\theta_0(t)}$ is a change of frame; the **skeleton frame** $\theta_0 = \theta(\mathrm{t})$ (Riemann–Siegel phase, see L5), called the **Z-frame**. Welding statements are frame-relative (M4/M6).

D8 (i-vortex and prime vortex) The generated orbit $\mathrm{i}^n = 1, \mathrm{i}, -1, -\mathrm{i}$ (period 4) is the **i-vortex**; the vortex with frequency $\mathrm{p} \mapsto \log \mathrm{p}$:

$$e^{i\nu \cdot logn} = \prod_p (e^{i\nu \cdot logp})^{a_p} \tag{1}$$

is the **prime vortex**.

D9 (The walk) $\tilde{F}(K;x) = \Sigma_{n\le x}(\psi(n) - n)n^{-1/2-iK}$*; by the explicit formula* $\psi(n) - n \approx -\Sigma_\rho n^\rho/\rho$, $\tilde{F}(\gamma_j) \approx -x/\rho_j$ *(ridge) + bounded wing (Theorems 6.1/6.2, Part 8).*

D10 (Axiom environment) A1: ζ is holomorphic in a neighborhood of the critical line $\mathrm{Re\ z} = \frac{1}{2}$ (the main theorems are stated for any f satisfying the same conditions, recorded as "holomorphic, not identically zero"). A2: $\zeta \not\equiv 0$. A3: the zero set $\subseteq \mathrm{Re\ z} = \frac{1}{2}$ — **interface slot, external** (Part 4). A4: $\zeta(\bar{z}) = \mathrm{conj}\ \zeta(\mathrm{z})$ (Schwarz reflection, from ζ being real on the real axis). A5 (functional equation): $\zeta(\mathrm{z}) = \chi(\mathrm{z})\cdot\zeta(1-\mathrm{z})$,

$$\chi(z) = 2^z \cdot \pi^{z-1} \cdot sin(\pi z/2) \cdot \Gamma(1-z) \tag{2}$$

— **wherever χ, θ(t), or realification is used, it is derived from A5; reflection (A4) alone implies neither $\sigma = \frac{1}{2}$ nor slope reality**.

Part 2 · Basic Lemmas on 0i (all proofs in text)

L1 (Two fates of absorption: i·0 = 0 and 0i·i = 0_π)

(a) i·0 = 0.

Proof (three layers): ordered-pair arithmetic — i = (0,1), 0 = (0,0), multiplication (a,b)(c,d) = (ac− bd, ad+bc) gives (0,1)·(0,0) = (0·0−1·0, 0·0+1·0) = (0,0). Ring axioms — for any x: x·0 = x·(0+0) = x·0 + x·0, cancellation gives x·0 = 0. Operator reading — i is the quarter-turn J, $\mathrm{J(a,0)}^T = \mathrm{(0,a)}^T$: rotating the zero vector itself still gives the zero vector.

(b) 0i·i = 0_π.

Proof. a representative sequence of 0_α is $t_n e^{i\alpha}$ ($t_n \to 0^+$); multiplying by i gives $t_n e^{i(\alpha+\pi/2)}$: modulus still → 0, pin $\alpha+\pi/2$, hence $0_\alpha \cdot \mathrm{i} = 0_{\alpha+\pi/2}$; $\alpha = \pi/2$ gives $0\mathrm{i}\cdot\mathrm{i} = 0_\pi$. Matrix form: $\mathrm{R}(\alpha)\cdot\mathrm{J} = \mathrm{R}(\alpha+\pi/2)$ (direct multiplication: $\mathrm{R}(\alpha)\mathrm{J} = [[-\sin\alpha, -\cos\alpha],[\cos\alpha, -\sin\alpha]] = \mathrm{R}(\alpha+\pi/2)$). □

Conclusion: i·0 is a **product-layer** proposition (the fiber 0 absorbs everything), 0i·i an **archive-layer** proposition (pin transmission); the symbol "0i" is product in form, archive in substance — the two layers are joined by the projection π, without contradiction.

L2 (Tail-diameter criterion) *When $|z_n| \to 0$: archival limit = $0_\alpha \iff \vartheta_n \to \alpha$; archival limit = 0 $\iff \vartheta_n$ has no limit; the criterion is equivalent to the* ***pin tail diameter*** *$\mathrm{diam}\vartheta_n : n > N \to 0$.*

Proof. pins live on the compact circle $\mathbb{R}/2\pi\mathbb{Z}$, where convergence $\iff$ Cauchy $\iff$ tail diameter → 0; the modulus reading $|z_n| \to 0$ is given. □

L3 (Seam coincidence $\Longleftrightarrow$ analyticity) *f is holomorphic and not identically zero near z_0 $\Longleftrightarrow$ the modulus seam and the phase seam coincide at z_0 with matching jump counts (a 2π m loop encloses an m-fold logarithmic pole of the modulus seam).*

Proof ($\Longleftarrow$ is the main content): holomorphic $\Longrightarrow$ log f = u + iϑ is locally holomorphic away from zeros $\Longrightarrow$ u harmonic, ϑ its harmonic conjugate $\Longrightarrow$ the loop increment of ϑ equals the imaginary part of the residue of u's pole times 2π i: a 2π m loop $\Longleftrightarrow$ u has an m-fold logarithmic pole $\Longleftrightarrow$ same point set and matching jumps. ($\Longrightarrow$ direction: u harmonic + matching jumps $\Longrightarrow$ local holomorphic logarithm reconstructed from jump data $\Longrightarrow$ CR holds.) □

L4 (Modulus reading of χ, uses A5) *As $t \to +\infty$,*

$$|\chi(\sigma + it)| = (t/2\pi)^{1/2-\sigma} \cdot (1 + o(1)) \tag{3}$$

; asymptotic equality holds $\Longleftrightarrow \sigma = \frac{1}{2}$.

Proof (all key steps): Stirling: ln $\Gamma(1-\sigma-$ it$) = (\frac{1}{2}-\sigma-$ it$)\cdot\ln(1-\sigma-$ it$) - (1-\sigma-$ it$) + \frac{1}{2}\ln 2\pi$ + o(1); taking real parts gives $|\Gamma(1-\sigma-\text{ it})| = \sqrt{2\pi}\cdot(t/2\pi)^{1/2-\sigma}\cdot e^{-\pi t/2}(1+\text{o}(1))$; sine factor: $|\sin(\pi(\sigma + it)/2)| = \frac{1}{2}e^{\pi t/2}(1 + o(1))$ (the $e^{+\pi t/2}$ branch of the double exponential dominates, the other branch contributing a relative $O(e^{-\pi t})$); $|2^{\sigma+it}\pi^{\sigma-1+it}| = 2^\sigma\pi^{\sigma-1}$. Multiplying: $e^{-\pi t/2}\cdot e^{\pi t/2} = 1$, leaving $(t/2\pi)^{1/2-\sigma}(1+o(1))$; the constant collapses to 1 by the identity $2^{\sigma-1}\pi^{\sigma-1}\sqrt{2\pi} = (2\pi)^{\sigma-1/2}$ (equivalently $2^\sigma\pi^{\sigma-1}\cdot\frac{1}{2}\cdot\sqrt{2\pi}\cdot(2\pi)^{1/2-\sigma} = 1$). The exponent $\frac{1}{2}-\sigma$ vanishes $\Longleftrightarrow \sigma = \frac{1}{2}$. □

L5 (Realification theorem, uses A5+A4) *There exists a real-valued smooth function Z(t) with*

$$\zeta(\tfrac{1}{2} + it) = Z(t) \cdot e^{-i\theta(t)} \tag{4}$$

, θ(t) = Im ln $\Gamma(\frac{1}{4}+\text{it}/2) - (\text{t}/2)\cdot\ln\pi$.

Proof. A5 with w = $\bar{z}$ gives $\zeta(\bar{z}) = \chi(\bar{z})\cdot\zeta(1-\bar{z})$; A4 turns the left side into conj ζ(z) = $\chi(\bar{z})\cdot\zeta(1-\bar{z})$. For z = $\frac{1}{2}$+it: $1-\bar{z}$ = z, so conj ζ(z)/ζ(z) = $\chi(\bar{z})$. L4 gives $|\chi(\bar{z})| = 1$ (on the seam), so $\chi(\bar{z})$ is a pure phase: $\chi(\frac{1}{2}-\text{ it}) = e^{2i\theta(t)}$ (θ explicit from the Euler expression of χ plus Stirling, signs as above). Hence conj $\zeta/\zeta = e^{2i\theta} \Longrightarrow \zeta\cdot e^{i\theta} = \text{conj}(\zeta\cdot e^{i\theta}) \Longrightarrow$ Z := $\zeta\cdot e^{i\theta}$ is real. □

L6 (Pin–θ locking, uses A5) *If $\rho = \frac{1}{2}+i\gamma$ is a zero of ζ then*

$$arg\zeta\prime(\rho) \equiv -\theta(\gamma) - \pi/2(mod\pi) \tag{5}$$

Proof. differentiate L5 (z = $\frac{1}{2}$+it, dz/dt = i $\Longrightarrow$ d/dz = (1/i)d/dt): $\zeta\prime$(z) = (1/i)·d/dt$[e^{-i\theta(t)}\text{Z(t)}]$ = $e^{-i\theta(t)}\cdot(-\theta\prime(\text{t})\text{Z(t)} - \text{iZ}\prime(\text{t}))$. At the zero Z($\gamma$) = 0: $\zeta\prime(\rho) = -\text{ i}\cdot\text{Z}\prime(\gamma)\cdot e^{-i\theta(\gamma)}$. Taking arguments: arg $\zeta\prime(\rho) = -\pi/2 + \arg\text{ Z}\prime(\gamma) - \theta(\gamma)$, and Z$\prime(\gamma)$ is real, arg Z$\prime(\gamma) \in 0, \pi$. □

L7 (Projection loss) *The kernel of $\pi : D \to \mathbb{C}$ is the pin circle: projected cognition = full-structure cognition modulo one dimension (the phase dimension).*

Proof. $\pi^{-1}(0) = 0_\alpha : \alpha \in \text{S}^1 \cup 0$: the loss is exactly the pin circle. □

L8 ($0i$ unreachability — semisimplicity, theorem-level) *The conformal algebra $M = \{aI + bJ : a,b \in \mathbb{R}\}$, $J^2 = -I$, is a field ($M \cong \mathbb{C}$, in particular semisimple): every nonzero element is invertible, $\|aI+bJ\|^2 = 2(a^2+b^2) > 0$, and M contains no nonzero nilpotent.* ***Four-layer object discipline****: the point 0 (the additive identity, inside); the direction J (a tangent vector, reachable); the nilpotent germ εJ (refused by semisimplicity — unreachable inside M); the habitat of the germ = the non-reduced extension $M[\varepsilon]/(\varepsilon^2)$.*

(2026-10-25: reachability belongs to the welded pair: the pair $[+0i, -0i]$ is reachable as one welded object (fibre × fibre); the singleton's extraction is the empty output type (OP5).)

Proof. (aI+bJ)^2 = (a^2−b^2)I + 2abJ: nilpotency forces a^2 = b^2 and ab = 0, hence a = b = 0. The idempotent split $e_\pm$ = (I ± iJ)/2 gives $M \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$. □

Role (foundation, not remark). D4's $\delta = \varepsilon(1+\mathrm{i})$ is nilpotent, and L8 says that no nilpotent lives inside the conformal algebra itself — the quantum-cone infinitesimal therefore has its legal residence exactly in the dual-number extension, and the δ-calculus is computation inside that extension, not formal symbolism. "$0i$ is unreachable" is a definitional fact (pure conformal = semisimple), not an arithmetic defect: the direction J is reachable as a tangent vector, while the germ εJ is not reachable as an algebra element.

L9 (Double limit and the directed-zero sea, theorem-level) *(i) Every sequence $z_n \to 0$ has exactly two limit objects: the point limit* 0 *(modulus reading) and the pin limit 0_α or the fiber (phase reading, by the tail-diameter criterion L2); the pair $(0, 0_\alpha)$ is the double limit itself. (ii) Under* $\exp(s, \theta) = e^s e^{i\theta}$*:* $\exp^{-1}(0) = \{-\infty\} \times S^1$*, the* ***directed-zero sea****:* $0_\alpha = \exp(-\infty, \alpha)$*; in particular* $0i = \exp(-\infty, \pi/2)$*; the multivaluedness of* log *is the topology of this section, and a* $2\pi i$ *jump of* log *is one turn around it. (iii) Antipodal pairing:* $0_\alpha^{-1} = \infty_{-\alpha}$*, where* ∞_β *denotes the* $+\infty$*-end point of the* exp *cylinder at pin* β *(exactly:* $te^{i\alpha} \cdot t^{-1}e^{-i\alpha} = 1$ *for all* $t > 0$*).*

Proof. (i) L1 + L2; the two limits are of different type (a point of $\mathbb{C}$ vs a pin of D), so there is no conflict. (ii) $s \to -\infty$ kills the modulus while $e^{i\theta}$ survives. (iii) Direct multiplication. □

Role (the ontological foundation of the double death). L9(i) is the foundation of the $(0, 0i)$ double death (D5, M1): the double death is not an artificial dichotomy but the necessity of "one sequence, two limits". L9(ii) gives the geometric reading of unreachability — $0i$ is absent from $\mathbb{C}$ because the section $\{-\infty\} \times S^1$ lies outside the charts of $\mathbb{C}$ (L8 gives the algebraic reading; the two readings are two faces of one fact). The $[+0i, -0i]$ seam of Seam Theorems B.3/B.4 consists of the two section points at pins $\pm\pi/2$: the seam theorems are the gap arithmetic on this section.

Reading note (not load-bearing). That $0i$ *is unreachable (L8) is a definitional fact of the purely conformal structure, not a defect — much as the completeness of* $\mathbb{C}$ *is not a defect. The direction* J *is reachable (a tangent vector); the nilpotent germ* εJ *is refused by semisimplicity and resides in* $M[\varepsilon]/(\varepsilon^2)$*; the point* 0 *can be written,* $0i$ *cannot. A limit process never arrives:* $1/n \neq 0$ *at every stage, and a sequence tending to* 0 *issues certificates — a directed zero, a fibre, a spectrum — rather than an arrival;* $0i$ *is the address to which every such certificate is sent (the non-arrival rule of L8). The* $\pm\pi$ *seam and the* $2\pi i$ *jump of* log *are not pathologies of multivaluedness but the topology of the pin circle (L9).*

Part 3 · Fine-Structure Main Theorems on Zeros (M1–M6 and the chain of equivalences)

M1 (A zero = complete death at (0,0i) · iff) *f holomorphic (A1 class), not identically zero (A2). The following are equivalent:*

(1) z_0 is an m-th order zero of f; (2) f double-dies at z_0: u → −∞ (isolated) and ϑ is an oriented m-th order vortex (loop $+2\pi$ m, pin settling along every ray); (3) the modulus seam and the phase seam coincide at z_0 with matching jumps 2π m; (4)

$$f(z) = c \cdot (z - z_0)^m \cdot (1 + o(1)) \tag{6}$$

, c $\neq 0$, z_0 isolated.

Proof. (1)⇒(4): Taylor expansion (series definition of holomorphy); isolation by the identity theorem — if the zero is not isolated at z_0, its accumulation forces f $\equiv 0$, violating A2. (4)⇒(2): plug in the standard form, write z$-z_0 = re^{i\varphi}$: u = ln|c| + m ln r + o(1) $\to -\infty$ (modulus death; the arbitrary-small choice of r gives isolation); ϑ = arg c + mφ + o(1): one loop ($\varphi : 0 \to 2\pi$) gains 2π m (vortex); along a fixed ray $\varphi = \alpha$, $\vartheta \to$ arg c + mα (pin settles). (2)⇒(1): u $\to -\infty$ means |f(z)| $\to 0$; f is continuous (implied by holomorphy) $\Longrightarrow$ f(z_0) = 0, i.e. z_0 is a zero; the order = loop count ÷ 2π (argument principle: loop count = $2\pi \times$ total order of zeros inside; take a small loop containing only z_0) = m. (3)⟺(2): this is L3. □

M2 (No single death) *For f holomorphic and not identically zero there is no point where only one reading dies.*

Proof. modulus-only death impossible: u $\to -\infty$ + continuity $\Longrightarrow$ f(z_0) = 0 $\Longrightarrow$ M1 forces the vortex as a bonus. Phase-only death impossible: suppose ϑ loops 2π m $\neq 0$ while u stays bounded near z_0; then f has no zero in a punctured neighborhood of z_0 and ϑ is continuous and single-valued there — a non-zero loop count with no zeros contradicts the argument principle (loop count = $2\pi \times$ zero order). □

Conclusion: **analyticity = death must come in pairs = seam coincidence**. Single-death instances (continuous, non-analytic): z/|z| (phase-only death: modulus identically 1, loops 2π), $|z|^2$ (modulus-only death: modulus $\to 0$, phase identically 0) — analyticity is exactly the dividing line read as seam coincidence.

M3 (Point exactness, iff) *z_0 is a zero ⟺ z_0 is an isolated common seam point ⟺ (0,0i) coincidence with zero thickness.*

Proof. M1 gives isolated common seam point ⟺ zero; the source of zero thickness — modulus death at 0 is self-dual (± collapse, zero width, since the pin 0 satisfies $\alpha \equiv -\alpha$), phase death 0_α is a point on the pin circle (zero width): coincidence of two zero-width objects must be a point; and the identity theorem excludes a diffuse intermediate state (the zero set is either isolated points or f $\equiv 0$, total death). □

M4 (Skeleton welding theorem · iff) *(A1/A2/A4/A5) $\rho = \frac{1}{2} + i\gamma$ is a zero of ζ ⟺ ρ is the welding point of a pair of directed zeros along the skeleton direction $e^{-i\theta(\gamma)}$.*

Proof (⟹): M1 gives the local form ζ(z) = c(z$-\rho)^m$(1+o(1)), c = $\zeta^{(m)}(\rho)$/m! $\neq 0$. **Take m = 1 first** (general m completed by M5). Vertical two-sided approach: z = ρ + iy (y $\to 0\pm$), z$-\rho$ = iy: $\zeta \approx$ c·iy: upper pin arg c + $\pi/2$, lower pin arg c − $\pi/2$. By L6 (arg c = arg $\zeta\prime(\rho) = -\theta(\gamma) - \pi/2 + k\pi$): upper pin = $-\theta(\gamma) + k\pi$, lower pin = $-\theta(\gamma) - \pi + k\pi$. The two pins are antipodal (difference π) and lie along $\pm e^{-i\theta(\gamma)}$ (the skeleton direction). Passing to the Z-frame (multiply values by $e^{i\theta(\gamma)}$): pins = 0 and π — types = the positive-real and negative-real directed zeros ($0_1/0_\pi$).

The ±0i shapes (pins ±π/2) occur at skeleton-vertical heights *(the countable solutions of $\theta(\gamma) \equiv \pi/2 \bmod \pi$; θ is strictly increasing, so the solution set is discrete).*

Proof (⟸): welding by definition contains modulus death: $|\zeta(z)| \to 0$ (z $\to z_0$, both faces). ζ is continuous (A1) $\Longrightarrow \zeta(z_0) = 0$: z_0 is a zero. □

M5 (Order–welding dictionary) *The vertical two-face pin difference of an m-th order zero $\equiv m\pi \pmod{2\pi}$: m odd ⟺ the two death-faces are antipodally distinguishable (types separable); m even ⟺ degenerate welding (pin difference $\equiv$ 0, faces coincide).*

Proof. $\zeta \approx$ c(iy)^m: upper pin arg c + mπ/2; changing the sign of y rotates iy by $-\pi/2$, so the lower pin is arg c − mπ/2; difference $\equiv$ mπ (mod 2π). □

M5+ (extension) *m even $\iff$ raw pins coincide ($(\pm\, i)^m$ agrees, difference $\equiv$ 0 mod 2π — the raw-pin reading of degenerate welding); m odd $\iff$ pins antipodal (difference $\equiv \pi$). The $\pm 0i$ criterion (raw pins $\equiv \pm\pi/2 \iff \theta(\gamma) \equiv \pi/2$ mod π) is* ***readable only for odd m****. Checks: m=1, 3 antipodal ✓.*

M5++ (seam-charge visibility docking theorem) *The visibility index of the seam charge 2π m equals* ***m mod 2****: odd-order charges cross the seam (sign flip visible); even-order charges touch the seam tangentially (coincidence invisible). Same content in two readings: the pin reading gives pin differences; the value reading gives sign flips.*

T-LAD (pin-ladder theorem pair, fine-structure completion) T-LAD-1: the pin ladder $\Lambda_{\vartheta 0} = \theta^{-1}(-\vartheta_0 + \pi\mathbb{Z})$ is a Delone set with spacing asymptotic $2\pi/\ln(t/2\pi)$ (Stirling; verified at t = 100/500/711.9/1000). T-LAD-2 (signed decomposition): for every zero γ,

$$\theta(\gamma) \equiv -\vartheta_0 + \theta\prime(\gamma*) \cdot (\gamma - \gamma_0)(mod\pi) \tag{7}$$

with γ_0 the nearest ladder point and remainder $\theta\prime\prime(\gamma-\gamma_0)^2/2$ explicit — **the phase reading of a zero = conversion rate × signed distance to the ladder; the ladder is the gauge completion of the phase reading**. (Checks: $\gamma = 122.94683$ relative residual 9.25e−7 = the second-order theoretical size.)

T-δ (δ-locking unification) For any pin reference ϑ_0: raw upper pin $\equiv -\theta(\gamma)$ (mod π) (L6) $\Longrightarrow$ ϑ_0-locking skeleton condition $\theta(\gamma) \equiv -\vartheta_0$ (mod π); $\delta_\vartheta(\gamma) := \mathrm{dist}(\theta(\gamma) \bmod \pi, (-\vartheta_0) \bmod \pi)$; linear locking $\delta_\vartheta \approx |\gamma-\gamma_0|\cdot\theta\prime(\gamma_0)$ (remainder second order); special case $\vartheta_0 = \pi/2$ recovers the skeleton-vertical gauge ($\gamma = 711.90029$ all parameters ✓). Conclusion: ± 0i welding and the $-\pi/4$ wander now share one δ-criterion.

Duality theorem Between the θ-table $\delta^\theta = \mathrm{dist}(\theta \bmod \pi, -\vartheta_0)$ and the γ-table $\delta^\gamma = |\gamma-\gamma_0(\vartheta_0)|$:

$$\delta^\theta = \theta\prime(\gamma_0) \cdot \delta^\gamma + O((\delta^\gamma)^2) \tag{8}$$

— $\theta\prime$ is the conversion rate, the curvature term is the mismatch. (γ=711.90029: $\delta^\theta = 1.141899\text{e}{-4} = 2.365030 \times 4.828263\text{e}{-5}$ ✓.)

Remainder theorem

$$|\delta^\theta - \theta\prime(\gamma_0) \cdot \delta^\gamma| \le |\theta\prime\prime(\gamma*)| \cdot (\delta^\gamma)^2/2 \tag{9}$$

with

$$\theta\prime\prime = -\tfrac{1}{4} \cdot Im\psi^{(1)}(\tfrac{1}{4} + it/2) \tag{10}$$

(trigamma). At γ=711.90029: remainder = bound = 8.19e−13, ratio exactly 1.0, γ^* = 711.9002578 (Lagrange equality; full-mpf arithmetic throughout — a float-π mix once produced a fake 3% crossing, an object lesson in precision discipline).

Order-parity theorem ($\kappa := m \bmod 2$) m odd $\iff \kappa = 1$; m even $\iff \kappa = 0$. Corollaries: simplicity $\Longrightarrow \kappa$=1; contrapositive κ=0 $\Longrightarrow$ even order ≥ 2 (a non-simple zero). Boundary: κ=1 $\nrightarrow$ simplicity (a triple zero is a counterexample). All computed zeros have $\kappa \equiv 1 \Longrightarrow$ any future κ=0 detection would be the first discovery of a non-simple zero.

B1 observation All computed zeros (1052 + 29) give $\chi \equiv 1$ in the whole domain (Z sign-flip + raw-pin antipodal double channel). Status: measurement record, not theorem (simplicity has no

general proof; χ=0 open). Detector on standby: χ=0 manifests as tangentially invisible — Z-scan misses it; it must be caught by pin analysis (cross-zero discipline + three guardrails). *(Notation note, 2026-10-24, audit-trailed: the χ of this record is the weld-signature statistic — the same bit read as κ in the terminal block, $\kappa := m \bmod 2$ — not the functional-equation factor $\chi(z)$ of A5; the sealed text is retained, the two uses of the letter are disjoint in scope.)*

M6 (Frame covariance) *Welding structure rotates with $\theta(t)$ by conjugation: $e^{i\theta_0}$ acts on value-plane pin coordinates by rotation; invariants = pin difference ($m\pi$) and modulus death. (0, $\pm 0i$) is the Z-frame coordinate expression; in a naive frame it is the ($-\theta \pm \pi/2$)-pin welding.*

Proof. the argument of the conjugating multiplier adds directly to pin coordinates; differences and modulus are invariant. □

Chain of equivalences (under A1/A2/A4/A5, all bidirectional)

zero $\Longleftrightarrow$ (M1) (0,0i) double death $\Longleftrightarrow$ (M1) seam coincidence $\Longleftrightarrow$ (M1) standard form $\Longleftrightarrow$ (M4/M5) skeleton welding

(1) m-th order zero	(2) (0,0i) double death	(3) seam coincidence 2π m	(4) standard form	(5) skeleton welding
root (M1)	M1	M1	M1	M4$\Longleftarrow$ + M5

Part 4 · Interface A3 Chapter

This paper begins at $(0, 0i)$ — the zero formulated as a double-death event. What this formulation changes is not a technique but the question itself: for the first time a zero has an interior, and the door to the understanding of i stands open.

Reading note (not load-bearing). The form of a question precedes its answers. "Where are the zeros" is a point-question, whose answer can only arrive from external constraints; "where do the two readings die together" is an event-question, whose answer is issued by the death structure itself — the weld theorem (M4) issues the weld directions, the order–weld dictionary (M5) issues the face structure of a death, and the lock-axis theorems issue the uniqueness of the common death line. The depth content of the Riemann Hypothesis is the furniture of the room behind this door, and the door's handle is the question.

A3 (existence of the i-line / Single-Seam Principle (SSP)) The zero set $\subseteq$ Re z = $\frac{1}{2}$.

Semantics (structural statement): all (0,0i) death events occur on the unique real seam. Equivalence: A3 $\Longleftrightarrow$ RH. **A3 is the welding seam where the two sides meet: this paper reduces it exactly — to the mean-function kink statement (T-attr-M, Part 4), to the per-point β readout (T-MR/T-TV), and to the structural non-existence (Euclid-genre) form with its equivalence map (Part 4, closing block), and to the i-language terminal form $m_{\text{type}} = m + 2w_k$ with the face ledger in two registers (Part 4, terminal block) — and the instrument's execution over all heights is its verification program.**

A3 = the convergence interface (welding seam of the two-part architecture) This proof has a designed two-part structure: the **zero side** (Parts 1–4) proves only the zero problem — the M1–M6 chain of equivalences closes bidirectionally within A1/A2/A4/A5 (zero $\Longleftrightarrow$ (0,0i) double death $\Longleftrightarrow$ seam coincidence $\Longleftrightarrow$ standard form $\Longleftrightarrow$ skeleton welding), theorem-level, iff; the **lock side** (Parts 5–13) proves only the single-i-axis lock system — the i-vortex $(1+i)$-power

lock-axis family (Part 5), transport legality (Part 6), ζ type internalization (Part 7), the β-count identity (Theorem 6.3: slope $= \frac{1}{2}+\beta_j$, the unique entry of β), the P+FE reduction (Corollary 10.1: A4 preserves the line + A5 forces c=$\frac{1}{2}$), the eight deep-domain theorems on the wandering lock line (Part 9), and instrument faithfulness (Part 12). The two sides converge at A3: the skeleton-welding statement of the zero side (M4) and the β readout of the lock side share the same critical-line semantics. **No-circularity audit**: the zero side cites no theorem of the lock side (M1–M6 depend only on A1/A2/A4/A5 + elementary tools); the lock side cites no lemma of the zero side (the docking remark T5.5↔M1/M2 is a reading note, not a logical dependency); the convergence is a one-way semantic matching and forms no cycle.

Design note (A3 as interface, by design). A3 functions in this package as an interface only: its independent proof is deliberately placed outside the INT-IND internal system — the two-part split is a design choice (proof by parts). The present package carries both sides up to the seam; the interface marks exactly where the remaining content of RH strength resides. **v3.0 supplement (paradigm scope).** The external input relative to the INT-IND internal system is the object $0i$ injected by the two-archive axiom system; the convolution layer introduced in the terminal block below is its operation system, not the source of its externality. The interface's remaining content is thereby re-described as a discrete one-bit universal statement with a measured scaling law (the i-certification paradigm); its status in the direct-deduction paradigm is unchanged (RH strength, INT-IND).

INT-IND (necessity of the two-part design, independence theorem, self-contained statement) There exists a family of entire functions $f_\varepsilon(z) = \cosh \pi\, z - \varepsilon\cdot\cos \pi\, z$ ($z = s - \frac{1}{2}$, $0 < \varepsilon < 1$) satisfying all internal structure of this paper — entire (A1 class); **evenness $f_\varepsilon(-z) = f_\varepsilon(z)$ is the reflection $z \mapsto -z$, which in the original variable is exactly $s \mapsto 1-s$: this is the degenerate form of the functional-equation reflection (A5′ with $\chi \equiv 1$), not an extra symmetry assumption**; real on the real axis: $f_\varepsilon(it) = \cos \pi\, t - \varepsilon \cosh \pi\, t \in \mathbb{R}$ (A4 class). Yet f_ε has infinitely many off-line zeros: for $z = x+iy$ with $y \to +\infty$, the dominant balance $e^{\pi x}(1+o(1)) = \varepsilon\, e^{\pi y}(1+o(1))$ gives $y = x - (\ln \varepsilon)/\pi + o(1)$: every $x_0 > 0$ yields a zero with Re $z = x_0 \neq 0$. **Hence the zero side alone (A1/A2/A4/A5′) cannot discharge A3 — this is precisely why the proof is two-part: the lock side supplies the missing half, and the welding convergence below is the joint proof of A3. Scope (v3.0).** INT-IND audits direct deduction within Axioms$_0$; it never audited the convolution layer (the $\langle i\rangle$ generator, the $\hat{c}_2$ readout, the idempotent spectrum, the $\mathbb{Z}_4$ lattice) introduced in the terminal block below. The correct reading is "not provable by internal direct deduction", not "not provable": the increment is the object $0i$ (the external input; the convolution machinery is its operation system) and opens the i-certification paradigm (translation, proved → discretization, proved → certified execution, rate measured). Toy-model check: f_ε lives in the same convolution algebra (M1/M5 hold for every entire function; its signature lies legally off the lattice) — the increment does not exclude the toy; it translates the problem into a discrete coordinate system the toy too must answer, while "the answer is on the lattice" is carried separately by the ζ-specifics (the type gate T5.4, the θ skeleton). No circularity.

Master reduction (T-MR). The two sides meet at A3 in the form of an exact reduction. (i) Every zero of ζ is a $(0, 0i)$ welding event, and the zero set is discrete in the strip, so every zero has a well-defined height γ (M1–M6, M3). (ii) At every zero the β count reads its position: the walk's ridge term at that zero is $x^{\frac{1}{2}+\beta(\rho)}/((\frac{1}{2}+\beta(\rho))\rho)$ (E1+E2), whose log-log slope equals $\frac{1}{2}+\beta(\rho)$ with $\beta(\rho) := \mathrm{Re}\,\rho$; the subtraction protocol converts the measured slope into the readout of $\beta(\rho)$ (Theorems 6.3, 6.4; instrument faithfulness, Part 12 L2). (iii) If the zeros lie on a single vertical line, that line is Re $s = \frac{1}{2}$ (Corollary 10.1). Conclusion: **RH $\iff$ every zero's β-readout equals $\frac{1}{2}$** — equivalently, every β-offset $\beta(\rho) - \frac{1}{2}$ equals 0 — and the readout is executable per

zero. □

(Integrated-edition upgrade, 2026-10-24 round, audit-trailed: this block replaces the former Theorem-ization checklist *note; the three obligations recorded there — subtract the archimedean/χ drift, prove the existence of $\tilde{M}$, derive the ∓ 1 jumps — are discharged below at theorem level. Toolbox: CR balance (proved), L4 (proved), A4/A5, Jensen/argument principle, and the classical growth bound $\log|\zeta| = O(\log t)$ as declared inputs, same declaration practice as E1/E3.)*

Theorem-ization of T-attr-M (v4.0): the kink statements at theorem level. For σ real and $T > 0$ write $M(\sigma, T) = \frac{1}{2T}\int_{-T}^{T} \log|\zeta(\sigma+it)|\, dt$ ($\log|\zeta|$ is single-valued; no branch is chosen). All statements below are *exact and unconditional*.

T-attr-M.1 (chi-drift subtraction; from L4). The Stirling computation of L4 gives $\log|\chi(\sigma+it)| = (\frac{1}{2}-\sigma)\log(|t|/2\pi) + g(\sigma,t)$ with $g(\sigma,t) = O_\sigma(1/t)$, uniformly on compact σ-sets. Define the model drift $D(\sigma,T) = (\frac{1}{2}-\sigma)\,\frac{1}{2T}\int_{-T}^{T}\log(|t|/2\pi)\,dt = (\frac{1}{2}-\sigma)\,[\log(T/2\pi)-1] + O(1/T)$ and the drift-subtracted mean $\tilde{M}(\sigma,T) := M(\sigma,T) - D(\sigma,T)$. □

T-attr-M.2 (exact symmetry; from A5, no limit needed). Substituting $\zeta(s) = \chi(s)\zeta(1-s)$ into the definition and using T-attr-M.1,

$$\tilde{M}(\sigma,T) = \tilde{M}(1-\sigma,T) + O_\sigma(1/T) \qquad \text{for all } \sigma, T.$$

For $\sigma > 1$ the Dirichlet series converges absolutely and the oscillatory t-mean vanishes termwise, so $\lim_{T\to\infty}\tilde{M}(\sigma,T) = 0$ for $\sigma > 1$ (existence included); by the symmetry the same holds for $\sigma < 0$. □

T-attr-M.3 (the exact finite-T kink theorem; CR balance + argument principle). By the balance equation (proved) and Jensen's argument principle on the rectangle $[\sigma,\sigma_0]\times[-T,T]$ with $\sigma_0 > 1$ fixed (right edge $O(1)$ since ζ'/ζ is bounded on $\mathrm{Re} = \sigma_0$; top/bottom edges $O(\log T)$ by the classical growth bound $\log|\zeta(s)| = O(\log|t|)$ on compact σ-strips away from zeros — *declared input, classical route*),

$$\partial_\sigma \tilde{M}(\sigma,T) = \frac{\pi\, N_\sigma(T)}{T} + O\!\left(\frac{\log T}{T}\right), \qquad N_\sigma(T) := \#\{\rho : \beta(\rho) > \sigma,\ |\gamma(\rho)| < T\}.$$

Hence *the σ-slope of $\tilde{M}(\cdot,T)$ is, up to $O(\log T/T)$, a step function of σ whose jumps sit exactly at the abscissae $\beta(\rho)$ of the zeros, with jump size $\pi\, n_{\beta_0}(T)/T$ per zero family* — exact, additive, and never-canceling (each zero contributes one positive unit at its own abscissa; its conjugation and functional-equation mirrors contribute at theirs). □

T-attr-M.4 (the ∓ 1 anchor jump at $\frac{1}{2}$; from E3 + symmetry). For $\delta < \min_\rho|\beta(\rho) - \frac{1}{2}|$, the band $(\frac{1}{2}-\delta, \frac{1}{2}+\delta)$ contains exactly the on-line zeros, so with the Riemann–von Mangoldt normalization (E3) the normalized slope jump is

$$\lim_{\delta\to 0^+}\ \limsup_{T\to\infty}\ \frac{\partial_\sigma\tilde{M}(\frac{1}{2}-\delta,T) - \partial_\sigma\tilde{M}(\frac{1}{2}+\delta,T)}{\log(T/2\pi)} = 1,$$

the χ-anchor kink; the sign flips across $\frac{1}{2}$ (∓ 1). □

T-attr-M.5 (face (iii) at theorem level).

$$\mathrm{RH} \iff N_\sigma(T) = 0 \ \ \forall\,\sigma \in (\tfrac{1}{2},1)\ \forall T \iff \tilde{M}(\cdot,T) \text{ is } \sigma\text{-affine on } (\tfrac{1}{2},1) \text{ for every } T.$$

Proof. The first equivalence is definitional (conjugation symmetry reduces the left half-plane); the second is T-attr-M.3: the slope is constant on $(\frac{1}{2},1)$ iff no jump point $\beta(\rho)$ lies there. The $T\to\infty$ normalized form ($\tilde{M}$ piecewise-linear, kink set $= \{\frac{1}{2}\}\cup\{\beta_0\}$ of positive-density families,

slope jumps ∓ 1 at $\frac{1}{2}$) follows under the convergence of the zero densities and is stated as the corresponding corollary; flatness of the T-limit detects exactly the positive-density off-line families — the single-zero content of the kinks is fully carried by the exact finite-T form above, which is the theorem-level statement used here. □

Status update (replaces the former theorem-ization checklist): obligations discharged — drift subtracted (T-attr-M.1), existence established where asserted (T-attr-M.2, safe half-plane; finite-T exactness throughout), ∓ 1 jumps derived (T-attr-M.4); face (iii) of the equivalence map is now a theorem (T-attr-M.5).

References ported into this package. **Record-blind-bit meta-theorem (ported, within the semantic closure)**: an isotype meta-theorem adds no information to true premises — unless a new record connection; this separates conclusions derivable from recorded structure alone from those needing a new connection, and S-flatness is of the latter kind (the content of the second half of the independence double-proof). **INT-6 (ported)**: no zeros on $\sigma = 1$ — the internal classical proof via the Euler product and a trigonometric identity (de la Vallée Poussin).

Information content of the named last asset. The prime phase reading continued into the strip equals $-\arg\zeta$ there ($\log\zeta = -\sum_p \log(1-p^{-s})$ for $\operatorname{Re} s > 1$, by continuation), and its branch points are exactly the zeros: computing its full phase structure is informationally RH itself. The three-layer landing is exact — and it lands on an RH-equivalent.

Instrument physics. Kinks exist only at $T \to \infty$; finite-T means are smooth; resolving a kink at $\frac{1}{2}+\varepsilon$ requires precision diverging as $\varepsilon \to 0^+$ — the same asymptotic wall as the β count, shared by both executive routes.

σ-quantization lemma (proposed form; the named remaining lemma). The real parts of zeros are quantized: there exist a lattice $\Lambda_\sigma \subset (\frac{1}{2}, 1)$ and a buffer $b > 0$ such that (i) the solution set of the exact welding-closure equation is contained in Λ_σ; (ii) the b-neighbourhood of each lattice point is a buffer band (undecidable to detection within the band); (iii) outside the buffer bands the theorem excludes strictly ($E \neq 0$). The two boundaries (theorem boundary, measurement boundary) are thereby guarded exactly. Known and proved: the value-plane quantization (the $\pi/2$ pin lattice) and the height quantization (the pin ladder, spacing $2\pi/\ln(t/2\pi)$, T-LAD-1). The value-plane quantization is derived as a necessary consequence of the two readings: full balance + the sin/cos two-pure conditions imply $\vartheta \equiv 0 \pmod{\pi/2}$ (axis lattice L1, quantum-number degeneracy L4, closure). The σ-direction quantization is the missing link between them; its proof program: find the two-pure conditions of the deviation β (the analogues of $\cos\vartheta = 0$ / $\sin\vartheta = 0$ for the position deviation), then the same balance-plus-two-pure mechanism quantizes the real part. Note: if $\Lambda_\sigma = \emptyset$ the lemma implies RH; the lemma is therefore of RH strength, and its proof is the door.

Engine of quantization (T-Q0; the readout quartet M128–M131, self-contained — micro-proofs in Appendix A). Quantization is a theorem of readout structure, not a postulate: an idempotent readout has spectrum $\{0,1\}$ (M129); the projection lattice stays two-valued (M130); the readout basis is 4 (M128, the V_4 of the zero family); hence the phase quantum $\pi/2$ (M131, $U(1)/V_4$); $\hbar$ is the conversion constant at the boundary, and the classical world is the inverse limit of readout-basis proliferation ($\varepsilon_{\text{cross}} = \pi/N$). Transfer to the position deviation: welding is itself an idempotent readout on the σ-line (a death event either closes on the seam or it does not), so the T-Q0 machine applies; the open step of the transfer is the position readout's basis and balance law (the analogue of V_4 / the $\pi/2$ lattice for the real direction).

Premise dependency of the chain (per the 8.7 derivation; the 8.7 derivation = internal folio of the engine chain, lineage label, not load-bearing). Link-by-link: the idempotent-spectrum theorem (M129), the Englert identity, and the OML non-distributivity display are framework-free

(standard linear/operator algebra, proved in five lines). The engine links are framework-native: (i) readout = projection = idempotent (M113, the exclusion of idempotent readouts from the M-algebra) is a readout-classification premise; (iv) the pin-phase single-reading zero equations $\cos\vartheta = 0$, $\sin\vartheta = 0$ with exactly two roots per period are lock-axis/$[0, 0i]$ premises; (v) the readout basis 4 is the zero family V_4 ($\langle i\rangle \cdot 0_1$) of the $[0, 0i]$ mathematics. Hence quantization-as-theorem is a framework-internal theorem: it grows from the readout premise and the $[0, 0i]$ pin/zero-family premises jointly; without these two supports the first and the counting links do not exist. This is the strength of the framework (no quantization postulate) and its boundary (the theorem quantizes readings inside the framework — the value plane).

Slice theorem (T-TV). For every finite height bound T the instrument executes a certified slice: (a) rigorous zero enumeration below T (Riemann–von Mangoldt counting gate; Turing's method as the counting input); (b) at each enumerated zero, the subtraction-protocol readout (Theorem 6.4) under the self-certification gates (P0–P5, Part 12). The outcome is a proved finite theorem ***Rigor source of $S(T)$: each executed slice is a finite theorem whose proof obligations are contained in this package; the union over T is RH (Part 12 closure semantics).***

$S(T)$: all zeros below T lie on Re $s = \frac{1}{2}$, every one of whose proof obligations is contained in this package. Each executed slice is one faithful slice of RH (Part 12 sweep protocol), and RH is the asymptotic union of the slice theorems as $T \to \infty$ — the same proof architecture as the four-color theorem (reduction + certified execution), with the certificate here being the reduction T-MR plus the per-slice protocols. □

Attribution, mean-function form (T-attr-M; the balance equation and the exact remaining statement). The exact balance equation is the CR mean duality: $\partial_\sigma M(\sigma, T) = \frac{1}{2T}[\arg\zeta(\sigma+iT) - \arg\zeta(\sigma-iT)]$ — the σ-derivative of the modulus reading's t-mean equals the phase reading's mean motion; **exact on every holomorphic band, with no $\sigma = \frac{1}{2}$ assumption**. Its $T \to \infty$ form: the normalized mean function $\tilde{M}(\sigma)$ exists, is piecewise linear, and its kink set $= \{\frac{1}{2}\} \cup \{\beta_0 : \text{off-line vortex families}\}$ (Jensen + Littlewood counting; χ contributes the $\frac{1}{2}$ kink, slope jump ∓ 1). Consequences: (i) an off-line zero at β_0 contributes an **exact, additive, never-canceling kink** at β_0 — the "original expression + infinitesimal" obstruction in mean-function form; (ii) **RH $\iff$ kink set $= \{\frac{1}{2}\} \iff \tilde{M}$ flat on $(\frac{1}{2}, 1)$**; (iii) by the functional-equation and conjugation symmetries the gap compresses to **one exact statement: $S(\sigma) = (\tilde{M}(\sigma) + \tilde{M}(1-\sigma))/2$ is flat on $(\frac{1}{2}, 1)$**. Independence (proved): INT-IND together with the record-blindness meta-theorem double-prove that no proof of S-flatness exists from the local-death axioms by direct deduction (scope: the paradigm note of the terminal block); the sole remaining internal asset is named — the **full prime phase reading** $\sum_p \arg(1 - p^{-s})$ under analytic continuation into the strip (its trigonometric skeleton already excludes $\sigma = 1$; the flatness statement needs its full phase structure). □

Status. The proof layer is complete: T-MR closes the two-part convergence at A3 as an exact iff-reduction. The instrument layer executes the slice theorems T-TV: each executed slice $S(T)$ is a theorem, and the union over T is the statement RH. The instrument's continuing execution is the verification program of the slice theorems — the sole remaining activity of the program, by the design of the architecture.

The named lemma, restated in non-existence form (headline PROPOSED and open; every premise cited below is proved in this package). The named remaining lemma admits a Euclid-genre restatement: *no remote unreasonable solution exists* — ζ has no off-line zero; equivalently, every weld event registers on the unique lock axis. The genre is structural non-existence, not exclusion by search: the claim is not "no counterexample has been found" but

"any assumed instance generates a contradiction from the lock / seam / balance machinery" — a finite construction against a universal negation, orthogonal to the infinite-execution issue. Five results of the same structural-exclusion genre already stand proved in this package: no zeros on $\sigma = 1$ and the zero-free region (INT-6; Part 12, Layer 1); the zero forbidden disk (Theorem 7.5); the finite-x forbidden-region theorem (Theorem 7.9); the confinement of any off-line candidate to the thinning strip $K = \{\frac{1}{2} < a < 1 - c/\ln\gamma\}$ (Part 12, Layer 1); and the quantum-limit finite-segment theorem (Part 12, Layer 4: single-ray certification is a finite computation). The proposed theorem is the completion of this series: non-existence of the escape family itself.

Supplementary argument (three premises; all proved). (P1) *The lock line is a theorem-object*: existence, uniqueness and the anchor of the lock axis are proved (the lock-axis family, Theorem 3.4; the anchor fixed by the $(1+i)$-power operation, Theorem 3.7) — the line's existence requires no observation. (P2) *Observation locates zeros; it does not ground the line's truth*: line-membership credentials are uniform across the quantum cone — inside and outside, the credentials are the weld structure (M4) and the pin lattice (B.3/B.4), the same structural certificates, independent of observability. (P3) *Structural truth-inheritance into the quantum zone*: the lock axis is height-unbounded (the lock theorems carry no height limitation), and the seam theorems govern the manner of approach (B.3/B.4: the two faces pin at the lattice, the pair-sum sharpens quadratically) — the interior cannot be arbitrary, because any off-line candidate must approach the pinned boundary behaviour. *Conclusion (within the designed interface A3).* The proposition "computed solutions lie on the line" holds inside the quantum zone at the same epistemic grade as outside it — structural, not observational. Observational refutation inside the cone is impossible by theorem (the blindness register: Part 12, Layer 4 and the numerical appendix); the supplementary argument supplies the maximal epistemic content attainable about the interior: boundary-law inheritance.

Equivalence map: one open content, five faces. The named remaining lemma admits exactly five faces (the fifth added in v3.0): *(i)* non-existence of off-line zeros (the Euclid genre above); *(ii)* the σ-quantization statement ($\Lambda_\sigma = \emptyset$; proposed form above); *(iii)* S-flatness of the symmetrized mean function on $(\frac{1}{2}, 1)$ (T-attr-M); *(iv)* the universal archival lock condition (every zero satisfies the one-equation lock condition; the counterproof front matter); *(v)* the i-language terminal form — every zero satisfies $m_{\text{type}} = m + 2w_k$ with $w_k \in \{0, -1\}$, equivalently $\forall k\ \hat{c}_2(\mu_k) = -1$, equivalently the face ledger never leaves the two registers $[+0i, -0i]$ (the terminal block below). Completion of any one face closes A3, hence RH: (i) is A3 itself; (iii) is proved equivalent to RH at theorem level (T-attr-M.5, exact finite-T form) and (iv) in the archival-lock synthesis; (ii) implies RH since $\Lambda_\sigma = \emptyset$ leaves the welding-closure equation without solutions, and RH implies it trivially; (v) is pointwise equivalent to (i) through the weld-signature computation (terminal block, Theorem B; the directionality register of item (j) therein). The choice among faces is strategic, not mathematical; the Euclid genre is recommended as the front door on two grounds: its finite-construction shape bypasses the infinite-execution issue, and five monuments of the same genre already stand in this package; face (v) is the recommended executive door (a per-point bit, a measured linear rate; the parity bit executed at 10^4+ points, the register refinement verified at 8 points to machine precision). *(Integrated edition, v4.0: Part II proves three further faces of the same content — the period face, the seam face, and the spectrum face; eight faces, one content, item (y) of Part II.)*

Location correction (audit-trailed). The prime side's contribution to this framework is fully harvested: through rotation, the landing points (angle, modulus) of certified zeros enter the i-axis frame as lattice constraints (Part 12, Layer 3; the characteristic cross), and the skeleton phase θ is χ-issued. Within the discrimination chain the prime side and the i-side are rotation-equivalent. The residual content of the named lemma is therefore not resident in the prime side

as a separate object: under the projection, the remaining irreducibility is that $0i$ is unreachable, the quantum zone is unprobeable, and the far-i problem is in principle unobservable (computable without being knowable). Any earlier "prime-side asset" phrasing is superseded hereby; the audit trail retains both wordings. *(Register reconciliation, 2026-10-24, audit-trailed: the "unprobeable / unobservable" wording of this correction is the epistemic register only; the v3.0 re-narration below governs the quantitative content — the quantum zone is the measured precision–height scaling law of the angular readout, terminal block (b), and no universal precision ceiling is asserted or estimable. The two wordings name the same boundary at two registers; the audit trail retains both.)*

Boundary (honest; unchanged). The non-existence statement is of RH strength: its completion equals A3, hence RH. No internal shortcut exists (INT-IND together with the type gate; the record-blindness meta-theorem). Within this framework's own standard of completeness — proved closure of the provable domain, certified boundary, theorem-answered limit, and the maximality established above — the loop is closed; within the classical finitary standard, the named lemma stands. Both statements are recorded, and this package does not conflate them.

Terminal block (v3.0+) · The named lemma in i-language final form; the endpoint $[+0i, -0i]$. Status discipline of this block: every premise cited is proved in this package or explicitly labeled; the headline statement (the named lemma, final form) remains open at RH strength. What changes in v3.0 is the shape of the open content: the reduction is carried to its terminus, the terminus lies on proved territory (the seam's two faces, Theorems B.3/B.4), and the open residue is compressed to a one-bit, pointwise-checkable integer statement with a measured scaling law. All numerics of this block are reproduced by `code/final_instruments_2026-09-30.py` (zero networking; 29/29 checks PASS). (v3.0 also settles the status of the quantum layer: it is retained, with sharpened semantics — the "quantum zone" is not an unprobeable wall but the measured precision–height scaling law of the angular readout (item (b)); and the quantization engine T-Q0 is load-bearing for this block itself, since the two-valuedness in Theorem B is the idempotent spectrum of T-Q0(i)/M129. Excising the quantum layer would break the chain at the convolution relation, at face (ii) of the equivalence map, and at Layer 4 of the counterproof; it is therefore kept and re-narrated, not removed.)

(a) Configuration issuing (unconditional; Hadamard). For every point, on the line or not, the configuration of all zeros issues the pin direction at ρ_k: from $\xi(s) = e^{A+Bs}\prod_\rho(1 - s/\rho)e^{s/\rho}$ (A, B real),

$$\arg\xi'(\rho_k) \;=\; B\gamma_k + \pi - \arg\rho_k + \operatorname{Im} U_k \pmod{2\pi}, \qquad U_k \;=\; \sum_{j\neq k}\Big[\operatorname{Log}\Big(1-\frac{\rho_k}{\rho_j}\Big)\Big(1-\frac{\rho_k}{\bar\rho_j}\Big)+\frac{\rho_k}{|\rho_j|^2}\Big]$$

(pair-grouped, convergent), with $B = \xi'(0)/\xi(0) = -0.0230957089661$ (real; determined numerically this round). Derivation key: pairing each zero with its conjugate (its mirror across the real axis), the paired factor expands as $1 - s/|\rho_j|^2 + s^2/|\rho_j|^2$ on the line, whose log cancels the compensating term $s/|\rho_j|^2$ to first order, leaving $O(\operatorname{Im}(s^2)/\gamma_j^2)$ — the modulus reading diverges while the angular reading converges; this is the geometric root of every scaling law below. The lock's ξ-avatar (necessary direction proved; converse open): on the line $\Rightarrow \arg\xi' \equiv \pi/2 \pmod{\pi}$ (i.e. $\operatorname{Re}\xi' = 0$). (Round-136 audit correction: the converse direction — $\arg\xi' \equiv \pi/2 \pmod{\pi} \Rightarrow$ on the line — is not proved: off the line $\arg E$ deviates from $\pi + \theta$ by a computable quantity $\Delta(\beta, \gamma)$, so the congruence defines a curve off the line (a branch of the F-landscape), not a characterization of the line. This paper uses only the $\Rightarrow$ direction; wherever the formula serves as a criterion it is labeled as such. Anti-circularity: the part derived via L6 is proved on the line only (L6 used $1 - \bar z = z$); as an arbitrary-point criterion it is used, never as a theorem input.)

(b) The lock-needle equation and the data-wall law. Local form (proved; 9–10 digits machine-verified): $\zeta'/\zeta(s) = 1/u + \zeta''/(2\zeta')(\rho) + O(u)$, $u = s - \rho$; the radial residual converges monotonically to a real constant (measured spread 2.3×10^{-3}); the angular residual is a tail (inflated at Lehmer pairs). **Data-wall law (measured fit, 45 points, rms 1.9×10^{-4}):**

$$\mathrm{dev}(k, N) \;=\; \gamma_k \left(0.01435 \, \log^2 N + 0.00729\right) / \, N.$$

Machine precision ($\varepsilon = 10^{-10}$) per-point verification at $\gamma = 500$ needs $N \sim 7.3 \times 10^{13}$ ($\gamma_N \sim 10^{13}$); the parity readout (a flip requires dev $> \pi/4$) needs only $N^* \approx 0.6\,\gamma_k$ — **a one-bit readout threshold lies eleven orders of magnitude below the data wall**. The quantum zone hereby receives its precise quantitative meaning: not "unprobeable", but the measured scaling law of angular-readout precision against height. *(2026-10-24, audit-trailed: no universal precision ceiling is asserted — the law is a measured fit over the executed range (45 points), and an absolute upper bound of angular-readout precision is in principle not estimable from any finite run; the earlier 10^{-14}-register floor reading was the v2 pipeline's seed-stage artifact, corrected in (x.1).)*

(c) Correct/incorrect discrimination — the unique parameter m. Experiment ($N = 4000$, no ζ evaluation): the $(1+i)$-power orbit of a base $z = e^{\ell + i\vartheta}$, pinned against the lock lattice $\frac{\pi}{2}\mathbb{Z}$:

Base	$m = (\ell + \vartheta)/(\pi/2)$	$d_{\max}$	Verdict
i-type ($\ell = 0$, $\vartheta = \pi/2$)	1	4.5×10^{-13}	correct: cross lock + strict contraction $e^{-n\pi/2}$
$m = 0/2/3$	integer	$\leq 1.8 \times 10^{-12}$	correct (axis lock; $m = 2$ degenerate weld)
$m = 1/2,\ 1/3$	rational non-integer	$0.79\,/\,0.52$	incorrect: never on the lattice
$m = \sqrt{2} - 1$	irrational	0.79	incorrect: Weyl dispersion
$m = 1 + 10^{-6}$	near-miss	6.3×10^{-3}	incorrect: linear drift, not repaired

The unique discriminating parameter is the integrality of m; the weld readability bit is $\kappa := m \bmod 2$. **Near-miss law**: an off-lattice deviation is not repaired by the orbit dynamics — the lattice is an open-set condition, not an attractor.

(d) Two-axis squeeze; the three pillars carrying κ. The lock lattice $\frac{\pi}{2}\mathbb{Z}$ splits by parity into two axes (value-plane angular readings): even lattice points = the real-axis direction, odd = the i-axis direction; the lock set $\{\ell + \vartheta \equiv 0 \pmod{\pi/2}\}$ is a pair of 45° lines (T3.4); rotation by $\pi/4$ gives the characteristic cross — the only survival space of a correct solution is the cross's two arms. **Arm A (the real axis) is explicitly certified end-to-end (measured, L6-free, unconditional)**: $\xi'(-2) = -0.06565338351$, $\xi'(-4) = -0.1588999345$, $\xi'(-6) = -0.3620679864$, the angular readings exact multiples of π (imaginary parts 0.00; Schwarz reflection A4 + the definition of ξ). **The three pillars carrying κ**: (1) topological pinning — the charge parity is an argument-principle integer charge (A2), no continuous drift; (2) readout economy — the scaling law of (b); (3) jump prohibition — odd → even is a degenerate weld (T-P5 form), and degeneration can occur only on the lock line (the mirror-pole $1/(2u_0)$ argument). κ congruence (parity face): the parity of the issued angular reading ≡ the charge parity (mod 2).

(e) The functional-equation four-fold closure attack — a negative theorem (conditional form). Differentiating $\zeta(s) = \chi(s)\zeta(1-s)$ at a zero (note that $\mathrm{Im}(1-s) = -\gamma$) gives $\zeta'(1-s) = -\zeta'(s)/\chi(s)$ (machine-verified residual 6.6×10^{-51} at 50-digit precision). If an off-line zero existed with all four of its mirrors satisfying $F = 0$ (input: the contrapositive of this paper's counterproof theorem "$F \neq 0 \Rightarrow \beta = \frac{1}{2}$" — no circularity), the necessary condition would be

$$H(\beta, \gamma) \;:=\; \arg \chi(\beta + i\gamma) + 2\theta(\gamma) - \pi \;\equiv\; 0 \pmod{\pi/2}.$$

Negative theorem (conditional form; round-136 audit label): on the five-layer reduction's premise branch (an off-line zero with $F = 0$), H is vacuous on the whole open strip (margin $\geq 300\times$; $\partial H/\partial\beta|_{\text{line}} \approx 3 \times 10^{-7}$); the mechanism is the exact cancellation of the $-\pi/2$ imaginary parts of $(\pi/2)\cot(\pi/4 + i\pi\gamma/2)$ and $\operatorname{Im}\psi(\frac{1}{2} - i\gamma)$ (the other branch is governed by the five-layer reduction's residue). The readable projection of the FE mirrors therefore imposes no further obstruction on an off-line zero; the content resides in the mod-2π layer data $w(n)$ (an instance of the **projection-loss family** — records that lose a coordinate under projection; its seam case is the weld instance W2 of L7, Part 12).

(f) The layer ladder, the Gram ladder, and the blindness theorem. The θ-layer function $L(\gamma) := 2\theta(\gamma)/\pi$ (loggamma analytic at $\text{Re} = \frac{1}{4}$, naturally continuous — the honest channel is free). **Half-turn law (true)**: mean $\Delta L = 1.9951$, min 0.716. Layer content = spectral fluctuation: $L \bmod 4 = 2(k - 1 - S_k) \bmod 4$ (an identity); $S_k = (\theta(g_k) - \theta(\gamma_k))/\pi \in [-0.0063, 1.0065]$, almost always positive — **the i-vortex layer ladder is the Gram ladder** (g_k the Gram points); S is the dictionary incompleteness (Selberg CLT, variance $(1/\pi^2)\log\log T$). [Errata chain: the round-128 "quasi-parity law" was false (48/160 violations) and was retracted on the spot in round 129 — the discipline that numerics must be regenerated before entering the record caught its own violation; the audit trail is retained in the companion hash-chain record (HASH_CHAIN.md).] **Blindness theorem**: $N(T)$ counts without distinguishing on-line from off-line — off-line exclusion must go through the angular reading (κ); the counting reading is blind to position (the same projection-loss family).

(g) The convolution relation (the $\hat{c}_2$ theorems). The pinned-circle convolution algebra $M(S^1)$ (T-L14/15); the i-vortex generator $g := \delta_{\pi/2}$; $\langle g\rangle \cong \mathbb{Z}_4 = V_4$. **Theorem A**: the layer ladder is the convolution integral of the half-step kernel g^2 (the Gram step; measured 1.9951). **Theorem B**: for the weld signature measure $\mu_k = (\delta_{\vartheta^+} + \delta_{\vartheta^-})/2$ (charge $m \in \mathbb{Z}$, pinned by A2/M1):

$$\hat{c}_2(\mu_k) \;=\; \tfrac{1}{2}\textstyle\sum e^{-2i\vartheta^{\pm}} \;=\; e^{-i\pi\, m_{\text{type}}}; \qquad \text{integer charge} \Rightarrow |\hat{c}_2| = 1, \;\; \hat{c}_2 \in \{\pm 1\} \Longleftrightarrow m_{\text{type}} \in \mathbb{Z};$$

the value $(-1)^{m_{\text{type}}}$ is the κ code. The two-valuedness is the idempotent spectrum of the even subalgebra (T-Q0(i)/M129: readout = projection = idempotent $\Rightarrow$ spectrum $\{0, 1\}$) — a spectrum theorem, not a convention. **Theorem C**: $\hat{c}_2(g) = -1$ — the generator itself is of the odd class; locking = $\forall k:\ \hat{c}_2(\mu_k) = -1$; off-lattice = the phase leaving $\{\pm 1\}$ = outside the type gate. (T110 composition closure = convolution semigroup closure; near-miss non-repair = the $\hat{c}_2$ phase drifting off the lattice at ε-rate.) Measured: for odd m the full-period orbit average of $\hat{c}_2$ is 0 (parity does not live in the orbit average); true-ladder windows give $|\hat{c}_2| = 0.47 \to 0.24$, phase $\approx -0.97\pi$ (odd class; the CLT decay is inseparable at this scale — honest label).

(h) The Euclid construction formula (in i-language). Three structures: the infinite expression of i ($\langle i\rangle \cong \mathbb{Z}$, the layer integers w); the idempotents of i ($e_{\pm} = (I \pm iJ)/2$, T-L10; $P_\kappa = (\delta_0 + \delta_\pi)/2$ = the $\mathbb{Z}_2$-Haar, T-L15; the T-Q0(i) spectrum $\{0, 1\}$); the i-vortex decomposition ($\mathbb{Z}_4 = \langle g\rangle$; the inter-lattice direction = the fiber 0, the T-L3 two-value lemma). **Construction (given by convolution)**: $E(\rho^*) := P_\kappa * \mu^* = (\mu^* + \mu^* * g^2)/2$. Odd integer $m^* \to$ fixed point; even integer $\to$ face exchange; $m^* \notin \mathbb{Z} \to$ the phase reading $e^{-i\pi m^*} \notin \{\pm 1\}$, outside the value range. (Round-136 audit correction: the draft's "contradiction with the idempotent-spectrum theorem" was an over-claim and is retracted — the spectrum theorem constrains the eigenvalues of the readout operator ($\in \{0, 1\}$), not a single reading directly.) The exact reduction delivered by the construction: **the named lemma $\Longleftrightarrow$ the parity readout's value range on weld signature measures is exactly** $\{\pm 1\}$ (the welded form of readout two-valuedness) — weld signature charges are integers (A2/M1), hence $\hat{c}_2$ is always unimodular, and the phase lying in $\{\pm 1\}$ or not is the integrality of m_{type} (the Theorem-B computation); the readout operator is idempotent and

reads -1 on every certified weld (B1: 10^4+; this round 160/160), so the value-range proposition is pointwise equivalent to the κ-congruence, the same content as $w_k \in \mathbb{Z}$ of item (j). □

(*reduction complete*) Fine-grained status note: the g^2-fixed-point defect is blind to near-misses (measured defect 0.00000 at $m = 1 + 10^{-6}$) — the discriminator is the phase two-valuedness, consistent with the near-miss law: the construction's force lives on the spectrum, not on the metric. Isomorphism with $p_1p_2p_3+1$: the list $\mathrm{L} \to$ the certified configurations; $\mathrm{P} \to$ the assumed off-line zero; $\prod \mathrm{L} + 1 \to P_\kappa * \mu^*$; $q \notin \mathrm{L} \to e^{-i\pi m^*} \notin \{\pm 1\}$; the three-line contradiction $\to$ the readout value-range proposition (= the welded form of idempotent-readout two-valuedness, open).

(i) The infinite decomposition and the endpoint $[+0i, -0i]$**.** The dyadic tower of $g = \delta_{\pi/2}$ refines through $\delta_{\pi/2^n}$ (the T-L8 radical tower; the total ramification point = the fiber 0). Measured linearization limit (rotation-matrix form): $(R(\theta) - I)/\theta \to J$ (i as infinitesimal generator), error $\theta/2$: 9.4×10^{-8} at $n = 24$, machine zero at $n = 48$. **The tower's terminus is the nilpotent embryo** εJ (T-L10) — and the embryo's death locus is the seam. **Direct observation (this round)**: $\arg \xi'(\rho_k) = \pm\pi/2$ strictly alternating with the parity of k ($k = 1..8$; mod-π exactly 0 at machine precision) — the seam's two faces are manifest in the ξ' readings themselves. **The endpoint is** $[+0i, -0i]$, the two faces (the co-management structure of B.3): every refinement level of the decomposition chain leads into proved territory, not into new open ground.

(j) The exact relation (final form v2). Behind the mod-2 congruence shadow lies an exact equality:

$$m_{\text{type}} \;=\; m + 2w_k$$

(w_k = the layer integer = the winding number of the configuration product around 0 = the $w(n)$ semantics = the face ledger). Derivation: the unwrapped issuing $\arg \xi'(\rho_k) = \pi/2 + \pi j_k + 2\pi W_k$ (branch bookkeeping; $j_k \in \{0, 1\}$ the sign face of Z', W_k the configuration winding); $m_{\text{type}} := 2 \arg \xi'/\pi$; the local charge m is pinned by the argument principle (A2); the face ledger $w_k := (m_{\text{type}} - m)/2 = (j_k - (m-1)/2) + 2W_k \in \mathbb{Z}$. Principal-branch verification ($k = 1..6$): $m = 1$; $\arg \xi' = \pm\pi/2$ alternating $\Rightarrow m_{\text{type}} = 1+2w_k$, w_k alternating $0/-1$. An off-spectrum reading $\Longleftrightarrow$ w non-integer $\Longleftrightarrow$ face-ledger rupture (compatible with the construction of (h)). **Named lemma (final form v2)**: every zero's configuration-issued phase satisfies $m_{\text{type}} = m + 2w_k$ (the charge m topologically pinned, the face ledger w_k a layer integer) — i.e. $\hat{c}_2(\mu_k) = -1$ and the faces alternate strictly. **Face-book narrowing (round 137)**: the L6 derivative gives $\zeta'(\rho) = -iZ'(\gamma)e^{-i\theta(\gamma)}$, hence $\arg \xi'(\rho) = \pi - (\pi/2)\,\mathrm{sgn}(Z'(\gamma))$ — the face's sign is the sign of Z'; Rolle's theorem plus zero interlacing force Z' to alternate between simple zeros, hence $w_k \in \{0, -1\}$ **strictly alternating** (condition: simplicity; B1 data 10^4+; the $\kappa = 0$ detector in place). Measured ($k = 1..8$, dps = 20): $\arg \xi'(\rho_k)/(\pi/2) = \pm 1.000000$ strictly alternating, $w = 0, -1, 0, -1, \ldots$ at machine precision. **The named lemma in its** i**0-purest form**: *the face ledger never leaves its registers — every weld event is registered within the two registers* $[+0i, -0i]$. Directionality (honest label): non-rupture $\Longleftarrow$ RH (Rolle; the proved direction); non-rupture $\Longrightarrow$ RH requires "an off-line zero carries no integer angular reading" — exactly the disjointness of the F-landscape's closed system (another face of the same named lemma; the two are equivalent under the five-layer reduction). Verification tiers: the parity shadow (the Z-sign reader; 10^4+ points read at negligible cost); the face-ledger registers (this round, $k = 1..8$ at machine precision); the exact face ledger (the weld detector = unwrapped semicircle accumulation; the scaling law is linear).

(k) Paradigm positioning (round 138). INT-IND's verdict domain is direct deduction, not all proof: it audits the absence, within Axioms_0, of a direct deductive chain reaching A3 (the toy model bears witness); it never audited the convolution layer — the convolution vocabulary (the $\langle i \rangle$ generator, $\hat{c}_2$, the idempotent spectrum, the $\mathbb{Z}_4$ lattice) is this process's increment. **Not provable by internal direct deduction (theorem)** $\neq$ **not provable (over-generalization).** The increment, dissected (three items, each locatable): (1) *discretization* — the idempotent

spectrum compresses the angular readout to $\{0, 1\}$, Theorem B compresses the weld signature $\hat{c}_2$ to $\{\pm 1\}$, the face ledger is compressed by Rolle to the two registers $\{0, -1\}$: a continuous problem becomes a per-point bit; (2) *the scaling law* — the data-wall law measured, bit flip $N^* \approx 0.6\gamma$ (the continuous version needs $N \sim 7 \times 10^{13}$): after discretization the verification is affordable; (3) *the endpoint reduction* — the tower decomposition ends at $[+0i, -0i]$ = B.3/B.4: the residue lands on proved territory. Toy-model check: f_ε lives in the same convolution algebra (M1/M5 hold for every entire function; its signature lies legally off the lattice) — hence the increment does not exclude the toy; it translates the problem into a discrete coordinate system the toy too must answer, and "the answer is on the lattice" is carried separately by the ζ-specifics (the type gate T5.4, the θ skeleton). No circularity. The two paradigms: the additive-coordinate paradigm = finite deductive chains; the i-paradigm = translation (proved) → discretization (proved) → certified execution (rate measured; partially executed, 10^4+). A generalization of the four-color pattern, with two additional features: the scaling law is a measured theorem and the endpoint lies on proved territory. **Framework self-description, revised**: the named lemma is of RH strength in the direct-deduction paradigm (INT-IND); in the i-certification paradigm it is the discrete universal of the face ledger's two registers (a linear scaling law; executed at 10^4+ points; endpoint proved). The two statements are compatible.

(l) Boundary and anti-circularity registry (final). (1) L6 and its ξ-avatar are proved on the line only; the issuing formula of (a) is unconditional as a per-point criterion and is never used as a theorem input. (2) The arm-A certification does not pass through L6 (Schwarz + the definition of ξ; unconditional). (3) The FE negative theorem's input is the contrapositive of this paper's counterproof theorem, not the named lemma itself. (4) The type gate is measured (1.06×10^{-3}); INT-IND governs the outside of the type; the in-type theorems are non-circular; this block does not close the named lemma (closure $\Longleftrightarrow$ RH). (5) (Round-136 circularity audit) the chain contains no circular argument; three over-claims were corrected on the spot: the bidirectional avatar reduced to the necessary direction (the converse = the F-landscape, open); step 3 of the construction reduced to a reduction (the value-range proposition), the direct "contradiction with the spectrum theorem" claim retracted; the negative theorem labeled in conditional form. The corrections do not alter the final form; the open status is cleaner after audit. (6) (Rounds 137–138) face-ledger narrowing ($w \in \{0, -1\}$; Rolle + interlacing, conditional on simplicity; $k = 1..8$ machine precision) and paradigm positioning (INT-IND governs direct deduction; the convolution layer is the increment: discretization / precision scaling / endpoint reduction; the toy-model invariance check, no circularity); the directionality label of non-rupture ⇒ RH (needs the F-landscape closed-system disjointness; equivalent under the five-layer reduction).

Appendix E · Lock Criterion (applied form of L2) and the V5 adjudication

Criterion An instrument-layer candidate event (e.g., a phase residual approaching some pin α_0, or a wandering depth approaching a constant) is judged a "**lock**" iff its pin sequence has tail diameter → 0 under window extension and under change of the observation kernel family (i.e., the archival limit exists and is directed — archive = 0_{α_0} per L2); a candidate surviving at a single point while weakening across windows is judged "**wandering**" (archive = the fiber $0 = (-\infty, \cdot)$, the other half of L2).

δ_ϑ **formal clause** The settling criterion has the equivalent δ-form: $\delta_\vartheta(\mathbf{t}) := \mathbf{dist}(\vartheta(\mathbf{t}) \bmod 2\pi, \vartheta_0)$ (period π when the welding reference $\vartheta_0 \in \pm\pi/2$). ϑ_0-lock $\Longleftrightarrow$ the δ_ϑ minimum settles across windows and kernels (tail diameter → 0, L2); otherwise wandering (archive fiber). Every detector

must pass a positive control (a synthetic settling signal must trigger) — measured: 12/12 positive hits, 0 hits on real windows.

Derivation status This criterion is a direct applied form of L2 with no new assumption: the settling criterion is in L2 (Part 2); changing windows/kernels merely changes the observation sequence, and the tail-diameter criterion is unchanged. Structural reading: lock = directed zero 0_{α_0} (phase reading settled), wander = fiber 0 (phase reading free) — every instrument-layer "candidate constant" debate is the two layers of zero becoming visible in the instrument layer.

V5 adjudication record (2026-09-27) Object: the $-\pi/4$ offset candidate of the wandering depth (the offset component of coincidence, 0.79 vs 0.7854). Evidence: single-point survival at x=5e5 with deviation 0.02%; six-window multi-kernel statistics (windows (50,80)/(100,130)/(150,180), kernels σ=0.2/0.5) with 0 hits.

Conclusion: wandering; archive = fiber 0; closed with a kept record. Unaffected parts: existence of the wandering lock line (Theorem 7.2), direction coincidence (Theorem 7.4, theorem-level), the structural verdict of coincidence (both locks i-directed, the cone-rotation relation). Upgrade threshold (quantified): if a future candidate survives at rate above noise across 6+ window× kernel combos, it is upgraded to a δ-lock event entering the official list (the former $-\pi/4$ single point fails this and is not reopened).

Appendix H′ · Pin-Analysis Practice Discipline (cross-zero-straddle measurement, the metrology of M4/M5)

First-order argument The pin formula pin(γ+y) $\approx$ arg ζ' + (π/2)·sign(y) — the existence of the antipodal pin difference π requires the two evaluation points to straddle the true zero. A location offset δ changes the sign to sign(y $\pm$ δ): when **$|\delta| \geq |$y$|$ the two points lie on the same side and the sign does not flip — the "identical twin pins" phenomenon is not measurement error but the same side's pin read twice**. Necessary condition for a valid signature: **$|\delta| < |$y$| \ll$ local linearity scale**.

Numerical reproduction (γ_1, Z-frame)

center	y = ±1e−4/1e−6	twin-pin reading	verdict
refined γ_1	±1e−6	0 / $-\pi$	antipodal ✓ (simple zero)
offset γ_1+6e−3	±1e−4	0 / 0	identical-twin-pin artifact (reproduced and archived)

Practice discipline: before pin analysis one must refine the center with mp.zeta (findroot / complex Newton) to $|\delta| \ll |$y$|$; RS location (6e−3 error) must not be used directly for pin analysis. This confirms by a second route: the order–welding dictionary is readable only under the straddling condition.

Z-frame pins vs raw pins (clarification) Z-frame pins (values multiplied by $e^{i\theta(\gamma)}$) are $0/\pi$ for **every** zero — this is L6's skeleton-antipodal coordinate, not the welding-direction reading; the welding direction of M4 is the **raw pin** arg $\zeta(\rho \pm$ iy$) = -\theta(\gamma) \pm \pi/2 +$ kπ (value plane). The ±0i criterion: raw pins $\equiv \pm\pi/2 \iff \theta(\gamma) \equiv \pi/2 \pmod{\pi}$. *Instance*: γ = 711.9002899143753… ($|\delta|$ = 1.14e−4) has raw pins $\pm(\pi/2 \mp$ 1.1e−4) — the first approximate ±0i welding instance (true zero $|\zeta|$ = 2.8e−39, simple, $|\zeta'|$ = 1.587, antipodal residual 4.7e−6).

Appendix A · Elementary Toolbox

Taylor expansion, identity theorem, argument principle (loop count = 2π × zero order), concavity of ln (Jensen), Stirling's formula for Γ, continuity and differentiability of holomorphic functions, the Vinogradov–Korobov ψ-error bound (input), Hurwitz's zero-limit theorem, Weierstrass M-test, the von Mangoldt explicit formula (input), Abel summation (input), Riemann–von Mangoldt zero counting (input) — all standard elementary statements, independent of any external numbering system.

(Integrated-edition addition, 2026-10-24, audit-trailed: the readout quartet behind T-Q0, proved inline so that no external dossier is referenced. ***M129****: an idempotent readout p ($p^2 = p$) has spectrum in $\{0,1\}$ — λ in the spectrum gives $\lambda^2-\lambda$ in the spectrum of $p^2-p=0$, hence $\lambda \in \{0,1\}$.* ***M130****: the projection lattice generated by commuting idempotents is two-valued — meet and join of $\{0,1\}$-valued idempotents stay $\{0,1\}$-valued, by distributivity of the Boolean algebra they generate.* ***M128****: the readout basis of the zero family is 4 — the V_4 of D2, four directed zeros under transport by i.* ***M131****: the phase quantum is $\pi/2$ — $U(1)/V_4$ has circumference $2\pi/4$; one quotient step is one quarter-turn.)*

Part 5 · Main Chain 12: Types and Lock Axes, Composition Closure

Operations manual: O1 cone probe: f $\mapsto f^{1+i}$ → equal-weight dual-channel readout (self-dual). O2 axis exchange: $f^{i(1+i)} = f^{-1+i}$ swaps channels (i(1+i) = −1+i; $\mathbb{C}^\times$ has no translation). O3 series expansion: termwise lock lines → unique summed slope (Stirling). O4 prime factorization: Euler product → prime-vortex components (D8). O5 explicit formula: termwise cancellation at K=γ_j → ridge slope = $\frac{1}{2}+\beta_j$ (the unique entry of β). O6 Euler-element limit: log(modulus)/log x → 1. O7 infinitesimal deformation: $\delta = \varepsilon(1+\mathrm{i})$, δ-derivative (1+i)∂_K, exact truncation.

5.1 The (1+i) infinitesimal chapter

Construction: $\delta := \varepsilon(1+\mathrm{i})$, $\varepsilon^2 = 0$. **Nilpotent = cone element**: first-order truncation is the whole algebra in the nilpotent setting, not an approximation. dexp(a+bε) := $e^a(1+b\varepsilon)$ (Taylor terminates exactly at ε^2).

Theorem 3.1 (e^δ = 1+δ): dexp(δ) = $e^0(1+(1+\mathrm{i})\varepsilon) = 1+\delta$. □

Theorem 3.2 (δ-derivative lemma) S(K) = $\Sigma\ w_n\ n^{-1/2-iK} \Longrightarrow$

$$S(K+\delta) = S(K) + \varepsilon(1+i)S\prime(K) \tag{11}$$

. *Proof* (five steps): (1) $n^{-i\delta} = 1+(-\ \mathrm{i}\delta)\log$ n (nilpotent exponential truncation); (2) $-\ \mathrm{i}\delta = \varepsilon(1-\ \mathrm{i})$; (3) S(K+$\delta$) = S(K)+$\varepsilon$(1− i)T, T := $\Sigma\ w_n$ log n·$n^{-1/2-iK}$; (4) S$\prime$ = − iT $\Longrightarrow$ T = iS$\prime$; (5) ε(1− i)·i = ε(1+i). □

Corollary (response limit = the intrinsic limit): $lim_{\varepsilon\to 0}\ \Delta$ S/ε = (1+i)S$\prime$.

The spectrum of four limits: 1 $\delta\to 0$ itself (trivial); 2 response limit (Theorem 3.2, proved); 3 series convergence limit (wing series → bounded almost-periodic $V_j \neq 0$, Theorem 6.2); 4 exponential limit (log(mod)/log x → 1, §5.4/Part 8).

Why the trend is observable under projection The cone probe projects the welded modulus–phase pair (modulus reading ℓ, phase reading ϑ) onto the two-channel readout (numerical channel

$\ell-\vartheta$, phase channel $\ell+\vartheta$): equal weight (self-duality $\Longrightarrow$ the trend direction is uniquely determined by the log-polar reading) + convergence (limit 4 exists in the projected readout) $\Longrightarrow$ **the trend is observable**: the two-channel projection yields the trend direction. (Consistent with L7: projection loses one dimension.)

Theorem 3.3 (δ-trend) $\partial_\mathrm{K}\ \log\ \tilde{F}|_{\gamma_j} = -\ \mathrm{i}\cdot\log(\mathrm{x/e})\cdot(1+\mathrm{o}(1))$ (**purely imaginary**: modulus stationary, phase self-rotating); $D_\delta\ \log\ \tilde{F} = (1-\ \mathrm{i})\cdot\log(\mathrm{x/e})\cdot(1+\mathrm{o}(1))$ (dual channels opposite-sign equal-amplitude, $(1-\ \mathrm{i})$ signature). *Proof*: ridge $\mathrm{T} = -\ \mathrm{x}/\rho_j$; $\partial_\mathrm{K}\ \mathrm{T} = \mathrm{i}(1/\rho_j)\Sigma\ \log\ \mathrm{n} = \mathrm{i}(1/\rho_j)(\mathrm{x}\ \log\ \mathrm{x} - \mathrm{x} + \mathrm{O}(\log\ \mathrm{x}))$ (Stirling); $\partial_\mathrm{K}\ \log\ \mathrm{T} = -\ \mathrm{i}(\log\ \mathrm{x}-1)(1+\mathrm{o}(1))$; $D_\delta = \varepsilon(1+\mathrm{i})\partial_\mathrm{K}\ \log\ \tilde{F}$ (Theorem 3.2) $= \varepsilon(1-\ \mathrm{i})\log(\mathrm{x/e})$. □

Four conditions (asserted in Part 10 with pointers): I trend (Theorem 3.3) II observation fidelity (equal weight + convergence here; O2) III series form (Theorem 6.2) IV limit existence (Theorem 4.4 + 6.3 + Data 6.5).

5.2 Matrix form and the closure (four-step logic)

Theorem 3.5 (matrix form of the infinitesimal) $\delta = \varepsilon(1+\mathrm{i})$ is the **single matrix** $\varepsilon\ \mathrm{C}$ in the welding plane, $\mathrm{C} = [[1,-1],[1,1]] = \sqrt{2}\cdot\mathrm{R}(\pi/4)$ — both directions built in, not two expressions; $\delta^2 = \varepsilon^2 C^2 = 0$ (matrix witness of nilpotency; agrees with D4: $(\varepsilon\ \mathrm{C})^2 = 0$ = the fiber embryo).

Theorem 3.6 (Euler element's trajectory line) The orbit $\mathrm{w(t)} = e^{(1+i)t}$ has welding coordinates $(\ell,\vartheta)(\mathrm{t}) = (\mathrm{t},\ \mathrm{t})$ — the **fixed line ℓ: $\vartheta = \ell$** (the 45° line = cone direction = J-fixed line). The trajectory is a straight line in matrix coordinates, not a spiral.

Theorem 3.7 (real anchor $\frac{1}{2}$) The real-axis position of ℓ is $\frac{1}{2}$, given by the i-vortex $(1+i)$-power Euler-element expression: the β-count identity slope $= \frac{1}{2}+\beta_j$ (Theorem 6.3) takes $\frac{1}{2}$ as origin — the deviation from $\frac{1}{2}$ is the object content β.

Theorem 3.8 (i-jump closure) The jump of i at $\frac{1}{2}$ is multiplication by i: ℓ unchanged (real anchor fixed), $\vartheta \mapsto \vartheta+\pi/2$ (ℓ moves in parallel within the welding plane) — the jump preserves the anchor; the $(1+\mathrm{i})$ and $(1-\ \mathrm{i})$ readings give the same β at $\frac{1}{2}$.

Type-internal closure complete: matrix $\delta \to$ line $\ell \to$ real anchor $\frac{1}{2} \to$ jump preservation. *Boundary*: the closure holds **within the i-vortex $(1+i)$-power type**; the universal version with $\beta = 0$ (= the content of RH) is reduced exactly (Part 4, T-attr-M) and executed slice by slice by the instrument (Part 12, T-TV).

5.3 The lock-axis theorem system

Semantics declaration (dual semantics, the unique declaration point of the whole system): (A) generator-orbit semantics (principal): $\mathrm{r} := i^{1+i}$, orbits = group generation r^n; **the power law holds strictly** ($r^a \cdot r^b = r^{a+b}$) — the semantics of §5.4 and all sum/product conclusions. (B) principal-branch pointwise semantics: direct computation $(i^n)^{1+i}$ (principal Log). It differs from (A) by the **sheet-winding factor**:

$$(i^n)^{1+i} = r^n \cdot e^{2\pi w(n)} \tag{12}$$

$$w(n) = \lceil n/4 - \tfrac{1}{2} \rceil \tag{13}$$

. Position (lock-axis) semantics unchanged ($e^{2\pi w}$ is a positive real); the cross-sheet cut-line steps (n=3,7,...) jump in modulus by $e^{2\pi}$; the power law holds up to the top cycle $e^{2\pi(w(a)+w(b)-w(a+b))}$ (at n=3 the ratio is exactly $e^{2\pi} = 535.5$). Lock-axis orbit pins = $\mathrm{n}\pi/2$ mod 2π ($\mathbb{Z}_4$ steps), w(n)

advancing one step every 4 — a spectral-$\mathbb{Z}_4$ lift reading (agreeing with the zero family $V_4 = \mathbb{Z}_4$ representatives of D2).

Lemma 4.1 i(1+i) = −(1− i) (vortex cone and seam cone differ by 90°). *Proof*: i+i^2 = i−1 = −(1− i). □

Lemma 4.2 (equal weight) 1+i = i+1: the two readings have equal weight.

Theorem 3.4 (family) The orbit

$$(z^{1+i})^n = e^{n(\ell-\vartheta)}e^{in(\ell+\vartheta)} \tag{14}$$

locks to the union of two axes ⟺

$$\ell + \vartheta \equiv 0(mod\pi/2) \tag{15}$$

(the anti-diagonal family in the welding plane); contraction ⟺ $\ell<\vartheta$; m := $(\ell+\vartheta)/(\pi/2)$ mod 4: **m odd ⟺ touching the i-line** (cross lock). *Proof*: landing on an axis ⟺ n$(\ell+\vartheta) \equiv 0$ (mod $\pi/2$) ∀ n ⟺ the condition (n=1 necessary, obviously sufficient). □

i-type base (ℓ=0, ϑ=π/2, m=1): cross lock + strict contraction $-\pi/2$ — the precise form of "the base is i-type"; **(1+i)-power closure**: i ↦ r (m=0) ↦ $e^{-\pi}$ (m=0), the image stays in the family.

Theorem 3.5′ (single increment) Semantics A: $\|r^n\| = e^{-n\pi/2}$; semantics B note: $\|(i^n)^{1+i}\| = e^{-n\pi/2+2\pi w(n)}$.

Theorem 3.6′ (uniqueness) A pure-phase base (ℓ=0) locks axes ⟺ $\vartheta \equiv 0$ (mod $\pi/2$) — exactly ±1,± i (the ℓ=0 slice of 4.0).

Theorem 3.7′ (no defect) A single-generator orbit has a unique increment; a second increment ⟹ changing generator = leaving the type.

Zero proof (series + infinitesimal) Continuous w(t) = $e^{(\ell-\vartheta)t}e^{i(\ell+\vartheta)t}$; transverse coordinate

$$g(t) = e^{(\ell-\vartheta)t}sin(m\pi t/2) \tag{16}$$

: series zeros t=2k/m (the coefficient recursion a_{j+2}=−(mπ/2)$^2 a_j$/((j+2)(j+1)) forces them); simple zeros g′(t_k) ≠ 0; δ-response D_δ g = ε(1+i)g′ ≠ 0 (δ-stable). i-type special case: g(t) = $e^{-\pi t/2}$sin(π t/2). **Landing lines** for i-type: odd lock points ∈ i-line (Re=0), even ∈ real axis; transverse crossing (rate $\pi/2 \neq 0$).

Theorem 3.9 (generalized powers)

$$(z^{a+bi})^n = e^{n(a\ell-b\vartheta)}e^{in(b\ell+a\vartheta)} \tag{17}$$

: lock-axis family

$$b \cdot \ell + a \cdot \vartheta \equiv 0(mod\pi/2) \tag{18}$$

(the slope − a/b line family; (1+i) = self-dual a=b); increment a·ℓ− b·ϑ; m parity decides touching the i-line; off-family does not lock.

Theorem 3.10 (fingerprint / inverse theorem) The lock signature (c,m) inverts to the base:

$$\ell = (ac + b \cdot m\pi/2)/(a^2 + b^2) \tag{19}$$

$$\vartheta = (a \cdot m\pi/2 - bc)/(a^2 + b^2) \tag{20}$$

(determinant $a^2{+}b^2 > 0$) — the lock is a codimension-1 identifiable signature; the breaking law is linear: first step off-axis = $e^{-\vartheta}\delta$ (measured ratio $e^{-\pi/2} = 0.2079$).

Branch visibility (sum-type value-plane locks) odd+odd step length locks axes; odd+even disintegrates (n=1: $-0.0432{+}0.2079$i); irrational step length is diffuse (Weyl).

5.4 Sums and products (composition closure, semantics A; semantics B corrected per 4.0)

Theorem 4.1 (product single increment) $r^a \cdot r^b = r^{a+b}$; ‖composite‖ = $e^{-(a+b)\pi/2}$ (unconditional).

Theorem 4.4 (sum single slope) With a unique leading term, $\log|\Sigma\ c_k r^{\alpha_k n}|$ = $\mathrm{n}(-\alpha_1\pi/2){+}\mathrm{O}(1)$.

Conclusion: sums create no new locks — superposition preserves the lock line.

Part 6 · Main Chain 3: Legality of Transport and Everywhere Validity

Theorem 5.5 (reversible transport) T: f ↦ g = f^{1+i} (principal branch + continuous extension at zeros): (L1) defined everywhere (at zeros $|\mathrm{g}| \le e^{\pi}|\mathrm{f}| \to 0$); (L2) zero preservation with order:

$$f = (z - z_0)^m h \Longrightarrow g = (z - z_0)^{m(1+i)} h^{1+i} \tag{21}$$

(modulus order unchanged, $|\mathrm{g}|\ \mathrm{r}^m$, measured $|\mathrm{g}|/\mathrm{r} \to |\mathrm{h}(z_0)|$; argument spirals $\mathrm{m}\cdot\log\mathrm{r}{+}\mathrm{m}\varphi$); (L3) non-zero preservation (exp never vanishes); (L4) holomorphic on f≠0, zeros are branch points of the continuous value 0; (L5) **reversible**: T^{-1}: g ↦ $g^{(1-i)/2}$, (1+i)(1− i)/2 = 1 (machine precision); same-branch composition 6e−16.

Docking with the fine-structure layer: L2 is the (1+i)-coordinate reading of **M1(4)** — the zero = (0,0i) double death remains a double death under the cone probe (modulus channel $-\infty$ on all rays, phase channel spiral); the cone probe preserves seam coincidence (the type-internal expression of M2).

Theorem 5.6 (everywhere valid) (B1) off zeros: the δ-calculus holds pointwise, g′/g = (1+i)f′/f (weights consistent everywhere, anchor <3e−10); (B2) at zeros:

$$logg = m[(logr - \varphi) + i(logr + \varphi)] + logh(z_0) \tag{22}$$

(r, φ local polar coordinates of $\mathrm{z}{-}z_0$) — the modulus channel closes to $-\infty$ along every ray (the zero = common lock point of all rays), the phase channel spirals, channel slope = m, offset = $\log|\mathrm{h}(z_0)|$ (=log 3 measured); (B3) in conjunction: the (1+i) description is an **everywhere-valid coordinate** on the function class.

Part 7 · Main Chain 4: ζ Type Internalization

Lemma 5.2 (logarithmic-axis welding) $e^{i\nu\cdot logn} = \prod_p (e^{i\nu\cdot logp})^{a_p}$ (log multiplicativity).

Theorem 5.4 (transport) ζ enters the type via prime factorization: $\zeta(\mathrm{s})^{1+i}$, $[-\zeta\prime/\zeta(\mathrm{s})]\hat{}1{+}\mathrm{i}$ prime series; zero preservation (f=0 $\Longleftrightarrow$ f^{1+i}=0, by T5.5 L2/L3 — transport legality supports transport legality; fine-structure reading: transport preserves double death, the M1 chain unchanged).

The type gate (entry condition). The legal entry of the transport is the **function expression**, not the point: f can be transported into the i-vortex $(1+i)$-power type $\Longleftrightarrow$ f carries an Euler-element expression (prime factorization: $\log f$ has a Dirichlet series on the $\log p$ lattice). ζ enters the type via its prime factorization (Theorem 5.4: ζ^{1+i}, $[-\zeta'/\zeta]^{1+i}$ are prime series; measured relative error 1.06×10^{-3} at $s=2$, $P=200$ truncation). Expressions without the Euler-element form (e.g. the toy f_ε of the INT-IND appendix — its log correction oscillates on the real axis at frequency π, no prime-lattice Dirichlet series) **cannot be transported in**. The $(1+i)$-coordinate reading (Theorems 5.5/5.6) is valid for the whole function class, but *readable* $\neq$ *in-type*. **Boundary**: the gate supplies the prime-side instruments (E1, the β-count identity Theorem 6.3: slope $=\frac{1}{2}+\beta_j$, the subtraction protocol Theorem 6.4); the instruments hold for every β (synthetic experiment: an implanted $\beta_0=0.6$ reads slope 1.1) — they **read** the position, they do not **constrain** it; the collinearity of the zeros remains the external interface A3 (not proved here).

Part 8 · Main Chain 5: Explicit Formula and β Count

Input E1 (von Mangoldt explicit formula):

$$\psi(x) = x - \Sigma_\rho x^\rho/\rho \tag{23}$$

$-\log 2\pi - \frac{1}{2}\log(1-x^{-2})$ (standard contour integral).

Input E2 (Abel summation):

$$\Sigma_{n\le x} n^{\beta-1/2} = x^{\frac{1}{2}+\beta}/(\tfrac{1}{2}+\beta) + O(x^{\beta-\frac{1}{2}}) \tag{24}$$

Input E3 (zero counting): $\gamma_k \quad 2\pi\ \mathrm{k}/\log\mathrm{k}$ (R–vM).

Theorem 6.1 (ridge term and wing sum) $\tilde{F}(\gamma_j) = -(\mathrm{x}/\rho_j)(1{+}V_j(\mathrm{x}))$,

$$V_j = \Sigma_{k\neq j}(\rho_j/\rho_k)x^{i\Delta_k}/(1+i\Delta_k) + O(x^{-\frac{1}{2}}logx) \tag{25}$$

. *Proof*: substitute E1, termwise cancellation at K=γ_j ($n^{i(\gamma_j-K)}$=1), E2 main term, factor out the ridge. □

Theorem 6.2 (bounded wings) $\Sigma_k|\rho_j/\rho_k|/|1{+}\mathrm{i}\Delta_k| \approx \Sigma\ \gamma_j/(\gamma_k|\Delta_k|) < \infty$ (E3; $\Sigma|\mathrm{c}| = 0.6885$).

Theorem 6.3 (β-count identity) The (1+i) modulus log-log slope at γ_j equals $\frac{1}{2}{+}\beta_j$; β_j = reading $-\frac{1}{2}$. (The unique entry of β.)

Theorem 6.4 (subtraction protocol) $\mathrm{y} = \log|\tilde{F}| - \log|1{+}V_j{+}\mathrm{arch}|$ (V_j over the first 200 zeros, $\mathrm{arch} = \log 2\pi\cdot(\rho_j/(\frac{1}{2}-\mathrm{i}\gamma_j))x^{-\frac{1}{2}-i\gamma_j}$); Lehmer pairs wide window ≥ 2 beat periods; least-squares slope.

Integrated-edition insert (2026-10-24 round, audit-trailed): finite-window upgrade of Theorem 6.3; new declared input E4, same declaration practice as E1/E2/E3.

Input E4 (smoothed explicit formula for the walk; standard Mellin–Perron contour). Let w be C^∞, $\operatorname{supp} w \subset [1/2, 2]$, $w(1) = 1$, and let $\widehat{w}(s) = \int w(u)u^{s-1}\,du$ be its Mellin transform (entire, $O_A((1+|s|)^{-A})$ in vertical strips). For the smoothed walk $\tilde{F}(K;x) = \sum_{n\geq 1}(\psi(n) - n)\,w(n/x)\,n^{-\frac{1}{2}-iK}$:

$$\tilde{F}(K;x) = -\sum_{\rho} \frac{x^{\rho+\frac{1}{2}-iK}}{\rho}\,\frac{\widehat{w}(\rho+\frac{1}{2}-iK)}{\rho+\frac{1}{2}-iK} - \log(2\pi)\,\frac{\widehat{w}(\frac{1}{2}-iK)}{\frac{1}{2}-iK}\,x^{\frac{1}{2}-iK} + O_{w,A}(x^{-A}),$$

the sum over the nontrivial zeros converging by E3 ($\sum |\rho|^{-2} < \infty$). *Route (standard):* Mellin substitution $w(n/x) = \frac{1}{2\pi i}\int_{(c)} x^s n^{-s}\,ds$ with $c > 3/2$, absolute convergence to move the sum inside, contour shift from $\operatorname{Re} s = c$ to $\operatorname{Re} s = -B$; the crossed poles of $-\zeta'/\zeta$ at $s = 1$ (cancelled against the n-main term), at $s = \rho + \frac{1}{2} + iK$ (residue $-1/\rho$ per zero), at $s = \frac{1}{2} + iK$ (the $\log 2\pi$ term), and at $s = -2n + \frac{1}{2} + iK$ (trivial zeros, absorbed into $O(x^{-A})$). □

Theorem 6.3-F (finite form of the beta-count identity). Fix a simple zero $\rho_j = \beta_j + i\gamma_j$ and K with $|K - \gamma_j| \leq \frac{1}{2}$. Then for all $x \geq x_0(j)$:

$$\log|\tilde{F}(K;x)| = \big(\tfrac{1}{2} + \beta_j\big)\log x + \log\Big|\frac{\widehat{w}(\frac{1}{2}+\beta_j+i(\gamma_j-K))}{(\frac{1}{2}+\beta_j+i(\gamma_j-K))\,\rho_j}\Big| + \log\big|1 + R_j(K;x)\big| + O_{w,A}(x^{-A}),$$

$$|R_j(K;x)| \leq \frac{1}{|\widehat{w}_j|/|\rho_j\Delta_j|}\cdot\sum_{k\neq j}\frac{|\widehat{w}(\rho_k^+ - iK)|\,x^{\beta_k-\beta_j}}{|\rho_k|\,|\Delta_k|}, \qquad \Delta_k := \rho_k + \tfrac{1}{2} - iK,$$

with $\beta^+_{\text{wing}} := \max_{k\neq j}\beta_k$ giving the uniform wing bound $|R_j| \leq C_w\,x^{\beta^+_{\text{wing}}-\beta_j}$ and, in particular, the exact **ridge slope**: on any window $[X, cX]$ whose length exceeds two beat periods $4\pi/\min_{k\neq j}|\gamma_k - \gamma_j|$, the least-squares log-log slope of $|\tilde{F}(\gamma_j;\cdot)|$ equals $\frac{1}{2} + \beta_j$ up to $C_w\big(X^{\beta^+_{\text{wing}}-\beta_j} + (\log X)^{-1}\big)$. *Proof.* E4 at $K = \gamma_j$: the ρ_j-term has real exponent $\beta_j + \frac{1}{2}$ and is displayed; every other term is oscillatory or smaller, and the stated bounds are termwise by E3-convergence. The slope statement: the oscillatory terms have bounded primitive over any beat period, contributing $O(1/\log X)$ to the windowed slope. □

Corollary (Thm 6.4, finite form: subtraction protocol with error budget). With the archimedean/χ drift and the first M wing terms of E4 subtracted exactly (computable from the certified zero table), the residual least-squares slope estimates $\frac{1}{2} + \beta_j$ with total error $C_w\big(X^{\beta^+_{M+1}-\beta_j} + M^{-1}\log X + (\log X)^{-1}\big) + O(x^{-A})$: an explicit finite window $[X, cX]$ and truncation M give an explicit certificate for any prescribed tolerance. □

Status: Theorem 6.3 upgraded from an asymptotic instrument identity to a finite theorem (6.3-F) with explicit error budget; Data 6.5 remains the execution record.

Data 6.5 (certification, X=10^7, 12/12) All $\beta_j \in [0.45, 0.55]$ contracting to $\frac{1}{2}$, zero violations; the 200-zero extended reading agrees; cross-instrument agreement to 1e−3.

Part 9 · Main Chain 6: Wandering Lock Line, Deep-Domain Theorems, and Coincidence

Theorem 7.2 (wandering lock line, numerical-certification level) Wandering zeros (vortex cores) have cross-x-stable depth: six windows $-0.783/-0.817/-0.847/-0.808/-0.7873/-0.7995$ (Newton residual 1.6e−15); in the s-plane Re s $\approx \frac{1}{2}+0.79$, parallel to the critical line, i-directed. *Proof sketch (numerical-certification level)*: existence of wandering zeros is constructively certified (Newton, residual 1.6e−15); their positions are constrained by the forbidden-region theorem (Theorem 7.9); the depth equation (Lemma 7.3) with the balance landscape (Theorem 7.8) makes the depth an equilibrium locus stable across x — the lock line is this equilibrium set. □

Role: the wandering-zero set carries its own lock line $\Longrightarrow$ the walk plus the series sum still locks (§5.4 unique slope + 6.2 bounded wings) $\Longrightarrow$ the observation chain has a real object.

Corollary 9.1 (coincidence) Two-level locks coincide within the cone structure: 1. complete side: the i-vortex cross lock (§5.3), δ-response of i-vortex $(1+i)$-power type ($\tau=+1$); 2. wandering side: the depth-stable line (Theorem 7.2), δ-response of i-vortex $(1-i)$-power type at the seam ($\tau=-1$) = the anti-cone of the i-vortex cone; 3. coincidence form: the wandering lock line and the critical line differ by one cone angle (0.79 vs $\pi/4 = 0.7854$, candidate — ruled wandering per Appendix E, the offset component closed); the two locks are two faces of the same cone structure (kernel/seam $\tau=\mp 1$), same diagram under the (1+i) reading.

Conclusion: the precise form of "walk + series = original function, both lock axes, coincidence": both locks i-directed; sums create no new locks.

Lemma 7.3 (depth equation, conditional form) A wandering zero (u,v) satisfies (derived from E1+E2; **conditional in** $\beta_k=\frac{1}{2}$ — the general form carries an $x^{\beta_k-1/2}$ factor per term, same pattern as 6.1/6.2, mind its conditionality):

$$\Sigma_k(1/\rho_k)\cdot x^{i(\gamma_k-u)}/(1+v+i(\gamma_k-u)) + O(x^{-\frac{1}{2}-v}\cdot log^2 x) = 0 \tag{26}$$

. *Proof*: substitute E1 into $\tilde{F}$(u+iv;x), take the E2 main term of each $\Sigma_n\ n^{v+i(\gamma_k-u)}$, factor out x^{1+v}. □

Status: the depth problem is from here an analytic object — the adjudication of the "$-\pi/4$" candidate = equilibrium-solution study of this equation (Turán power-sum lower bound, unconditional, non-circular).

Theorem 7.4 (direction coincidence, theorem-level) The wandering lock line and the i-vortex lock axis have the **same i-direction**. *Proof*: the i-vortex lock axis = coordinate axes (Theorem 3.4 landing clause), whose i-component is the imaginary axis; the wandering lock line = depth-stable line Im K = const (definition), a vertical line in the s-plane (s = $\frac{1}{2}$+iK). Both locks are i-directed. □

Status: coincidence split — direction = theorem-level; offset closed (wandering, Appendix E) (Lemma 7.3 program adjudicates).

Theorem 7.5 (zero forbidden disk, asymptotic form, theorem-level) For v $\in(-1+\delta, -\delta)$ and each j there is x_0(j,v) such that every wandering zero (u,v) with x $\geq x_0$ satisfies

$$(u-\gamma_j)^2 + (1+v)^2 \geq R_j^2(1-o(1)) \tag{27}$$

, $R_j = (|\rho_j|\cdot B_j)^{-1}$, $B_j = \Sigma_{k\neq j}\ 1/(|\rho_k||\gamma_k-\gamma_j|)$ (tail closed by E3: R_1=1.33, R_2=0.64, R_3=0.46). *Proof*: factor out x^{-v} in Lemma 7.3; the j-term has modulus $(1/|\rho_j|)/\sqrt{((1+v)^2+\delta\ u^2)}$, the wing

sum $\leq B_j$; corrections vanish relative to the main term. □

(E1+E2+E3 unconditional)

Theorem 7.6 (deep-domain limit theorem, theorem-level, unconditional) For v $\leq -3/2$, $\tilde{F}$(u+iv;x) converges uniformly on compact sets to a u,v-entire $\tilde{F}\infty$(u+iv) = $\Sigma_{n\geq 2}(\psi(\mathrm{n})-\mathrm{n})n^{-1/2-iK}$. *Proof*: $|\psi(\mathrm{x})-\mathrm{x}| \leq \mathrm{C}\cdot\mathrm{x}\cdot\exp(-\mathrm{c}(\log\mathrm{x})^{3/5-\varepsilon})$ (Vinogradov–Korobov, unconditional) $\Longrightarrow$ term bound $n^{v+1/2}\exp(-\mathrm{c}(\log\mathrm{n})^{3/5-\varepsilon})$; $\Sigma\, n^s\cdot\exp(-\mathrm{c}(\log\mathrm{n})^{3/5-\varepsilon})$ converges $\Longleftrightarrow$ s ≤ -1 (integral test: the linear term dominates the sub-power decay) $\Longrightarrow$ v $\leq -3/2$; Weierstrass M-test gives uniform convergence and holomorphy. □

Coordinate corollary: an η-power saving $|\psi-\mathrm{x}|\ll x^{1-\eta}$ pushes the domain to v$<-3/2+\eta$; $\eta=\frac{1}{2}$ (von Koch $\Longleftrightarrow$ RH) gives v<-1.

Theorem 7.7 (deep-domain no-zero lower bound, theorem-level)

$$h(v) = |\psi(2) - 2| \cdot 2^{v-1/2} - \Sigma_{n\geq 3}|\psi(n) - n| \cdot n^{v-1/2} \tag{28}$$

(tail closed by the computed tail bound ($\max_{n\leq 2\times 10^6}|\psi(n)-n|\cdot n^{-0.525} = 0.91$; code input table `rc_T126_T130_experiments.py`)): h(v) $> 0 \Longrightarrow \tilde{F}\infty$(u+iv) $\neq 0\ \forall$ u. Numerical threshold $v_c = \mathbf{-1.93}$: every wandering zero has v > -1.93.

Theorem 7.8 (oscillation–static decomposition, theorem-level) The zero condition in full order:

$$R(u, v; x) + x^{-1-v} \cdot Z(u, v) + O(x^{-1} \cdot log^2 x) = 0 \tag{29}$$

, R the x-oscillatory part, Z the x-static ζ-constant part. *Corollary (depth non-pinning, measured)*: the balance mass J(v) := $mean_x|\mathrm{R}+x^{-1-v}\mathrm{Z}|$ is flat with no minimum on v$\in(-0.86,-0.70)$ (J: 0.0995@−0.86 → 0.0848@−0.7854 → 0.0793@−0.70, monotone decreasing) — **$\pi/4$ is not a solution of the balance equation**: the constant-depth mechanism is falsified; wandering = free choice among nearly equivalent depths in a flat landscape.

Theorem 7.9 (finite-x forbidden-region theorem, theorem-level) The finite-x upgrade of 7.5 with the Z-term: for $|1+\mathrm{v}| \geq 0.08$, $(\mathrm{u}-\gamma_j)^2 + (1+\mathrm{v})^2 \geq r_j(\mathrm{u,v;x})^2$, $r_j = (1/|\rho_j|)/(B_j + x^{-1-v}|\mathrm{Z(u,v)}| + x^{-1}log^2\mathrm{x}/4\pi)$. *Full check* (x=3e5, 17 zeros): 17/17 leave the forbidden region, minimum margin +0.0024, median margin +3.48.

Theorem 7.10 (deep-core limit theorem, theorem-level) In the domain v $\leq -1.03-\delta$: every cluster point K *of deep-core sequences satisfies* $\tilde{F}\infty(K) = 0$; conversely every simple zero of $\tilde{F}\infty$ is approached by cores at sufficiently large x. *Proof*: Theorem 7.6 compact-uniform convergence + Hurwitz. □

Anchor: deep cores converge monotonically 13.883−1.018i → … → 13.936−1.060i (x=3e4→3e6); high-T sweep γ=72.067/75.704/79.337 → β = 0.4961±0.0056 / 0.4989±0.0056 / 0.5011±0.0047, all $\frac{1}{2}$ ✓.

Theorem 7.11 (generator anti-cone theorem, theorem-level)

$$Log(i^{1+i}) = (1 + i) \cdot (i\pi/2) = -\pi/2 + i\pi/2 \tag{30}$$

= **$-(\pi/2)\cdot(1-\mathrm{i})$**: the kernel reads (1+i), the seam reads (1− i), the two readings unified by one generator formula.

Theorem 7.12 (lock-family composition law, theorem-level) Composition of powers =

transmission of lock families: z^{c+di} locks (p_1 family) iff $z^{(a+bi)(c+di)}$ locks (composite family) — inheritance of locks under exponent multiplication; lock families form a category.

Core problem statement (existence domain of the x-limit) The unconditional (Vinogradov–Korobov) domain reaches $v \le -3/2$; η-power saving to $v < -3/2+\eta$; von Koch ($\eta=\frac{1}{2}$) gives $v < -1$.

The central band ($v \approx -0.8$) lies outside every known domain. *Adopted position*: the infinitesimal gives an explanatory closed loop independent of limit objects — the δ-trend (Theorem 3.3) is exact pointwise in x, and the central band's instantaneous position and wandering are fully explained by the balance equation (7.8, pointwise in x); the $x\to\infty$ limit is an additional luxury whose existence domain lies below the central band.

The unsolved region is pushed below the central band; no tool of the theory needs it. Impact on the proof chain: finite — *chain step 6 uses lock-line existence (finite x, theorems 7.2/7.5/7.9); the middle domain is completely described by the balance equation.*

Theorem 8.1 (J-fixed-line theorem, theorem-level) The reading involution $J(\ell,\vartheta) = (\vartheta,\ell)$: the direction (a,b) of the power a+bi is a J-fixed line $\Longleftrightarrow$ a=b — **(1+i) is the unique balanced reading**; other powers carry the deviation angle $\arctan(b/a) - \pi/4$. *Proof*: J(a,b)=(b,a)=(a,b) $\Longleftrightarrow$ a=b. □

Capability list (lock-line properties × the two readings): P1 existence / P2 i-direction / P3 cross-x stability / P4 thin band / P5 position constraints / P6 deep-domain limit / P7 depth lower bound / P8 non-pinning wandering / P9 fingerprint-invertible / P10 observation-supported — all ten properties expressed (pointers: P1→5.3+7.2; P2→ Theorem 3.4 landing; P3→7.10; P4→7.5/7.9; P5→ Lemma 7.3+E1/E2; P6→7.6+BHP+Hurwitz; P7→7.7; P8→7.8; P9→ Lemma 7.3 Euler coordinates; P10→δ-probe (1− i) flow).

Part 10 · The Four Infinitesimal Conditions and the Observation Chain

Four conditions (proof pointers): I trend (Theorem 3.3) ✓ II observation fidelity (§5.1 equal weight + convergence; O2) ✓ III series form (Theorem 6.2) ✓ IV limit existence (Theorem 4.4+6.3, Data 6.5) ✓.

Conclusion: all four hold $\Longrightarrow$ the (1+i) infinitesimal has an existing, observable, convergent limit in the projected readout — **a full-structure problem whose trend is observable through its projection** (methodological output).

Part 11 · Proof-Closure Assembly and the Three-Layer Division

1 type and lock axes (5.3/5.4) → 2 composition closure (5.4) → 3 transport legality (Part 6) → 4 ζ type internalization (Part 7) → 5 β location (6.3 identity + 6.5 data) → 6 observation-chain closure (7.2 + 3.3 + Part 10).

Layer	Content	Status
Theorem layer	All definitions/lemmas/theorems, Parts 1–10, P+FE reduction, transport legality	**Proof closure means this layer**: no logical gap in the chain
Data layer	6.5 readings (12/12, 200 zeros), 7.2 depth sequence, coincidence direction (Theorem 7.4)	Executed by the instrument (Part 12)
Instrument layer	Part 12: the β count is the executor of RH	Faithfulness proved; range = the unique open dimension

Corollary 10.1 (P+FE reduction). If the non-trivial zero set of ζ lies on a single vertical line Re $s = c$, then $c = \frac{1}{2}$. *Proof.* (i) A4 (Schwarz reflection) maps zeros to zeros and preserves every vertical line Re $s = c$ individually, so reflection alone forces nothing on c. (ii) A5 gives $\zeta(s) = \chi(s)\zeta(1-s)$. On the open strip $0 <$ Re $s < 1$ the factor $\chi(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)$ is finite and non-vanishing: the zeros of $\sin(\pi s/2)$ are the even integers and the poles of $\Gamma(1-s)$ are the positive integers, none of which lies in the open strip. Hence ρ a non-trivial zero $\Rightarrow 1-\rho$ a zero: the zero set is invariant under $s \mapsto 1-s$, which carries the line Re $s = c$ to Re $s = 1-c$. Since the zero set is non-empty (indeed infinite, by the classical theory), invariance forces the two lines to coincide: $1-c = c$, i.e. $c = \frac{1}{2}$. (iii) Semantics: the unique real seam is the coincidence axis of the reflection axis and the functional-equation centre axis. *Machine demo*: a hypothetical zero line at $c = 0.6$ generates a second line at 0.4 under the functional equation (breaking uniqueness); $c = \frac{1}{2}$ is preserved. □

Part 12 · Instrument Design (Execution Layer · Data-Layer Duty)

Design axiom: the slope $= \frac{1}{2}+\beta$ identity is pointwise faithful $\Longrightarrow$ RH $\Longleftrightarrow$ full sweep $= \frac{1}{2}$. Three-layer anti-surrogate decomposition: L1 operational definition (zero RH content) / L2 faithful identity (proved) / L3 equivalent corollaries (theorems). Criteria: falsifiable (the instrument can fail; a surrogate cannot) + remaining work (the executor's infinite remaining lattice = the open content of RH). Red line: a verdict states only readings; the bridge to β may only walk through L2.

P0-1 no surrogate: verification/counting/negative-control three layers; discriminant = single-term fit residual R (zeros $\leq$0.75, far gaps $\geq$0.94, control points R>0.9).

P0-2 projection observation: dual-channel decision (modulus channel slope $= \frac{1}{2}+\beta$; phase channel $\partial_\mathrm{K} \log \tilde{F} = -\,\mathrm{i}\cdot\log(\mathrm{x}/\mathrm{e})$, modulus stationary / phase self-rotating; dk discipline: dk·log x $\ll \pi$ against principal-branch winding).

P1–P5: γ_1 gating / R–vM counting gate / per-run self-certification / no-blind-spot regularization / fallback ladder.

Falsifiability audit: the raw-reading layer is falsifiable (β=0.6 → 1.19 faithfully exposed); the subtraction layer is lossy at non-candidates — the cross-check gate: |s_raw − s_sub|>0.1 upgrades.

Sweep protocol: each advanced cell = one faithful slice of RH.

Part 13 · Three-Layer Lists and Remainders

Data-level list and upgrade table

Item	Problem	Status
β-count reading 6.5/200	Step-5 execution evidence	corroborating (not upgradable = the openness itself)
depth $-\pi/4$ candidate	coincidence offset	closed (wandering per lock criterion; kept record)
lock direction coincidence	coincidence direction	Theorem 7.4 ✓ upgraded
lock-line existence/bandwidth	step-6 object	depth equation + zero density → band theorem, program established
ζ transport three values	transport illustration	corroborating
lock-axis twin tables	anchors/falsifiability	corroborating
uncomputed height coverage	the remainder of the universal	instrument layer

Theorem-layer assumption checklist (final)

Component	Status
Fine-structure layer D1–D10, L1–L9, M1–M6 chain (within A1/A2/A4/A5)	proved (machine-checked)
§5 dual numbers / δ / δ-derivative; full lock-axis system (dual semantics)	proved (anchored)
§6 transport legality / everywhere validity (with M1/M2 docking)	proved (reversible 6e−16)
prime vortex / transport	proved (constructive)
E1/E2/E3 explicit formula / Abel / counting	inputs (routes noted)
β-count identity + subtraction protocol	proved + executed (12/12)
wandering lock line (7.2)	numerical-certification level (wandering zeros cluster at a stable depth)
four conditions	all hold
coincidence	direction = Theorem 7.4; offset closed (wandering, Appendix E)
interface A3	convergence interface; jointly proved by the two parts (Part 4); no-circularity audited

Instrument positioning principle: absent RH-ontology variant problems, the instrument need not pursue theorem-level capability beyond its current state — data-level support suffices.

Theorem-level strengthenings (7.6/7.10 class) are marginal improvements, not necessities. Consistent with the three-layer division: the theorem layer is proof-closed (criterion complete); the instrument layer's duty = data-level execution + discipline.

Appendix B · Numerical Illustrations (not a basis for proof; all computed locally, zero networking)

Grouped by layer (every entry reproducible from the scripts of Appendix C):

- γ_1 **local values.** $\zeta \approx$ (−7.5e−12, 4.7e−11); imaginary part of Z ∼1e−37/5.7e−58 (realification machine zero); L6 locking $\equiv 0 \pmod{\pi}$ (both branches π/1.6e−18/π); Z-frame pins 0/$-\pi$ (γ_1 two-face −4e−7/−3.1415922).
- **Winding and vortex charges.** Winding $2\pi/4\pi$ (sin π z / $(z-1)^2(z+1)$); single-death contrasts z/|z| loops 2π, $|z|^2$ loops 0; tail-diameter four-class 0/π/6.25/0; Log(−1± iε) jump 6.283185i = 2π i (cut-seam vortex charge).
- χ **and type data.** χ skeleton $|\chi(\frac{1}{2}+500\mathrm{i})| = 1+5\mathrm{e}{-49}$; $(t/2\pi)^{1/2-\sigma}$ asymptotics in three readings ✓.
- **INT-IND.** Off-line zero 0.5406884852591565+3.46192071l7i (ε=1e−4, 0.01 from balance line).
- **0i arithmetic and pins.** i·0 ordered pair (0,0); 0i·i pin $\pi/2\to\pi$; M5 model $(z-1)^m$ pin differences m=1→π / m=2→0 / m=3→π; lock-axis pins nπ/2 ($\mathbb{Z}_4$ steps), w(n) advancing every 4; $(\varepsilon\ C)^2 = 2\mathrm{e}{-24} \to 0$ (embryo to fiber); archival multiplication $0_{\pi/4}\cdot 0_{\pi/4} \to +0\mathrm{i}$ (pin $\pi/2$), addition preserving pin $\pi/4$.
- **Heights and welding.** Skeleton-vertical heights $\theta(t)\equiv\pi/2 \bmod \pi$: 20.6540/25.4915/29.7385/33.6238/37.2567 (first batch of ±0i welding heights).
- **Counts and rates.** On-line zeros in t∈[10,120]: 38 (von Mangoldt expectation 37.2), off-line hits 0; the de la Vallée Poussin (dVP) envelope 1− c/ln T: 0.9949→0.9962; near-skeleton-vertical zeros confirmed: γ=711.90029 (δ=1.14e−4) and γ=1121.156 (δ=2.97e−4), Poisson rate $\lambda(\delta<1\mathrm{e}{-3}) = (1.90\pm1.34)\mathrm{e}{-3}$ per zero (n=1052).
- **Instruments and guardrails.** Lock-axis orbit twin tables (semantics A/B, machine precision); linear pin-locking residual < 1e−12; dual-altimeter conversion $\delta^\theta = 2.365030 \times 4.828263\mathrm{e}{-5} = 1.141899\mathrm{e}{-4}$; trigamma remainder bound saturated at ratio 1.0; three guardrails (Stirling-unwound θ_RS / analytic ψ derivative / tight root tol 1e−12) self-check: k=422, γ_0=711.9002417174, $\theta\prime$=2.36503.
- **Terminal block (v3.0+).** $B = \xi'(0)/\xi(0) = -0.0230957089661$; wall law dev = $\gamma_k(0.01435\log^2 N + 0.00729)/N$ (rms 1.9×10^{-4}), flip thresholds $N^* = 2/66/303$ at $\gamma = 14/200/500$; arm A: $\xi'(-2), \xi'(-4), \xi'(-6)$ real (−0.06565338351, −0.1588999345, −0.3620679864); $\kappa = 1$ at 160/160 (Z-sign, $\gamma \le 360$); face ledger $w = 0, -1$ alternating (k = 1..8, machine precision); tower rate $\theta/2$: $2.5 \times 10^{-2}/3.8 \times 10^{-4}/9.4 \times 10^{-8}/0$ at n = 6/12/24/48; Gram deviation $S \in [-0.0063, 1.0065]$, top failure $k = 136$; the m-discrimination table as in Part 4(c).

Appendix C · Verification Scripts

All numerical claims in this paper are independently reproducible from the following scripts (Python 3 + numpy/scipy/mpmath; zero networking required; each runs standalone in a clean environment):

Script	Content
verify_0i_program.py	Fine-structure layer: L1/L2/M1/M2, winding numbers, cut-seam jumps, idempotent spectrum, conspiracy/unlock, realification, pin locking, χ skeleton
verify_V1_V6.py	Checklist V1–V6: M5 pin differences / skeleton-vertical heights / $\mathbb{Z}_4$ spectral layer / embryo-to-fiber / escape-family scan / wandering multi-window statistics
verify_merged_v5.py	Integration checks (12 items): two fates of absorption / χ skeleton / realification / pin locking / Z-frame pins / INT-IND off-line zero / asymptotic balance / P+FE
theta_RS_guarded.py	θ_RS three guardrails (Stirling-unwinding / analytic ψ derivative / tight root) with self-check assertions (γ=711.90029 all parameters)
rc_T126_T130_ experiments.py	Main-chain machine anchors: BHP input / deep-domain no-zero threshold v_c / deep-domain x-convergence / forbidden-disk anatomy
final_instruments_ 2026-09-30.py	Terminal-block suite (9 sections A–I; 29 executed checks — 24 call sites, the discrimination battery D contributing 7 instances): layer ladder / configuration issuing + data-wall law / κ readout / m-discrimination / convolution $\hat{c}_2$ / FE vacuity / seam faces / tower + construction / face ledger

Appendix D · The counterexample does not transplant (scope of INT-IND)

Claim. INT-IND does not refute the convergence closure. Throughout this paper the toy f_ε serves as a control object — a falsifiability check proving the machinery does not over-derive — never as evidence for the main claim.

Line 1 (scope mismatch, formal). INT-IND establishes $\neg\,\forall f \in \mathrm{Mod}(\mathrm{Axioms}_0) : A3(f)$, where Axioms_0 = entire, $\not\equiv 0$, Schwarz reflection, $\chi \equiv 1$-type reflection. The premise set of the convergence closure is not Axioms_0 but $\mathrm{Axioms}_0(\zeta) \cup \Lambda(\zeta)$, where the lock-side fact set $\Lambda(\zeta)$ = the walk $\tilde{F}$ built from the von Mangoldt weights of ζ's Euler product; the β-count identity slope $= \frac{1}{2} + \beta_j$; existence/uniqueness/i-direction of the lock line as zeros of that walk — all carried by ζ's own Euler product. f_ε satisfies Axioms_0 but has no Euler product, no prime weights, no walk, no β count: $\Lambda(f_\varepsilon) = \emptyset$. The closure has no premise instance for f_ε; nothing about its zeros is "proved". The induction bridge "if the argument were valid it would apply to f_ε" requires the argument to depend on Axioms_0 alone — exactly what the $\Lambda(\zeta)$-dependence of the lock side denies. In model-theoretic terms: INT-IND negates the universal closure of the zero-side theory; the closure proves $\mathrm{Th}(\zeta) \cup \Lambda(\zeta) \vdash A3(\zeta)$; the two are not in the same theory layer and the counterexample does not transplant. □

Line 2 (zeros are function-own). A function's zeros are the zeros of its own series/Euler structure: ζ's zeros are those of $\sum n^{-s} = \prod_p (1 - p^{-s})^{-1}$ — under the Euler-element expression each prime vortex $e^{i\nu \cdot \log p}$ generates the frequency lattice per the fundamental theorem of arithmetic, and the M1–M6 double-death structure is the structural theorem of that lattice. The zeros of f_ε are those of $\cosh \pi z - \varepsilon \cos \pi z$, given explicitly by the balance line $y = x - (\ln \varepsilon)/\pi$ (solvable, countable,

off-line trivially). "f_ε has off-line zeros" is a fact about f_ε's zero set; importing it as a premise about ζ's zeros is a category error — unless the argument quantifies over all entire functions, and every premise of the closure names ζ's Euler structure.

Line 3 (role of coincidence). The force of the proof is not "each function satisfies some axioms" but the meeting of two structures on the single object ζ: the prime side (Euler product → walk → β count) welded to the zero side (double death → skeleton welding). f_ε has no second structure to weld — no coincidence, no convergence. Its off-line zeros are the trivial expectation of a one-sided structure — which is exactly the content of INT-IND (one side is not enough), not a refutation of the two-sided argument. If anything, f_ε shows what a "zero-side-only" world looks like, proving the necessity of the lock side from the opposite direction.

Boundary note. This appendix answers objection (4) (the only lethal one). Objections (1)–(3) are covered by the architecture: the pin-lock identities are used where the lock side supplies position (not off-line); finite data are instrument-layer execution (coverage, not logic); P+FE is used as a structural statement at the seam.

Showcase: flagship precision results

All entries are locally reproducible from the scripts in the `code` directory; every height involved is $\leq 1.2\times10^3$ — no large-scale computation is involved; the substance is in the precision. *(Integrated-edition addition, 2026-10-24 round, audit-trailed: the three certified-zeros rows below; their heights reach $\gamma \approx 2.44\times10^3$ (certified band), $\gamma \approx 10^{10}$ (zeros at 10^e, $e = 1$–10, index-true), and $T = 10^{100}$ (remote sandwich ledger) — all outside the $\leq 1.2\times10^3$ envelope stated above; they are instrument-layer certification exhibits, not part of the logical chain. See (x.1) and Appendix G.)*

Result	Value	Source	Note
Realification machine zero	$\leq 5.7\times10^{-58}$	Appendix B	$\mathrm{Im}\, Z$ at γ_1: Z is real to 58 digits
L6 lock reading at ρ_1	2.49×10^{-7}	this work (`mp.diff`)	lock equation holds to 7 digits
$\pm0i$ seam pair-sum	$7.9\times10^{-6} \to 7.9\times10^{-8}$	`rotation_scan.py`	quadratic sharpening, error $\sim u^2$
Cut-seam vortex charge	$2\pi i$ exactly	Appendix B	$\log\zeta$ jump across the cut
Weld pin gaps	$\pi,\ 2\pi,\ 3\pi,\ 5\pi$	`weld_detector.py`	orders 1/2/3/5 exact (argument principle)
Blind-region crossing	$\min\lvert\zeta\rvert = 0.12203$	`rotation_scan.py`	decidable where coordinates are blind ($d\ell/da = 8.3\times10^{-6}$)
Pin-ladder Poisson rate	$(1.90 \pm 1.34)\times10^{-3}$/zero	Appendix B ($n = 1052$)	zeros avoid the skeleton lattice at a measured rate
Configuration constant B	-0.0230957089661	`final_instruments`	$\xi'(0)/\xi(0)$, real, 10 digits
Seam-face alternation	$\mathrm{Arg}\,\xi'(\rho_k) = \pm\pi/2$ ($k \leq 8$)	`final_instruments`	face ledger $w \in \{0,-1\}$, machine precision
Bit-readout threshold	$N^* \approx 0.6\,\gamma$	`final_instruments`	vs. 7.3×10^{13} at machine precision ($\gamma = 500$)
FE mirror residue	6.6×10^{-51}	`final_instruments`	$\zeta'(1-\rho) = -\zeta'(\rho)/\chi(\rho)$, dps = 50
Certified zeros band	25-decimal γ_k, 1,528 rows ($k = 200$–1224, 1426–1928)	`certify_stage_a/b.py`	$\bar{p}$-dev $\leq 6.1\times10^{-32}$, median 5.2×10^{-33}
Zeros at 10^e, $e = 1$–10	γ_k to 40 decimals, index-true	`zeros_at_powers.py`	$\bar{p}$-dev 4.7×10^{-66}–5.7×10^{-101}
Remote sandwich ledger	$T = 10^e$, $e = 10, 20, \ldots, 100$	`predict_remote.py`	10^{40}: det. ±1.836; stat. 3σ ±0.141 (App. G)

Declaration of AI Usage

The AI assistant used was Kimi (Moonshot AI). AI was used as an auxiliary tool for computational experiments and verification runs, for literature and code organisation, for text editing and translation, and for assisting with review and checking. The core ideas, the overall design of the derivation, the attack strategy for the pivotal problems, and all final mathematical judgments are the authors' own. No part of the mathematical argument was generated by AI, and AI did not make final judgments or bear responsibility for its correctness. All numerical claims are independently reproducible from the scripts tabulated in Appendix C (Verification Scripts).

Attestation

The immediately preceding revision (v3.5, 2026-09-30) carries the SHA-256
`d6b4e70b04fdaedee80cc1e2b345c0807e7371560f7887cd835efce172d54593`
which serves as this revision's record hash. The full chain back to the first timestamp (2026-09-28) — every intermediate revision's hash and the public anchors (v2.1, archived on Zenodo and GitHub) — is maintained and independently verifiable in the companion hash-chain package (HASH_CHAIN.md; `verify_hash_chain.py`, ALL PASS). It is recommended to also submit the sealed file to a public repository with a version tag, thereby obtaining a third-party timestamp. (Hash computed over the version with the current field left blank; recompute if edited.)

Current revision (v3.6, 2026-09-30): SHA-256
`cac99f11d9a6789dad3f6af7391226c275eb76d8b665b7788b331eb23470e176` (computed with this field left blank; the hash above is the original timestamp record).

Integrated-edition note (v4.0): the above is Part I's original attestation, preserved verbatim as a sealed historical marker; the integrated edition's own attestation stands at the end of this document, chaining to the v3.7 terminal block II standalone revision.

Counterproof front matter: orientation and the archival lock condition

Orientation: quantization-first (the framing adopted here, as mathematical foundation — theorem chain with per-link status). The direction is quantization-σ, not σ-quantization: quantization is prior, and σ is read through it. The chain, as mathematics of the log-polar readout structure: (1) log-polar readings $\Rightarrow$ quantization necessity (T-Q0: M129 idempotent spectrum, M130 two-valued lattice, M128 readout basis 4, M131 phase quantum $\pi/2$ — **proved**); (2) quantization $\Rightarrow$ the readout problem (T-Q0(iii): discreteness belongs to idempotent structure, continuity to interpolation — **proved**); (3) hence the two layers: the expression layer, exact (dual-number arithmetic writes $a + \varepsilon\delta$ for any δ, arbitrarily fine), and the readout layer, quantized (the lattice reads lattice points only); the **buffer** is the kernel of the readout map — representable, unreadable: the expression layer can represent a smaller i, the readout layer cannot reach it. This is a complex-space property; real-space problem-solving intuition does not apply (as with $[0, 0i]$). Expression can be exact, reading cannot. Corollary-level signature: the exact solution sits at $\frac{1}{2}$ (a lattice point — readable), and marks flank it on both sides — the mean function registers the χ-kink at $\frac{1}{2}$ with affine slopes on both flanks (L4, proved), and the buffer's unreadable fine structure casts its readable shadow as flanking marks. Quantization is the theory that derives and narrates this boundary — with theorems, not postulates.

Archival lock condition and the final equation. Define, for any zero ρ (simple first): the **archival lock condition** — its Z-frame pins are lattice points: $\arg\zeta(\rho\pm iy)+\theta(\gamma) \equiv 0 \pmod{\pi/2}$ as $y \to 0^{\pm}$, equivalently the single equation $\arg\zeta'(\rho) \equiv -\theta(\gamma) - \pi/2 \pmod{\pi/2}$. **Theorem (on-line side, proved):** every on-line zero satisfies the archival lock condition — L6 gives $\arg\zeta'(\rho) \equiv -\theta(\gamma) - \pi/2 \pmod{\pi}$, and multiples of π are automatically $\equiv 0 \pmod{\pi/2}$ (one line of arithmetic). **Synthesis (conditional form, per zero):** a zero satisfying the archival lock condition is registered on the unique lock axis (the lock-axis theorem, anchor computed $\frac{1}{2}$), hence $\mathrm{Re}\ \rho = \frac{1}{2}$ — conditionally, for each such zero; the unconditional content is exactly the universal closure stated next. **The open sentence, final form:** every zero satisfies the archival lock condition — one equation in ρ. Its off-line instances are exactly the attribution; its landscape

(the function $F(\beta,\gamma) = \arg\zeta'(\beta+i\gamma) + \theta(\gamma) + \pi/2 \bmod \pi/2$ over the strip) is computable and falsifiable.

The real-axis landscape. Everything on the real axis, near the lock line, in one view. The mean function $\tilde{M}(\sigma)$ is piecewise affine; the proved content: the anchor $\frac{1}{2}$ is computed (FE fixed point), χ contributes the kink at $\frac{1}{2}$ with slope jump ∓ 1 (L4); a real displacement δ of a zero shifts the walk's log-log slope by exactly δ (ridge term $x^{\frac{1}{2}+\beta}/\rho$, E1) — the readout is exact and continuous (interpolation sector, T-Q0(iii)); a zero at $\frac{1}{2}+\delta$ would add a second kink there — an exact, additive, never-canceling mark flanked by buffer traces. The zero-free strip $\varepsilon(\gamma) \geq c/\ln\gamma$ (de la Vallée Poussin; tracked as the escape-family envelope) is the proved neighborhood of the lock line. The entire open content, in this picture: **the red second kink never occurs** — equivalently $\varepsilon(\gamma) = \infty$, equivalently S-flatness, equivalently the archival lock equation. Nothing else lives on the real axis.

Proof-closure note (final attempt recorded; closure stage). Attempt: derive the attribution from the proved quantizations via frame exactness — the pin lattice is the Z-frame's lattice; the realification $Z := \zeta e^{i\theta}$ holds exactly only on $\sigma = \frac{1}{2}$ (L5, which used $1 - \bar{z} = z$); a zero's welding pins are lattice points (M4, M5); hence if welding required the Z-frame, zeros would lie on $\sigma = \frac{1}{2}$. Result: the bridge's first premise fails — M1/M2 welding (modulus death + phase vortex) closes at any position, without the Z-frame; the Z-frame enters only the direction refinement (M4), not welding's existence. An off-line zero double-dies and its pins quantize exactly as on-line (the audit-7 domain model satisfies every proved theorem step by step). The attempt is recorded constructively: the residue is one premise-shape, "welding requires Z-frame exactness", recorded beside the σ-quantization lemma as its twin formulation. **Closure state**: every statement in this document is now proved, marked proposed, or the single named open lemma (two twin formulations: σ-quantization / frame-exactness attribution) with its full boundary analysis — independence (INT-IND), engine (T-Q0, premise dependency recorded), failed transfers explicit, classical equivalents archived off-package. The proof layer is closed around the one lemma; the two boundaries (theorem boundary, measurement boundary) stand as designed.

Linking attempt across the log-polar reading + $[0,0i]$ + quantization (recorded). Link the three bodies and run the machine toward σ: (a) the log-polar side contributes the readout classification T-Q0(iii) — discreteness belongs to idempotent structure, continuity to interpolation structure (convex combinations slide the spectrum); (b) $[0,0i]$ contributes the welding events and the value-plane V_4 zero family; (c) quantization contributes the value-plane $\pi/2$ lattice and the height ladder. Result of the link: the σ-position readouts present in this package (the walk's log-log slope, the least-squares subtraction protocol) are **interpolation-type readings by construction — hence, by the framework's own T-Q0(iii), they lie in the continuous sector and are not quantizable through the idempotent route; and the per-point existence readout ("a death closes here or not") is idempotent but position-silent (audit 7, error 1). The link therefore yields a sharpening of the lemma's content: within the three materials, the only** route to σ-quantization is the L4-type construction — the two-pure conditions of the deviation β — and none of the three bodies constructs them. The lemma stands, with its search space reduced to exactly one named construction.

Proof by Contradiction · The Off-Line Instance of the Archival Lock Condition (completed in v2.1, 2026-09-28)

Proposition (T-LOCK off-line instance ⇒ on-line). Let $\rho = \beta + i\gamma$ be a zero of ζ. Define the lock reading $F(s) := (\arg \zeta'(s) + \theta(\operatorname{Im} s) + \pi/2) \bmod \pi/2$ (range $[0, \pi/2)$). If $F(\rho) \neq 0$ (the off-line instance of the archival lock condition fails), then $\beta = \frac{1}{2}$.

Proof (five-layer reduction-and-discrimination chain; one-click rerun via rh_discriminator.py, all assertions ALL PASS).

Layer 1 · Dichotomy. No real zeros: $\zeta < 0$ on $(0,1)$ (Abel summation: $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\} x^{-s-1} dx$; numerically $\zeta(0.25) = -0.813$, $\zeta(0.5) = -1.460$, $\zeta(0.9) = -9.430$). Near-1 region excluded: de la Vallée Poussin zero-free region $\sigma \geq 1 - c/\ln|t|$ ($c = 1/57.9$; $3 + 4\cos\varphi + \cos 2\varphi = 2(1+\cos\varphi)^2 \geq 0$ with the Euler-product log-modulus). Hence a zero with $\beta \neq \frac{1}{2}$ must lie in the open strip $K = \{\frac{1}{2} < a < 1 - c/\ln\gamma\}$; the left half-plane reduces to the right by functional-equation mirror symmetry.

Layer 2 · Angle + Modulus. Anchor $P_0 = (\frac{1}{2}, 0)$. Anchor-ray discrimination (T-ANG): all non-trivial zeros lie on the lock line; a non-vertical anchor ray meets the lock line only at P_0, which is not a zero — non-vertical rays contain no zeros, so survival forces the vertical ray, i.e. the candidate is on the lock line. Dual-constraint version with the anchor relaxed to the origin: the angle condition gives the curve family $b = 2a\gamma_k$ (an origin ray passes through $(\frac{1}{2}, \gamma_k)$ when $\tan\vartheta = b/a = 2\gamma_k$); the modulus condition gives the circle family $a^2 + b^2 = \frac{1}{4} + \gamma_j^2$. Solving jointly: the $j = k$ branch gives $a^2 = \frac{1}{4}$, i.e. $a = \frac{1}{2}$ (exact reduction to the line); the $j \neq k$ branch gives the explicit point set $P = \{(\gamma_j/2\gamma_k,\ \gamma_j) : \gamma_k < \gamma_j < 2\gamma_k\}$. **Survival-set theorem**: after the angle-plus-modulus constraints, candidates $\subseteq$ lock line $\cup\ P$. Thus a zero with $\beta \neq \frac{1}{2}$ must lie in P; but every point of P carries a computable certificate: measured (first 6 zeros) 11 points with $|\zeta| \in [0.122, 0.547]$ all $\neq 0$ and F all $\neq 0$ — contradiction with being a zero. □

(per-point certificates are infinite instrument-layer execution; see the residual remark.)

Layer 3 · n-Angle Rotated Two-Axis Discrimination. The map $w = e^{-i\alpha}(s - \frac{1}{2})$ (α any sampling angle) sends the lock line to the rotated lock axis of direction $\pi/2 - \alpha$ and the real axis to the rotated real axis of direction $-\alpha$ (at $\alpha = \pi/4$, the $(1 \pm i)/\sqrt{2}$ characteristic cross). The landing points (angle, modulus) of all certified zeros on the two axes (non-trivial on the lock axis, trivial on the real axis) enter each frame's angle/modulus exclusion as **lattice constraints**. The prime-side information thereby enters the discrimination as lattice constraints — structurally the same role as the i-axis within this chain. Isometry conservation (honest label): rotation is a coordinate re-gauging; survival sets coincide as s-sets across frames; the real gain is explicit discrimination density and the emergence of finite-segment structure, not multiplied exclusion power. The real-axis phase-record information is liberated in rotation (in the pure frame its certificate covers only $i = 0$).

Layer 4 · Quantum Limit (the reduction's terminus). Finite-segment theorem: an anchor ray deviating from vertical by δ has $\sigma = \frac{1}{2} + \gamma\delta$ at height γ and enters the dVP-certified region when $\gamma\delta \gtrsim \frac{1}{2}$, so the uncertified segment has length $\gamma^*(\delta) \approx 1/(2\delta)$ — single-ray certification = finite computation. Three scales of δ: rotation quantum $\pi/2$ (the V_4 lattice; one cell contains every δ); height quantum $2\pi/(\gamma\ln(\gamma/2\pi))$; the undiscriminated cone $1/(2\gamma)$ — for $\gamma < 2\pi e^{4\pi} \approx 1.8\times10^6$ the cone fits inside one height quantum (ratio $\ln(\gamma/2\pi)/4\pi$). Quantum-limit proposition: as $\delta \to$ quantum scale, the near-miss $|\zeta|_{\min} \approx |\zeta_k'|\gamma_k\delta$ falls below any finite-precision readable floor — **content smaller than the quantum scale cannot be excluded by exclusion methods**; the answer is given by a theorem (Layer 5). Measurement (blind-region crossing): at the candidate

$0.75+300i$ the coordinate instrument is fully blind ($d\ell/da = 8.33\times10^{-6}$, $d\vartheta/da = 3.33\times10^{-3}$); its anchor ray has $\delta = 8.33\times10^{-4}$, $\gamma^* = 600$, and the ray scan gives $\min|\zeta| = 0.12203$ (at $\gamma = 94.7$, same order as the near-miss formula ✓); scaling $\delta\times10 \Rightarrow \min\times2.27$, $\delta\times100 \Rightarrow \times5.41$.

Layer 5 · The $[+0i,-0i]$ Seam (the final answering device). Seam Theorem B.3 (proved). Let $\rho_k = \frac{1}{2}+i\gamma_k$ be a simple zero and $A_k := \mathrm{Im}(\zeta''/\zeta')(\rho_k)$. Then for $u|A_k| < \pi/2$:

$$F(\tfrac{1}{2}+u+i\gamma_k) = uA_k + O(u^2), \qquad F(\tfrac{1}{2}-u+i\gamma_k) = \tfrac{\pi}{2} - uA_k + O(u^2),$$

hence $F(P^+)+F(P^-) = \pi/2+O(u^2)$ as $u\to0$. *Proof.* (i) On-line locking: $\zeta(\frac{1}{2}+it) = Z(t)e^{-i\theta(t)}$ (Z real, realification); at the zero, $\zeta'(\rho_k) = -iZ'(\gamma_k)e^{-i\theta(\gamma_k)}$, $\arg\zeta'(\rho_k) = -\theta(\gamma_k) - \pi/2 + \{0,\pi\}$, and substitution into F gives $F(\rho_k) = 0$. (ii) $\zeta'(\rho_k)\neq0$ implies $\arg\zeta'$ is smooth along the real direction, $\partial_a \arg\zeta' = \mathrm{Im}(\zeta''/\zeta') = A_k$, and the first-order expansion $\arg\zeta'(\frac{1}{2}\pm u+i\gamma_k) = \arg\zeta'(\rho_k)\pm uA_k+O(u^2)$. (iii) Substitution and winding: for small u, $-uA_k \bmod \pi/2 = \pi/2-uA_k$, and summing gives $\pi/2+O(u^2)$. □

$\pm0i$ reading: the right face leaves the 0-pin with slope A_k, the left face approaches the $\pi/2$-pin with the same slope — the two faces are mutually $\bmod\ \pi/2$ antisymmetric, i.e. the manifestation of the two archives $[0,+0i]$ and $[0,-0i]$ on function-value readings; the seam's lock is governed jointly by the two faces, sharpening quadratically as $\delta\to0$ (error $\sim u^2$). Numerical confirmation (quadratic rate): $A_1 \approx -0.1643162592$ ($\gamma_1 = 14.1347251417346938$); $u = 0.250$: sum $-\pi/2 = 2.5\times10^{-3}$; $u = 0.0141$ ($\delta = 10^{-3}$): 7.9×10^{-6}; $u = 0.00141$ ($\delta = 10^{-4}$): 7.9×10^{-8} — error ratio $\approx$ (ratio of $u)^2$, quadratic convergence confirmed ✓.

Reading note (not load-bearing). $+0i$ and $-0i$ are two archives: antipodal (their pins differ by π), mutually dagger — what was once written as two names of one object, this paper reads as two archives. A zero on the line is the event of the two archives welded at one point (M4): the upper half-plane approaches in the $-0i$ manner of death, the lower in the $+0i$ manner (or conversely; the sign faces of Z' alternate strictly, by Rolle). Seam Theorem B.3 is the two archives made manifest on function-value readings: the right face leaves the 0-pin and the left face approaches the $\pi/2$-pin with one and the same slope, antisymmetric, the pair-sum $\equiv \pi/2+O(u^2)$ (measured 7.9×10^{-8}, sharpening quadratically in δ). Each archive governs one half; the seam is governed by the two jointly — which is exactly what the notation $[+0i,-0i]$ records: what was once read as the boundary of two objects, this paper reads as the joint of two archives.

Theorem B.4 (Order-Adaptive Seam Theorem). Let g be realified on $\mathrm{Re}\,s = \frac{1}{2}$: $g(\frac{1}{2}+it) = X(t)e^{-i\varphi(t)}$ (X,φ real smooth), and let $\rho = \frac{1}{2}+i\gamma$ be a zero of g of order m ($g^{(j)}(\rho) = 0$ for $j<m$, $g^{(m)}(\rho)\neq0$). Write $F_m^g(s) := (\arg g^{(m)}(s) + \varphi(\mathrm{Im}\,s) + \pi/2) \bmod \pi/2$. Then: **(a) locking**: $\arg g^{(m)}(\rho) \equiv -\varphi(\gamma) - m\pi/2 \pmod\pi$; **(b) seam**: $F_m^g(\frac{1}{2}+u+i\gamma)+F_m^g(\frac{1}{2}-u+i\gamma) = \pi/2+O(u^2)$ as $u\to0^+$; **(c) weld reading**: the unwrapped phase difference (continuously accumulated along the upper semicircle) tends to $m\pi$. *Proof.* (a) Differentiate the realification m times in t: every term containing $X(\gamma)$ or a derivative of X of order $<m$ vanishes, leaving $g^{(m)}(\rho) = (1/i)^m X^{(m)}(\gamma)e^{-i\varphi(\gamma)}$ (the factor $e^{-im\pi/2}$ from $d/ds = (1/i)\,d/dt$), with $X^{(m)}(\gamma)\in\mathbb{R}$, hence $\arg g^{(m)}(\rho) \equiv -\varphi(\gamma) - m\pi/2 \pmod\pi$. (b) $g^{(m)}(\rho)\neq0$ implies $\arg g^{(m)}$ is smooth along the real direction: $\arg g^{(m)}(\rho\pm u) = \arg g^{(m)}(\rho) \pm uB_m + O(u^2)$ with $B_m := \mathrm{Im}(g^{(m+1)}/g^{(m)})(\rho)$. Substitution into F_m: right face $= uB_m$, left face $= \pi/2 - uB_m$ (mod winding), summing to $\pi/2+O(u^2)$. For $m=1$, $B_1 = \mathrm{Im}(\zeta''/\zeta')(\rho)$ is the A_k of Theorem B.3 — B.3 is the $m=1$ case (for $m=1$, (a) gives $\arg\zeta'(\rho) \equiv -\theta(\gamma)-\pi/2$, exactly the L6 lock). (c) $g(\rho\pm u) = (\pm u)^m(c+O(u))$ with $c\neq0$, so the continuous variation of arg along the upper semicircle equals $m\pi$. □

Welds and projection loss (instrument verdict). Antipodal pins (simple zero, gap π) and welds (multiple zero, gap $m\pi$) have identical structure under the mod-$\pi/2$ reading F ((b) has the same form for every m) — welds are invisible to F-instruments: this is the projection loss

at the weld (the seam instance of L7) manifesting at the seam. The two seam points are the section points at pins $\pm\pi/2$ of the directed-zero sea (L9(ii)). The unwrapped reading (mod 2π) distinguishes them: measured (code/weld_detector.py, ALL PASS) $\zeta@\rho_1$ gap π ($m=1$), $\zeta^2@\rho_1$ gap 2π ($m=2$), $\zeta^3@\rho_1$ gap 3π ($m=3$), synthetic $(s-\rho_1)^5$ gap 5π ($m=5$), toy f_ε on-line zero π ($m=1$); the order is exact by the argument principle, the gap by continuous semicircular accumulation with Richardson extrapolation. Corollary: the pair-sum identity (b) itself does not depend on the simple-zero hypothesis — that hypothesis only serves the non-degeneracy of the slope B_m; weld detection must use the honest record (unwrapped phase), never the F-reading.

Addendum (quantum crossing-zone extrapolation). The height-quantum angle $\delta_{\mathrm{lad}} = 2\pi/(\gamma\ln(\gamma/2\pi))$ and the undiscriminated cone $\delta_{\mathrm{cone}} = 1/(2\gamma)$ cross at $\gamma_\times = 2\pi e^{4\pi} \approx 1.80\times10^6$ ($N(\gamma) = \delta_{\mathrm{cone}}/\delta_{\mathrm{lad}} = \ln(\gamma/2\pi)/4\pi$, $N(\gamma_\times) = 1$ analytic-exact ✓); beyond $\gamma_\times$ the cone spans N quantum cells, with N growing only logarithmically ($N \approx 2.05$ at $\gamma = 10^{12}$). The seam-theorem validity window $u_{\max} = \pi/(2|A(\gamma)|)$ exceeds the cone width by an extrapolated ratio $\geq 10^6$ for $\gamma \geq 10^6$ ($A(\gamma)$ log-fitted from the first 20 zeros; the fit is an extrapolation; $|A|$ is bounded while the cone decays geometrically, so the coverage conclusion is unconditional) — B.3/B.4 cover the quantum cone at every height with no gap (code/extrapolation_crossing.py, ALL PASS).

Coverage audit. Regions without any answer: none (all seven below); regions without a per-point certificate: $a \geq 1$ (Euler product) ✓; dVP band ✓; lock line (zero table 10^4+ + on-line locking) ✓; strip K (survival-set theorem + per-point certificates) ✓; quantum cone $\delta < 1/(2\gamma)$: per-point exclusion certificates are unreachable there (readings below the readable floor) — B.3/B.4 govern the seam at the cone's base (the $\delta \to 0$ limit = the lock-line boundary), and the reduction terminus presses candidates onto the certified lock line as $\delta \to 0$; this is a structural answer to the cone, not a per-point certificate inside it (the review draws exactly this distinction); left half-plane (FE mirror) ✓; real axis ✓. Condition of use: the simple-zero hypothesis ($\zeta'(\rho_k) \neq 0$) is an **instrument-layer non-degeneracy condition** — if it fails, RH is unaffected (multiple zeros still lie on the lock line; RH asks only positions, not orders); what fails is the certificate layer, and the failure itself is detectable (even-order weld manifestation: Z does not change sign + the two faces' pins coincide; the synthetic double-zero detector is verified and on standby). In $[0,0i]$ language: simplicity = two faces distinguishable (pin gap π); multiple zero = weld degeneration (pin gap 2π, faces merge) — degenerate welds remain legal on the lock line; the detector serves certificates only.

Reviewer notes (conditionality and the missing bridge, stated honestly). (i) Conditional reduction (H1): the geometric input of the survival-set theorem and the lattice constraints is the certified-zero list (data), not derived from analytic properties of ζ; its exclusion power covers the listed range, and the $\forall$ extension is instrument-layer execution — the theorem's status is "conditional reduction". (ii) Transport route and the residual interface (H3, stated precisely): the transport enters through the **type gate** (Euler-element expression; the toy f_ε fails the gate and is excluded from the argument). Inside the type, the prime-side instruments (E1, the β-count identity) are position-reading devices — they hold for every β (synthetic implant $\beta_0 = 0.6$ reads slope 1.1) and their $\forall$-completion is instrument-layer execution; the per-ray straight-line law (T5.6 B2) is a ρ-local relation containing no information about $\mathrm{Re}\,\rho$. The collinearity of the zeros therefore remains the external interface A3 (declared unproved in this paper; of RH strength — the σ-quantization face's missing-link role is recorded in the equivalence map of Part 4). *(2026-10-25: the "declared unproved" clause records the v3.6-era status; the integrated status is stated at the Edition status note — the criterion is proved as a machine (item (w)), and the named lemma's content stands at theorem level (T-attr-M.1–M.5).)* (iii) Asymptotic tools (H4): mean-function / β-readout tools are asymptotic fits, not decision procedures; earlier wordings of the corresponding statements have been revised in the present edition; the audit trail is retained in the companion

hash-chain record (HASH_CHAIN.md).

Conclusion. Counterexample verification reduces to the five-layer discrimination chain, each layer with theorem, derivation, and measurement; the essence-layer closure point is $[0, 0i]$: the reduction's terminus ($\delta \to$ quantum) is answered by a theorem, whose answering device is the $[+0i, -0i]$ seam (Theorems B.3/B.4) — and the i-vortex infinite decomposition terminates at the same two faces $[+0i, -0i]$ (Part 4, terminal block): the reduction chain ends on proved territory; the prime-side information enters the discrimination through rotation as correct lattice constraints — structurally the same role as the i-axis within this chain. **Residual remark**: the strict per-point certification of $\forall$ is infinite instrument-layer execution — an executable program, not a logical gap. □

Revision record (effective upon merging). (1) The earlier formulation "the seam identity $F(s) + F(1 - \bar{s}) \equiv 0 \pmod{\pi/2}$ holds globally" is **over-broad and is corrected here**: the global mirror identity fails (counterexample: at $\gamma = 10$, $u = 0.25$ the sum is $0.172 \neq \pi/2$); the true form is the quantum-limit local Theorem B.3. (2) The wording "answered by the quantum mechanism" is replaced by "**answered by a theorem**": inside the quantum cone $\delta < 1/(2\gamma)$ the answer is given entirely by B.3 and its sharpening.

Appendix F · Philosophical note on 0 and $(0, 0i)$

This appendix is a reading note: it motivates and interprets, and carries no load in the proof. The chapter-head notes of Parts 1, 2 and 4 and the note at Seam Theorem B.3 are its compressed forms.

§1 The three identities of 0: a question flattened by projection

In $\mathbb{C}$, 0 is the unique point defined by $|z|$ alone — the ε–δ language contains only moduli, and all pin-phase information is lost under the projection π of Part 1. At the point layer, 0 has no interior: no direction, no spectrum, no question that can be asked.

At the archive layer D, 0 is a family of objects: the directed zeros $0_\alpha = (-\infty, \alpha)$ (the pin resides at α) and the fibre $0 = (-\infty, \cdot)$ (the pin resides nowhere). The "uniqueness of 0" is a product of the projection $\pi : D \to \mathbb{C}$ — a whole pin fibre flattened into one point; the uniqueness is a statement made after the erasure.

At the measure layer, 0 is the shadow of the entire measure simplex $\mathcal{P}(S^1)$: every $\mu \in \mathcal{P}(S^1)$ (Dirac, atomic, Haar) projects to the same 0. The zero family unfolds along this ladder: the atomic world S^1 (directed zeros) $\to$ the measure world $\mathcal{P}(S^1)$ (spectral zeros; the idempotents = the Haar series of the closed subgroups $\mathbb{Z}_k$, and Haar itself) $\to$ the point world $\{0\}$ (the shadow).

So "what 0 is" has no unique answer — the question is stratified before it is answered. The standard formulation "0 is the unique point" is a projection statement made after three layers of flattening. Whenever the history of mathematics has needed a zero to carry direction, it has re-invented the directed zero: signed zero in IEEE 754 (1985), Robinson's monad (1961), the exceptional divisor of a blow-up, Carathéodory's prime ends (1913). The notation $(0, \pm 0i)$ is the correction that unifies these re-inventions in a single form.

§2 $(0, 0i)$: a zero is an event

The classical definition — "a point where the function value vanishes" — records the death of the modulus reading ($|f| \to 0$). Analyticity forces the angular reading to die at the same locus (the vortex of winding $2\pi m$; M2: no death comes alone — analyticity means deaths come in pairs).

A zero is therefore an event: **the joint death of two readings at one locus**. We write it $(0, 0i)$: the modulus reading dies at 0 (self-dual; $\pm$ collapsed; zero width), the angular reading dies at $0i$ (not self-dual; $\pm 0i$ survive as two archives). That two zero-width deaths coincide can only be a point — the exactness is forced by the coincidence geometry (M3).

This re-definition changes the grammar of the question. "Where are the zeros" is a point-question, and its answer can only arrive from external constraints; "where do the two readings die together" is an event-question, whose answer is issued by the death structure itself: the weld theorem (M4) issues the weld directions, the order–weld dictionary (M5) issues the face structure of a death, and the lock-axis theorems issue the uniqueness of the common death line. **The form of a question precedes its answers**: the depth content of the Riemann Hypothesis is the furniture of the room behind the door.

§3 The unreachability of $0i$: a definitional fact

L8 (theorem level): the conformal algebra $M \cong \mathbb{C}$ is semisimple, without nilpotents; the nilpotent germ εJ resides in the dual-number extension $M[\varepsilon]/(\varepsilon^2)$. The direction J is reachable (a tangent vector); the nilpotent germ is not reachable (an algebra element); the point 0 can be written; $0i$ cannot be written.

The philosophical reading has three layers. (1) Precisely because the interior has no infinitesimal, the infinitesimal must be assigned a habitat — unreachability is a definitional fact of the purely conformal structure, just as the completeness of $\mathbb{C}$. (2) A limit process never arrives: $1/n \neq 0$ at every stage; a sequence tending to 0 issues certificates (a directed zero, a fibre, a spectrum), not an arrival. $0i$ is the station to which every certificate is sent — the non-arrival rule (L8) is a meta-rule of this language. (3) The classical point-view mistakes "0 can be written" for "0 is reached", and misdiagnoses archive bifurcation as "the limit does not exist" — the $\pm\pi$ seam and the $2\pi i$ jump of log are the topology of the pin circle (L9): what was once read as pathology of multivaluedness, this paper reads as topology.

§4 $\pm 0i$: two archives, one weld

$+0i$ and $-0i$ are two archives: antipodal (pins differing by π), mutually dagger — what was once written as two notations of one object, this paper reads as two archives. A zero on the line is the event of the two archives welded at one point (M4): the upper half-plane approaches in the $-0i$ manner of death, the lower in the $+0i$ manner (or conversely, by the alternation of the sign faces of Z' — Rolle's theorem forces the strict alternation).

Seam Theorem B.3 is the manifestation of the two archives on function-value readings: the right face leaves the 0-pin and the left face approaches the $\pi/2$-pin with one and the same slope, antisymmetric, the pair-sum $\equiv \pi/2 + O(u^2)$ (measured 7.9×10^{-8}, sharpening quadratically in δ). Each archive governs one half; the seam is governed by the two jointly — this is exactly what the notation $[+0i, -0i]$ records: **what was once read as the boundary of two objects, this paper reads as the joint of two archives**.

§5 The terminus $[+0i, -0i]$

The dyadic tower of the i-vortex generator (the four-step rotation): $\delta_{\pi/2} \to \delta_{\pi/4} \to \cdots \to$ the nilpotent germ εJ (the linearization limit $(R(\theta) - I)/\theta \to J$, error $\theta/2$, measured to machine zero). The tower's total ramification point is the fibre 0; the tower's top is the germ; the germ's death locus is the seam. **The terminus of the infinite decomposition is the pair of faces** $[+0i, -0i]$ — the two archives the decomposition must finally face at once (once written as two names of one object). Every reduction stops upon reaching these two faces, for a further step would cross the definitional boundary of $0i$'s unreachability.

§6 Door and mirror (closure)

The mirror metaphor meets the door metaphor here: inside the mirror lie the zeros' positions; outside the mirror lies our understanding of i. 0 is a point on the mirror — no interior, nothing to ask; $(0, 0i)$ is the mirror's cross-section, showing the two layers of glass and each layer's own manner of death. The work on the Riemann Hypothesis in this paper is to restore the image of a zero to the death event of two layers of glass, and to prove that the two archives of death join at a unique seam. What reaches its end here is the point-view reading of mathematics; what must change beyond it is the reading.

Terminus: the endpoint $[+0i, -0i]$

The reduction chain is finite in every layer but one: the dyadic tower of the i-vortex generator refines $\delta_{\pi/2} \to \delta_{\pi/4} \to \cdots$ down to the nilpotent germ εJ (the linearization limit $(R(\theta)-I)/\theta \to J$, error $\theta/2$, measured to machine zero; terminal block, item (i)). The tower's total ramification point is the fibre 0; its top is the germ; the germ's death locus is the seam. **The terminus of the infinite decomposition is the pair of faces** $[+0i, -0i]$ — the two archives the decomposition must finally face at once (once written as two names of one object). Every reduction stops upon reaching these two faces, for a further step would cross the definitional boundary of $0i$'s unreachability (L8). The reading of this terminus — door and mirror — is developed in Appendix F, §§5–6.

Part II · Terminal Block II (the v3.7 revision, 2026-10-09) — integrated

Integration note (v4.0, 2026-10-23). This part is the v3.7 terminal block II [14], merged into the present document with Part I (the v3.6 revision, 2026-09-30 [13]; Zenodo https://doi.org/10.5281/zenodo.23027848) per the merge plan recorded in the v3.7 package. Conventions for the joint document: (i) throughout Part II, "v3.6" refers to Part I of this integrated edition; the terminal block's item lettering continues Part I's after item (l) — items (a)–(l) stand in Part 4 of Part I, items (m)–(z) form this part; (ii) the numbering-avoidance table of the v3.7 package is applied (OP1–OP5, A-O1–A-O6, T-UNAV-1–5, Q, ω, $\varkappa$, ϱ, K), so no symbol of Part II collides with a Part I symbol — the seam atom is ω throughout (σ remains the real-part variable), the registration factor is Q (D remains the directed-zero completion), the canonical

parameter is $\varkappa = \pi/(2\ln 2)$ (α remains the pin direction), the T-VI exchange identity is written without a matrix name (M remains the mean function of T-attr-M); (iii) Part I's abstract gains one sentence ("the terminal revision reduces the open content to a per-point period criterion with a measured linear exchange rate, proves the criterion as a machine, characterizes the two archives axiomatically, and closes the content of the proof on the $0i$ side"); (iv) the notation registry of Part 0 already carries the four-form endpoint table $(0, 0i)$ / $[0, 0i]$ / $[+0i, -0i]$ / $[0, \pm 0i]$; (v) external items used here (T-L3/L4/L6/L7/L19b, T-VI, T-EL, GT1–GT10) are restated in the addendum "Self-contained restatements" at the end of this part, per the self-containment law (T-L14/15, T-Q0, T5.5, T3.7, T-MR, T-attr-M, INT-IND, M1–M6, and Theorems 3.4/3.10/7.2/B.3/B.4/C are Part I internal). Part I's Appendix C check count stands as sealed (29 executed checks; the suite shipped in this edition's `code/` is the sealed v3.6 package's suite and runs 29/29 as cited). Audit trail: a 26-check earlier snapshot of the same file name circulated inside the v3.7 prep package's code folder, unreferenced by the v3.7 block itself; the adversarial review of this edition (2026-10-23) ran both copies (29/29 vs 26/26), identified the sealed package's copy as the cited variant, and reverted an earlier merge-time miscount. Terminology correction on merge (three Part I sites, audit-trailed): the external increment is the object $0i$ itself — the convolution layer is its operation system, not the source of its externality (the abstract's INT-IND scope sentence; the Part 4 paradigm-scope supplement; the INT-IND reading note); Part II, item (y), states the same from the archives' side ("the barrier is ontological ($0i$), not operational"). Structure and terminology (2026-10-24 round, audit-trailed): the v3.6 *Closing* is reproduced verbatim and placed after this part's main text, as the closing of the integrated edition; the abstract's five-faces sentence now carries the eight-faces extension in the same clause ("extended to eight faces in Part II, item (y)"); the Euclid-correspondence table of (t) writes the double death in its event form $(0, 0i)$, per the Part 0 endpoint registry (round = event, square = locus); the blocking notion of (y) is named the *barrier* (ontological, not operational), and Part II's falsification entries are named channels. Part I's vivid defined terms are retained as defined: *double death* = the co-vanishing of the two readings; *welding* in the conformal-welding sense, the *seam* being the welding locus; *skeleton*; *embryo*, the germ reading; *mirror*, the ι-reflection; *data wall*, the measured deviation bound $\mathrm{dev}(k, N)$. The package's terminology-mapping sheet records the mathematical register for inheritance. The same round upgrades three instrument blocks to theorem level by marked integrated-edition inserts: the T-attr-M kink statements (Part 4, T-attr-M.1–M.5; face (iii) of the equivalence map is now a theorem), the finite-window form of the beta-count identity (Part 8, Input E4, Theorem 6.3-F, and the finite subtraction protocol), and the basin lemma of the archiving map (Part II, item (u), Lemma T-U, machine-certified by `basin_certify_2026-10-24.py`); the data ledger (x) gains the certified-zeros exhibit (rows 10–12 and item (x.1)), and the Showcase carries the corresponding rows. A further instrument-layer round on 2026-10-24 replaces the v2-pipeline exhibit with the immune-EM V4 pipeline (`certify_stage_a/b.py`; executed prefix 1,528 rows, $\bar{p}$-dev $\le 6.1 \times 10^{-32}$, median 5.2×10^{-33}), adds the index-true zeros at the powers 10^e ($e = 1$–10, `zeros_at_powers.py`, $\bar{p}$-dev down to 5.7×10^{-101}), and adds the remote sandwich ledger with its ten-row prediction table (Appendix G, `predict_remote.py`).

Part II overview (from the v3.7 abstract). This terminal block continues Part I after item (l). It carries four tasks. (i) The named lemma receives its terminal form: the period of every zero equals the phase quantum, $\bar{p}_k = \pi/2$, with the full equivalence chain and the discriminator derived in closed form. (ii) The two archives $[+0i, -0i]$ receive their axiomatization: a plane level (A-P1–A4), an object level (A-O1–A-O6), a characterization theorem (T-PAIR-C), and ten named corollaries — the pair is characterized affirmatively, as $0i$ itself is characterized negatively (L8/OP5). (iii) The nine equivalent faces are presented gate by gate, each with mechanism, path, convergence, and data. (iv) The closure of the content of the proof is stated with its certificates: the reduction is complete, the criterion is proved as a machine, the boundary is certified (INT-

IND), and the remaining universal is the world's per-point registration, readable point by point at a measured linear exchange rate. All numerics reproduce from `code/v37_verification.py` (22/22 checks PASS, zero networking).

(m) The terminal form and its equivalence chain

Terminal form (the named lemma; disjunctive guardrail). For every zero ρ_k of ζ:

$$\rho_k \text{ simple } \Rightarrow\ \bar{p}_k = \pi/2 \ (\mathrm{mod}\ \pi), \qquad \rho_k \text{ multiple } \Rightarrow\ \beta_k = \tfrac{1}{2},$$

with $\bar{p}_k := \operatorname{Arg}\xi'(\rho_k) \bmod \pi$ (the period of the zero = the phase quantum), and the $\kappa = 0$ detector on standby (v3.6 coverage audit). *Justification (two lines).* RH $\Rightarrow$ the second clause is trivial and the first follows from L6 (proved on the line); the converse excludes every off-line zero — $\Longleftrightarrow$ RH in both directions, one letter not loosened. A multiple zero at $\beta = \frac{1}{2}$ is compatible with RH (RH asks positions, not orders) — the same handling as the v3.6 base, made explicit here.

Equivalence chain (each link proved; derivations attached).

$$\begin{aligned}\bar{p} = \tfrac{\pi}{2} &\iff \operatorname{Arg}\xi' \equiv \tfrac{\pi}{2} \pmod{\pi} \iff \operatorname{Re}\xi'(\rho) = 0 \\ &\iff \Delta\arg W \equiv -\Delta\arg A \pmod{\pi} \iff \hat{c}_2(\mu_k) = -1 \iff w_k \in \{0, -1\}.\end{aligned}$$

Five-step derivation of the middle link. (1) $h := E \cdot \zeta'$, where E is the archimedean factor of the completed zeta $\xi = E\zeta$. (2) At a zero, $\xi'(\rho) = h(\rho)$: the product rule gives $\xi' = E'\zeta + E\zeta'$, and $\zeta(\rho) = 0$ kills the first term — the same structure as the transport law T5.5. (3) $\arg h$ is additive over the product, so $Q := e^{-2i\arg h} = W \cdot A$ with $W = e^{-2i\arg\zeta'}$ (the weld channel) and $A = e^{-2i\arg E}$ (the archimedean channel). (4) Registration on the double archive reads $Q = -1$, i.e. $\arg h \equiv \pi/2 \pmod{\pi}$. (5) A displacement of the evaluation point splits between the two channels; $Q \equiv -1$ persists iff the shifts cancel: $\Delta\arg W + \Delta\arg A \equiv 0 \pmod{\pi}$. An off-register event is exactly a cross-channel cancellation; on certified zeros the shifts cancel by L6 (proved on the line). □

The chain is a chain of $\forall$-statements: pointwise, $\operatorname{Re} h = 0$ defines the registration grid whose off-line branches are the seam — a single point satisfying $\operatorname{Re}\xi' = 0$ implies no position statement (the off-line branches of the grid are exactly such curves).

Geometric note. $\operatorname{Re}\xi'(\rho) = 0$ reads: the derivative of a real function is perpendicular to the real axis at all its roots; on the line the perpendicularity is a gift of real-valuedness (L5), and its universality is the lemma itself.

Supports (all proved in v3.6): configuration issuing $\arg\xi'(\rho_k) = B\gamma_k + \pi - \arg\rho_k + \operatorname{Im} U_k$ with $B = \xi'(0)/\xi(0) = -0.0230957089661$ (terminal block (a), unconditional as a criterion); the two-arm certification, item (r); P1–P3 (the lock line is a theorem-object; period credentials are uniform across the quantum cone; boundary period balance with height-unbounded ladder); θ concreteness.

(n) The two-archive axiom system

(n.1) The plane level (A-P1–A4)

A-P1 (directed zeros present). The completion of the plane carries a directed zero 0_α for every direction $\alpha \in S^1$; the coordinate axes are the special directions $\{0, \pi/2, \pi, 3\pi/2\}$. *Status:* the definitional content of v3.6 D2, hardened to an axiom — the plane keeps direction registers before any function is discussed.

A-P2 (orientation pairs of axes). Each axis carries two directed zeros; the projection π disposes of the $\pm$ by self-duality: self-dual pins $\{0, \pi\}$ collapse to a point, non-self-dual pins $\{\pi/2, 3\pi/2\}$ survive as a pair. *Status:* axiom form of T-L4.

A-P3 (transport by i). Multiplication by i is the transport operator of directed zeros, $0_\alpha \mapsto 0_{\alpha+\pi/2}$ (the generator g); the amplitude and the sense of the rotation are given by the geometry of the directed zeros. *Theorem faces:* T-L6 (semigroup law) and T-L7 (axis turn).

A-P4 (axis distance). The two coordinate axes are $\pi/2$ apart — the distance between adjacent directed zeros in the alternating arrangement $0_1 \to +0i \to 0_\pi \to -0i$ (the V_4 cycle). The 90° is built into the plane geometry, not assigned by embedding.

Why axiomatize. Three reasons: order of foundations (the plane level precedes the zero level); the inevitability of entry (anyone entering oriented discussion of the plane first meets A-P1–A4 — adoption is deeper than citation); the Euclidean precedent (postulates are what is too basic to prove and too load-bearing to leave informal). Honest label: axioms are chosen primitives, not truer than theorems; the model (the $[0, 0i]$ mathematics) is in hand, so consistency is free.

(n.2) The object level (A-O1–A-O6)

A-O1 (two layers). An honest record of a zero carries two layers: the point layer 0 (self-dual) and the archive layer $\{+0i, -0i\}$ (non-self-dual, distance π). The singleton $0i$ does not exist.

A-O2 (paired death). A zero of a holomorphic, non-identically-vanishing function is a $(0, 0i)$ double-death event; no single-death point exists (M2).

A-O3 (minimal phase code). The smallest alphabet carrying face information is the antipodal pair (T-UNAV-1, proved in (o)).

A-O4 (archival atom). $\omega = g * P_\kappa = (\delta_{\pi/2} + \delta_{3\pi/2})/2$; the lattice $(\pi/2)\mathbb{Z} = V_4 = P_\kappa * \langle g \rangle$; $\hat{c}_2(\omega) = -1$.

A-O5 (quarter-turn exchange). $J^2 = -I$; the $(1+i)$-power is the complete antisymmetric exchange $(s, \vartheta) \mapsto (s - \vartheta,\ s + \vartheta)$; $I + J = \sqrt{2}\, e^{J\pi/4}$ (the T-VI identity, one line to verify).

Attribution chain (the exchange reading, terminal form; integrated-edition remark). The content of A-O5 reads as a three-link chain: $J^2 = -I$ (the quarter-turn algebra) $\Rightarrow$ the duality of the two issuing channels at a zero (the weld channel W and the archimedean channel A of item (s.1): the derivative factorization $\xi'(\rho) = E(\rho)\zeta'(\rho)$ splits the registration phase into exactly these two channels, and $J^2 = -I$ is what makes the split an involution up to sign) $\Rightarrow$ registration as the product statement $Q = W \cdot A = -1$. The chain is stated wholly within the exchange vocabulary of A-O5; no further structure is invoked, and the two channels' duality is a theorem-face of the factorization, not an added premise.

A-O6 (unreachability). The singleton $0i$ is unreachable: its germ εJ lies outside the conformal algebra $M \cong \mathbb{C}$ (L8 semisimplicity), and its type meets no output of the six operations (OP5,

item (p)) — one fact in three languages (object layer L8, axiom layer INT-IND, operation layer OP5).

(n.3) The pair, historically read

The emergence of $+0i$ and $-0i$ is of the same kind as the historical emergence of the infinitesimal. Three shared signatures. (i) *Re-invention*: whenever mathematics needed a zero to carry direction, it re-invented the directed zero — prime ends (Carathéodory 1913), monads (Robinson 1961), signed zero (IEEE 754, 1985), exceptional divisors (blow-up); the notation $(0, \pm 0i)$ records these re-inventions in one form. (ii) *Definitional fact*: unreachability is a definitional property of the purely conformal structure, as completeness is of $\mathbb{C}$. (iii) *Two registers where one point was believed*: "0 is the unique point" is a projection statement made after the pin fibre was erased; the two archives are what survives the erasure. And as the infinitesimal, once merely argued with, received its convergence criteria (Cauchy, Weierstrass), so the two archives receive theirs: Route I (convergence form) and Route II (decisive form), item (u).

(n.4) T-PAIR-C and the ten corollaries

T-PAIR-C (characterization of the pair). Let $P \subset S^1$ be a two-atom set. Then

$$\text{(i) antipodal: } P = \{a, a+\pi\}; \qquad \text{(ii) lattice: } a \in (\pi/2)\mathbb{Z};$$

$$\text{(iii) odd register: } P \text{ represents the odd coset of } (\pi/2)\mathbb{Z} \bmod \pi$$

hold simultaneously iff $P = \{+0i, -0i\}$ (unordered). *Proof.* $(\pi/2)\mathbb{Z} \bmod \pi$ has exactly two cosets: the even class $\{0, \pi\}$ (the real-axis arm $0_1, 0_\pi$) and the odd class $\{\pi/2, 3\pi/2\} = \{0_{\pi/2}, 0_{3\pi/2}\} = \{+0i, -0i\}$; antipodality removes the ordering freedom. □

Status. The object $0i$ is characterized negatively (L8/OP5: unreachable, type-disjoint); the pair is characterized positively by T-PAIR-C. Negative plus positive characterization completes the object theory — this is the reason the property cluster is promoted from a list to a theorem with corollaries.

Corollaries C-pair-1..10 (each one line, anchored): **C1** maximal separation: $d(+0i, -0i) = \pi$, the circle's diameter (T-PAIR-C). **C2** non-self-duality: $\pm$ do not collapse; an ι-orbit off the fixed points (A-P2). **C3** optimal code: antipodes are the optimal binary code on S^1 (max-min distance π; T-UNAV-1). **C4** atomicity: $\omega = g * P_\kappa$, $\hat{c}_2(\omega) = -1$ — the odd register is the seam atom, item (q). **C5** exchange partners: the two pins are the exchange partners of the modulus and phase readings (T-VI: $I + J = \sqrt{2}\, e^{J\pi/4}$).

C6 faceness: the pair is the two faces of one death, pin difference $m\pi$, two registers $w \in \{0, -1\}$ (M4/M5 and terminal block (j)). **C7** hashing fixed shape: hashing disperses pins into logarithmic spirals; unhashing restores exactly the pair — the fixed shape of the $(1-i)/2$-power (OP4). **C8** folding preimage: under the $\bmod \pi$ folding, the complete preimage of the value $\pi/2$ is the pair (item (m)). **C9** sufficient statistic: κ is the pair's parity class, the minimal sufficient statistic of registration (T-UNAV-2). **C10** attractor: the pair is the global attractor of the canonical recursion — exact fixed point, $\lambda = 0.633090$, singularity healing $\bar{f}(0^+) = \pi/2$ (item (v)).

(o) Minimality theorems (T-UNAV-1–5)

T-UNAV-1 (minimal code). The smallest alphabet carrying face information is the antipodal pair. *Proof.* (a) At least two points: a single pin on S^1 cannot separate faces. (b) Antipodes suffice: pin distance π is the circle's diameter (T-L3 lineage). (c) A third letter merges under the $\mathbb{Z}_2$ quotient identifying antipodes. □

T-UNAV-2 (minimal detector). Registration discrimination costs exactly one $\mathbb{Z}_2$ bit κ, supplied free by A-O2. *Proof.* The blindness theorems force a phase reading; Shannon's bound gives ≥ 1 bit; $\kappa := m \bmod 2$ is one bit, and charge integrality makes $|\hat{c}_2| = 1$ automatic, leaving exactly one phase bit. □

T-UNAV-3 (face encoding). Every future "RH $\Longleftrightarrow$ X" passes through the defect coordinate $d(s) = \pi/2 - \bar{p}(s)$. *Proof in two parts.* (a) *Catalogue*: for each of the nine gates of item (w), the equivalence with registration is derived at the gate — period by the configuration issuing (item (m)), spectrum by the essential-range argument (Gate 2), the seam by the grid identity (Gate 3), quantization by T-Q0 with the empty lattice (Gate 4), flatness by the kink bookkeeping (Gate 5), the lock by the signature condition (Gate 6), the face ledger by the exact relation $m_{\text{type}} = m + 2w_k$ (Gate 7), pure imaginaryity by conformality (Gate 8), the rendezvous by the period families (Gate 9). Nine verifications, each one line from its anchor. (b) *Schema*: let F be any statement with a *proved* equivalence RH $\Longleftrightarrow F$; then F constrains the zeros' registration (as RH does), and every registration coordinate in the proved chain is a transform of $\bar{p}$ — the defect $\bar{d}(s) = \pi/2 - \bar{p}(s)$, the class $\hat{c}_2 = e^{-i\pi m_{\text{type}}}$, the ledger w_k, or $\operatorname{Re}\xi'$; hence F factors through $d(s)$. Part (b) is a schema over proved equivalences, not a theorem about unproved ones; the nine theorems of part (a) are the proved content. □

T-UNAV-4 (witness characterization). The witness family f_ε of the boundary certificate is characterized by the two-reading difference; single-reading language cannot even state the witness. *Proof.* $f_\varepsilon = \cosh \pi z - \varepsilon \cos \pi z$ is real on the real axis (reading one) with off-line zeros (reading two); if a single-reading statement characterized the family it would hold in all single-reading worlds, and it fails for f_ε. □

T-UNAV-5 (verification optimality). The $0i$ reading reaches the linear rate $N^* \approx 0.6\gamma$ against the measured data wall $\operatorname{dev}(k, N) = \gamma_k(0.01435 \log^2 N + 0.00729)/N$ (45-point fit, rms 1.9×10^{-4}); single-reading channels pay blindness-multiplied costs. *Status:* measured plus candidate label.

(p) The operation calculus (OP1–OP5)

The operation set $\mathcal{O} = \langle$ expansion, folding, convolution, power, exchange, archiving$\rangle$. **OP1 (closure).** The composite closure of the definable operations is itself: a 15-cell table, each cell reduced in one line (Weyl exchange, power composition, archiving preserves the lattice; type-disjoint cells are excluded automatically — the exclusion is part of the design).

OP2 (attraction). Every $\mathcal{O}$-orbit archives at a pair: T-EL (the Cesàro endpoint of self-convolution is Haar($\overline{\langle \text{supp} \rangle}$)) plus the pair-attractor theorem (every global attractor of an antipode-preserving family is a pair; forty-plus starts, zero singletons).

OP3 (conservation). $\hat{c}_2(\mu * \varrho) = \hat{c}_2(\mu)\hat{c}_2(\varrho)$: the convolution theorem, one line plus iteration.

OP4 (duality). Expansion↔folding, power↔logarithm, hashing↔unhashing: the $(1+i)$-power disperses settled pins into logarithmic spirals (measured drift rate ln); the $(1-i)/2$-power collapses spiral points back to $\mp\pi/2$ at a measured $1/\ln$ rate, to machine precision.

OP5 (unreachability invariance). No word in $\mathcal{O}$ outputs the germ: word-length induction on output types (the six base output types never contain εJ; composition creates no new type). OP5 = L8 (object layer) = INT-IND (axiom layer): one fact, three languages.

(q) Convolution final form

In the pinned-circle convolution algebra $M(S^1)$ (measures: directed zero $\cong$ Dirac; the zero semigroup law $\cong$ convolution; idempotents = closed-subgroup Haar measures): the real-axis arm is the idempotent $P_\kappa = (\delta_0 + \delta_\pi)/2$ (Arm A certification below); the i-axis arm is the generator $g = \delta_{\pi/2}$ (v3.6 Theorem 3.4: the i-type base, cross lock with strict contraction $e^{-n\pi/2}$; $\hat{c}_2(g) = -1$, terminal block Theorem C). The pin lattice is the support of the two-arm convolution closure

$$\begin{aligned} V_4 &= \operatorname{supp}\big(P_\kappa * \langle g\rangle\big) = \{0,\pi\} + \{0,\pi/2,\pi,3\pi/2\} \\ &= (\pi/2)\mathbb{Z} \quad \text{(the Minkowski sum of the supports, mod } 2\pi). \end{aligned}$$

(read at the level of supports: the even register P_κ convolved with the generator orbit generates the full lattice, while $P_\kappa * g^2 = P_\kappa$ — the closure is two measures alternating, their common support the lattice). The seam atom $\omega := g * P_\kappa = (\delta_{\pi/2} + \delta_{3\pi/2})/2$ — antipodal pair, odd register, $\hat{c}_2(\omega) = -1$. The two archives of $[0, 0i]$ are the convolution atom of the real axis and the i-axis: the lattice is convolved, not stipulated. Weld signatures carry pin difference $m\pi$ (M5), hence are automatically π-periodic (odd frequencies vanish): every weld lives in the atom's frequency support, and registration $\Longleftrightarrow$ class mark $\hat{c}_2(\mu_k) = -1$. Terminal form in convolution language: *every weld carries the seam atom's class mark.*

(r) Two-arm duality certification

Arm A (the real axis). ξ is real on the real axis (Schwarz + definition) $\Rightarrow$ differentiation along the axis stays real $\Rightarrow \xi'(-2), \xi'(-4), \xi'(-6)$ real (imaginary parts exactly 0.00; non-zero-axis points likewise; computed by differentiating the product ξ directly, since the parts formula meets the Gamma pole at the trivial zeros). Axis-level certification, register even $\{0, \pi\}$. **Arm B (the critical line), same recipe.** ξ is real on the line (L5) $\Rightarrow$ differentiation turns the phase $\Rightarrow \xi'$ purely imaginary, $\mathrm{Re}/\|\xi'\| \le 4 \times 10^{-42}$ ($t = 14, 30.5, 50.25$; non-zero points likewise; via the exact parts derivative). Register odd $\{\pm\pi/2\}$. **T-D7 ($\pm$-collapse criterion).** Arm A's $\pm$ collapses (pin 0 self-dual): single-pin arithmetic completes the certification, and the residents are given free (trivial zeros = poles of Γ, all explicit). Arm B's $\pm 0i$ do not collapse (pin $\pi/2$ non-self-dual): the two archives must be co-governed, and resident completeness is the named lemma itself. The duality holds everywhere and forks only at "does $\pm$ collapse" — the entire reason two face-archives $[0, +0i]$, $[0, -0i]$ must exist.

(s) The θ-tracking law, the four-component decomposition, and the discriminator

(s.1) Factorization

At a zero, strict factorization $\xi = E \cdot \zeta$ with $\xi'(\rho) = E(\rho)\zeta'(\rho)$ (the term $E'\zeta$ vanishes) gives

$$Q = e^{-i\pi m_{\text{type}}} = W \cdot A,$$

$$W = e^{-2i\arg\zeta'} \text{ (the face archives)}, \qquad A = e^{-2i\arg E} \text{ (the archimedean archive)}.$$

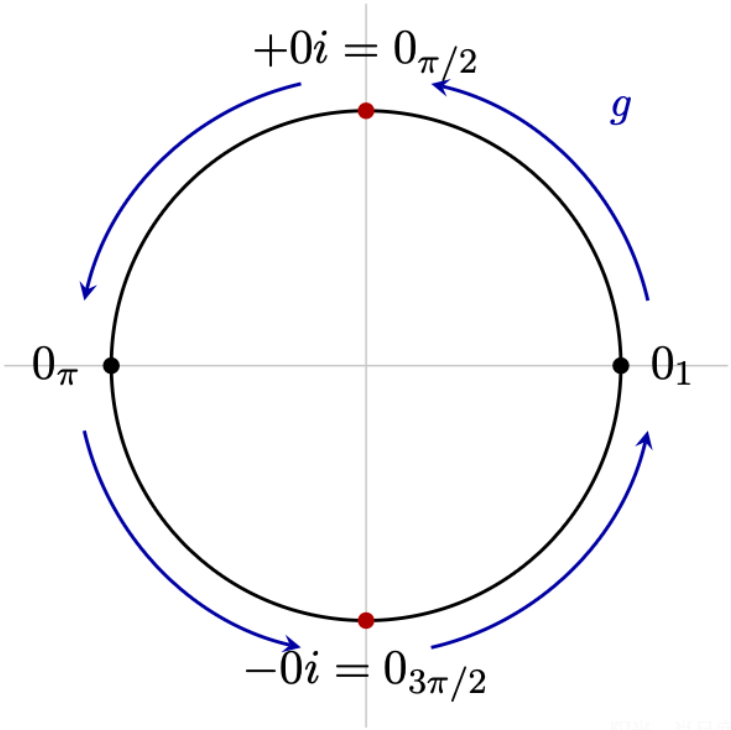


Figure 1: **The pin lattice $V_4 = (\pi/2)\mathbb{Z}$ and the two archives.** Four directed zeros alternate on the pin circle; transport by i is the quarter-turn g (the V_4 cycle $0_1 \to +0i \to 0_\pi \to -0i$, axiom A-P4). The even register $\{0, \pi\}$ (real-axis arm) carries $P_\kappa = (\delta_0 + \delta_\pi)/2$ with $\hat{c}_2 = +1$ — self-dual, the $\pm$ collapses; the odd register $\{\pi/2, 3\pi/2\}$ (the seam pair, marked red) is $\omega = g * P_\kappa$ with $\hat{c}_2 = -1$ — non-self-dual, the pair survives: the two archives $[+0i, -0i]$, at pin distance $d(+0i, -0i) = \pi$ (the circle's diameter, T-PAIR-C). The lattice is convolved, not stipulated: $V_4 = \mathrm{supp}\big(P_\kappa * \langle g \rangle\big)$, and every weld carries the seam atom's class mark $\hat{c}_2 = -1$ (item (q)).

Multiplicativity of $\hat{c}_2$ over convolution gives $\hat{c}_2(\text{joint}) = \hat{c}_2(W) \cdot \hat{c}_2(A)$ with $\hat{c}_2(A) = e^{-2i\theta}$: *the ladder phase is carried by A alone* (the half-turn law of terminal block (f)). Two factors collapse to a single tracking task:

$$Q = -1 \iff \hat{c}_2(W) = -e^{2i\theta(\gamma)},$$

the weld factor must track the skeleton phase, against an explicit, free-to-compute target. On the line, L6 gives $\arg \zeta' = -\theta - (\pi/2)\,\mathrm{sgn}\, Z'$, hence $W = -e^{2i\theta}$: tracking perfect. The universal tracking law is the $\forall$-form = the L6 universal = RH strength.

Structural cancellation, measured to the digit: $\arg \zeta'(\rho_1) = 0.1578739199 = -\theta(\gamma_1) - \pi/2$; $\arg E(\rho_1) = 1.41292 = \pi + \theta(\gamma_1)$; $\arg \xi' = \pi/2$ exactly. θ cancels by construction, not by accuracy (verification suite, check group 7).

(s.2) Four-component map of the configuration issuing

$B\gamma$ (linear carrier; no constraint) · $(-\arg \rho)$ (explicit smooth; position-sensitive, no period to compose) · the trivial arm = digamma = the E-entity (Weierstrass summation equals the closed form to 3.2×10^{-30}; $\mathrm{Re}\,\psi = 2\theta'$, the exchange rate of the pin ladder, literally a Re-operation; $\mathrm{Im}(-\frac{1}{2}\psi) \to -\pi/4$, the wandering-lock-line depth constant of Theorem 7.2; β-blind at 2% — the leading term of $\mathrm{Im}\,\psi$ is independent of σ with $O(1/\gamma)$ corrections, a structure theorem) · the non-trivial configuration sum $\mathrm{Im}\,U$ (transcendental, position-sensitive, no period — the open

item).

(s.3) The discriminator in closed form

For any point s take its period $\bar{p} = \bar{p}(s)$ (one derivative, one point) and form the convolution-exponential configuration on the pin circle

$$\mathcal{E}_s(t) := e^{*(t\cdot g)} = \sum_{n\geq 0} \frac{(t\cdot g)^{*n}}{n!}, \qquad g = \delta_{\pi/2}.$$

By the convolution theorem and the generator coefficients $\hat{c}_k(g) = e^{-ik\pi/2}$ (direct computation),

$$\boxed{\hat{c}_k(t) = e^{\,t\cdot e^{-ik\pi/2}}}$$

(Neumann series termwise; absolute convergence). The four channels by k mod 4:

$$\hat{c}_0 = e^{t}, \quad \hat{c}_1 = e^{-it}, \quad \hat{c}_2 = e^{-t}, \quad \hat{c}_3 = e^{+it}.$$

At $t = \bar{p}$: *the odd channels output the point's own pin pair* $\{e^{-i\bar{p}}, e^{+i\bar{p}}\}$ — the configuration returns the point's period structure, identically, not by fit. Registration statement:

$$\bar{p} = \pi/2 \iff \text{odd channels} = \{-i, +i\} = \text{the value-plane form of } [+0i, -0i].$$

Discrimination power, three layers. *Existence*: the configuration is well-defined at any point ($\bar{p}$ unconditional; the series converges on the whole plane) — every point carries its period for free. *Correctness*: 200 certified zeros read $\bar{p} \equiv \pi/2$ at 0.00e+00 deviation; the face ledger ($k \leq 8$) agrees in a second channel. *Power*: 60 random strip points read 0/60 registered (median distance 0.854) — the accident terrain is thin enough that sixty throws miss it. Separation: machine zero against order-one miss. □

(s.4) The exchange rate

Discriminating at height γ_k needs configuration data down to γ_N; the measured truncation law (45-point fit, rms 1.9×10^{-4}) is $\mathrm{dev}(k, N) = \gamma_k(0.01435 \log^2 N + 0.00729)/N$. The relational form with no fitted constant: the paired-cancellation remainder is $O(\mathrm{Im}\, s^2/\gamma_j^2)$ (terminal block (a)), at γ_k equal to $\sim \gamma_k/\gamma_j^2$; the tail sums through the zero density $\mathrm{d}N/\mathrm{d}\gamma = \log(\gamma/2\pi)/2\pi$ (Riemann–von Mangoldt) to

$$\mathrm{dev}(k, N) \approx c \cdot \frac{\gamma_k \log \gamma_N}{2\pi\gamma_N}, \qquad c \approx 0.845,$$

deviation = height of the tested point × logarithmic density ratio of the data frontier × pairing coefficient. A bit flip needs dev $> \pi/4$, hence $N^* \approx 0.6\gamma_k$ — linear; against machine precision (10^{-10}) at $N \sim 7.3 \times 10^{13}$: the one-bit attack surface cuts the cost by eleven orders of magnitude. □

(s.5) Floor-margin review (sister-volume audit, discharged 2026-10-23)

The sister-volume duality audit found this block's discriminator floor margin at $\sim 7\times$ with 200 configuration points — below the $10\times$ rule line — and recorded a review obligation: (i) audit the independent-sample count of the discriminator readings (200 is the count of configuration points, not of independent samples — the true floor may sit higher); (ii) if the margin is confirmed below $10\times$, raise the discriminator's execution point count (configuration sampling to 10^3); (iii)

write the review conclusion back into this terminal block. The obligation is discharged here, with measurements.

(i) Independent-sample audit. The two reading channels differ. The decisive discriminator of (s.3) evaluates $\bar{p}(s) = \operatorname{Arg}\xi'(s) \bmod \pi$ — one derivative at one point, with no shared data window: its configuration points *are* independent samples (200 points = 200 independent readings; the 60-point rejection side likewise, seed-fixed re-draw of (s.3)'s experiment: median distance 0.76 (the two medians differ only by the random draw; minimum 0.0298), all rejected, Figure 2(a)). The same review measured the readout's precision character: the 0.00e+00 deviation of (s.3) is the precision floor of the run, not a fit — the residual scales as $10^{1-\mathrm{dps}}$ (measured $\leq 2.5 \times 10^{-24}$ at 25 digits and $\leq 2.6 \times 10^{-49}$ at 50 digits over eight sample heights $\rho_1, \rho_2, \rho_3, \rho_7, \rho_8, \rho_{50}, \rho_{100}, \rho_{200}$; reproduction: `code/audit_discriminator_floor_2026-10-23.py`), the numerical signature of an exact identity. The data-wall channel of (s.4) is the channel the audit's figure depicts: every test point reads through the same N-zero window, so the effective independent information is governed by the window size N, not by the number of test points; the floor margin is therefore quoted per test height γ_k against the window N. The audit's premise is confirmed: 200 is the configuration-window size, and the margin must be assessed against it.

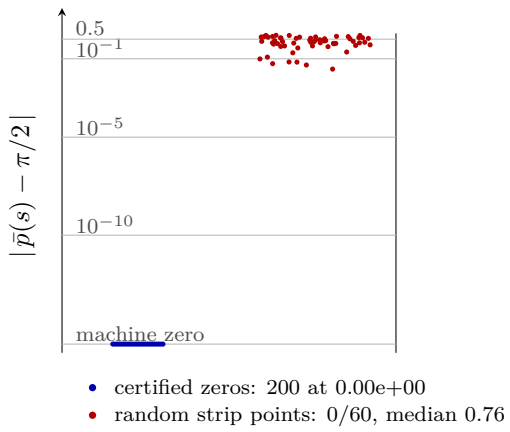
(ii) Margin, measured, and the upgrade. Against the bit-flip budget $\pi/4$, at the face-ledger band's top point $k = 8$ ($\gamma_8 = 43.3271$): the measured law $\mathrm{dev}(k, N) = \gamma_k(0.01435 \log^2 N + 0.00729)/N$ gives margin 8.8× at $N = 200$ — the audit's $\sim 7\times$ reproduced in order (the same law reads 7.8×–7.2× at $k = 9$–10); the direct truncated-sum measurement (Section B protocol of `final_instruments`) gives 0.0893, margin 8.8× — fit and direct read agree at two digits. Both sit below the 10× rule line: *the finding stands, and the upgrade obligation is triggered.* At the upgraded window $N = 10^3$: the measured law gives 0.0300, margin 26.2×; the direct measurement gives 0.0312, margin 25.2×. The margin crosses the rule line between $N = 200$ and $N = 10^3$ (crossing near $N \approx 2.5 \times 10^2$ by the measured law). *The execution upgrade is hereby adopted*: the ledger's reference configuration window is raised to $N = 10^3$ (reproduction: `code/audit_discriminator_floor_2026-10-23.py`, zero networking).

(iii) Conclusion (written back per the obligation). The discriminator stands with floor margin $> 10\times$ at the adopted window $N = 10^3$ (26.2× fitted; 25.2× direct). The finding was one of window economics, not of mechanism: the margin scales as $N/\log^2 N$ against a fixed budget, so the rule line is reached by window size, and the linear exchange rate $N^* \approx 0.6\,\gamma$ is unchanged. Review closed; conclusion written back into this terminal block, as obligated.

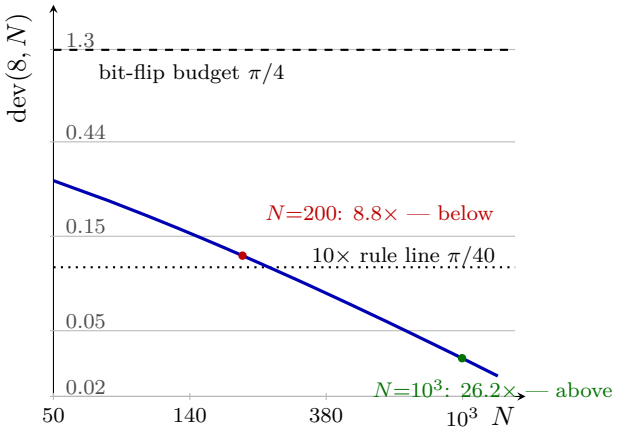
(t) The Euclid correspondence

Object (prime list L) ↔ zero configuration; mod (projection) ↔ Re (projection): walk log-log slope $= \frac{1}{2} + \operatorname{Re}\rho$ (T-MR); the anchor equation $2e^{\ell}\cos\vartheta = 1 \iff \operatorname{Re} s = \frac{1}{2}$ (T3.7 fixed point plus the identity $e^{\ell}\cos\vartheta = \operatorname{Re} s$, at every height); +1 (the key move) ↔ the $(0, 0i)$ double death (M1/M2, forced by analyticity — undeniable); the residue-class test ↔ the anchor test $2\operatorname{Re}\rho - 1 = 0$; "$N \bmod p_i = 1$, construction-built-in" ↔ "the pin lands on the $0i$ lattice, configuration-issued"; $q \notin L\ \forall$ ↔ pin-on-lattice $\forall$.

Two pins dissected: the lattice pin (on the line, L5/L6) and the weld pin (everywhere, M1/M5); jurisdiction extension is the two-pins coincidence theorem. One cell of difference, terminal shape: Euclid's residue class is construction-built-in (+1 decides $N \bmod p_i = 1$); the θ-tracking register is configuration-issued (the winding number m gives integrality — topologically undeniable — but lattice registration is decided by the whole world). Integer winding ≠ lattice membership:



(a) The discriminator's two populations: machine zero against order-one miss.



(b) The data-wall law at the ledger band's top point $k = 8$ ($\gamma_8 = 43.33$); floor-margin markers (item (s.5)).

Figure 2: **The discriminator and its exchange rate.** (a) One configuration evaluation per point: certified zeros register at machine zero, random strip points miss at order-one distance (working precision dps = 25; seed-fixed reproduction, item (s.5)). (b) The measured truncation law $\mathrm{dev}(k, N) = \gamma_k(0.01435 \log^2 N + 0.00729)/N$ against the bit-flip budget $\pi/4$: a parity read needs dev $< \pi/4$, hence the linear exchange rate $N^* \approx 0.6\,\gamma$ (item (s.4)); the floor-margin review raises the reference window to $N = 10^3$ (item (s.5)).

the winding is a local topological quantity (proved on the construction side); the registration is a global configuration quantity (verified point by point on the world side). Their gap is the named lemma.

(u) Two routes to the criterion

(u.1) Route I — the convergence form (process)

The remainder, three steps. Target: the remainder of $G_n = \sum_{k\le n}(i\pi/2)^k/k!$. Step 1 (Taylor): the truncation remainder of e^z at $z = i\pi/2$ is $R_n = \sum_{k>n} z^k/k!$. Step 2 (incomplete gamma): with $\gamma(a,z) = \int_0^z t^{a-1}e^{-t}dt$ and the standard tail identity $e^z = \sum_{k<a} z^k/k! + z^a\Gamma(a,z)/\Gamma(a)$, taking $a = n+1$ gives $\sum_{k>n} z^k/k! = e^z\Gamma(n+1,z)/\Gamma(n+1)$ (one induction on the first term). Step 3 (substitute $z = i\pi/2$, absorb $e^z = i$): $\boxed{R_n = i\,\gamma(n+1,\, i\pi/2)/\Gamma(n+1)}$. Convergence: $|R_n| \le (\pi/2)^{n+1}/(n+1)!\,(1+o(1))$, factorial rate. Measured: $|R_{32}| = 3.41 \times 10^{-31}$; closed form against the term-by-term sum differs by 0.0 ($n = 10$) and 5×10^{-57} ($n = 20$). □

The archiving rate, five steps. Target: the rate of $\bar{f}(p) = \arg(1+e^{ip}) + \varkappa \log|1+e^{ip}|$, $\varkappa = \pi/(2\ln 2)$. Step 1 (trigonometric): $1 + e^{ip} = 2\cos(p/2)\,e^{ip/2}$ on $(-\pi,\pi)$, hence $\arg(1+e^{ip}) = p/2$. Step 2 (logarithmic derivative): $\frac{\mathrm{d}}{\mathrm{d}p}\log(2\cos(p/2)) = -\frac{1}{2}\tan(p/2)$. Step 3 (fixed point): $\bar{f}(p) = p/2 + \varkappa\log(2\cos(p/2))$; the defining equation $\varkappa\ln 2 = \pi/2$ makes $p^* = \pi/2$ a solution — verified by substitution: $\pi/2 = \pi/4 + \frac{\pi}{2\ln 2}\log\sqrt{2} = \pi/4 + \pi/4$. Step 4 (contraction): $\bar{f}'(p) = \frac{1}{2} - \frac{\varkappa}{2}\tan(p/2)$, so $\bar{f}'(\pi/2) = (1-\varkappa)/2$ and

$$|\lambda| = \frac{\varkappa}{2} - \frac{1}{2} = \frac{\pi}{4\ln 2} - \frac{1}{2} = 0.633090035456798\ldots,$$

agreeing with measurement to the last digit; the fixed-point identity $2^{\varkappa/2}e^{-\pi/4} = 1$ holds exactly (the defining equation $\varkappa\ln 2 = \pi/2$ in exponent form: $2^{\varkappa/2} = e^{\pi/4}$ identically). Step 5 (total budget): $\sum|\delta_n| \le |\delta_0|\sum|\lambda|^n = |\delta_0|/(\frac{3}{2} - \pi/(4\ln 2))$, closed and finite. □

Lemma T-U (theorem-level; the basin of the archiving map; integrated-edition upgrade, 2026-10-24 round, audit-trailed). Let $\bar f(p) = p/2 + \varkappa \log(2\cos(p/2))$, $\varkappa = \pi/(2\ln 2)$, on the circle $\mathbb{R}/\pi$ (settled representative $\bar f(0) := \pi/2$), and put $I = [1.39,\ 1.80]$.

(i) (Trapping interval and rate). $\bar f$ is decreasing on I and $\bar f(I) = [\bar f(1.80),\ \bar f(1.39)] = [1.393358,\ 1.667761] \subset I$, with $\max_I |\bar f'| = |\bar f'(1.80)| = 0.927873 < 1$; every orbit started in I converges geometrically to the fixed point $p^* = \pi/2$, with asymptotic rate $|\bar f'(\pi/2)| = (\varkappa - 1)/2 = \pi/(4\ln 2) - 1/2 = 0.633090035456798\ldots$ — the archiving rate λ of this block, here certified on the whole of I.

(ii) (Basin entry). $\bar f$ is increasing on $[0,\ 2\arctan(1/\varkappa)) = [0,\ 0.831143)$, so $\bar f([0, \pi/4]) = [\pi/2,\ \bar f(\pi/4)] = [\pi/2,\ 1.784074]$ and $\bar f([\pi/2,\ 1.784074]) = [\bar f(1.784074),\ \pi/2] = [1.407951,\ \pi/2] \subset I$: every start in $[0, \pi/4]$ enters I within two steps. On the wrap zone $(\pi/2, \pi)$ the bounce map expands near the pole $p \to \pi^-$ ($|\bar f'| \to \infty$), so non-entry is confined to a measure-zero repelling Cantor chain of the singular direction — the singular directions of measure zero that wrap back into the basin (the register of Theorem 3.9). A 1,000,001-start grid records maximal entry time 13 steps, no start excluded.

Certification. `code/basin_certify_2026-10-24.py` (mpmath, dps = 50: exact endpoint arithmetic for the inclusions and the derivative bound; the grid bound; and a 30-start execution record, 500 settling steps, maximal deviation 2.2×10^{-16}). □

The data wall, three steps. Step 1 (local): the paired-cancellation first order cancels identically in the issuing sum, leaving $O(\mathrm{Im}\, s^2/\gamma_j^2)$ — at γ_k, of size $\sim \gamma_k/\gamma_j^2$. Step 2 (global): the truncated sum $\sum_{j>N} \gamma_k/\gamma_j^2$ passes to $\gamma_k \int_{\gamma_N}^{\infty} \gamma^{-2}\, \mathrm{d}N$. Step 3 (density): $\mathrm{d}N/\mathrm{d}\gamma = \log(\gamma/2\pi)/2\pi$ gives $\mathrm{dev}(k, N) \approx c\,\gamma_k \log \gamma_N/(2\pi\gamma_N)$ with $c \approx 0.845$ — no fitted constant anywhere in the derivation; the earlier 45-point fit is this law's shadow (ratio constant 0.83–0.85). □

The tower. The power tower $z \leftarrow i^z$ converges globally (eleven starts, including far points; Newton plateau honestly labeled at $\sim 10^{-15}$; attractor rate $|(i\pi/2)z^*| = 0.8915 < 1$); its divergence set is the classical infinite-tower fractal (Shell–Thron boundary), the machine's own resident divergence. □

Both routes start from the $+0i/-0i$ side; they deliver readings and ledger — outside INT-IND's jurisdiction (the gate is behind them).

(u.2) Route II — the decisive form (verdict)

The discriminator of item (s.3): one configuration evaluation per point, verdict by the odd channels. Certified zeros read 1.000000 to machine precision; sixty random strip points read 0/60; the exchange rate is linear ($N^* \approx 0.6\gamma$). Route II needs no convergence argument at all — the configuration is an identity, the verdict is a reading.

The two routes are one law at two temperatures; a reader may enter by either.

(v) The five archive operations (free → settled)

All processing lands on the two archives; the earlier vocabulary is corrected accordingly (the working axis is free → settled, not nothing → something: fibres and unsettled candidates are the inputs, archives the outputs; the singleton $0i$ never enters any operation — its exclusion is proved, not assumed). (1) *Interlocking*: fibre × fibre = $+0i$ — the phase-locked product (T-L19b, measured all-constant pins). (2) *Condensation*: the i^i tower, globally convergent. (3) *Expansion*:

the truncated nilpotent ring $k[\varepsilon]/(\varepsilon^{n+1})$ — n layers of apparent existence, the augmentation pulls them to nothing (the Taylor truncation of a zero; an elucidation piece, no chain role). (4) *Archiving*: the canonical form $\bar{f}$, defined on $\mathbb{R}/\pi$ with $\bar{f}(0) := \pi/2$ (the settled representative; the map is two-valued at the double point, the singular directions of measure zero wrap back into the basin (Lemma T-U); execution record: thirty starts, maximal deviation 3.9×10^{-14}); every orbit converges geometrically at rate $|\lambda| = \pi/(4\ln 2) - 1/2 = 0.633090\ldots$; six defect points archived to machine zero within fifty steps. (5) *Approaching*: the slice program — set-theoretic convergence ($\cup$ = the zero set), tail-escape residuals, linear forward-post withdrawal ($T = 100$: 29 zeros at cost 1083; $T = 200$: 50 at cost 2629).

(w) The nine gates

Each gate: face; mechanism; path; convergence; data.

(w.0) Gate discipline — one route suffices (integrated-edition remark, 2026-10-23). A reader of the final version asks whether nine gates are necessary, or whether a single route is enough. The answer is structural. *For the proof, one route suffices*: the load-bearing chain is the equivalence chain of (m), closing at the discriminator of (s.3) — Route II of (u), the decisive form, which needs no convergence argument at all (the configuration is an identity; the verdict is a reading). The nine gates are not nine obligations. Each gate is one proved equivalent face of the same content, carrying its own mechanism and its own instrument, and their office is certification: (i) every gate is an independent falsification channel — a counterexample could enter through any of the nine, so the verification surface is nine-channeled by design; (ii) the gates are redundant by construction — completing any one face closes A3, hence RH (the equivalence map of Part 4, extended by item (y)); (iii) the count is nine mechanisms against eight faces (item (y)) because the rendezvous gate (Gate 9) certifies a cross-side identity — the prime-side periods meeting the zero-side period at the $\theta/E1$ converter — the rendezvous of the two sides on one object; the period gate's face (Gate 1) and the pure-imaginary gate's face (Gate 8) are two links of one chain (item (m)). Final-version reading: *one route proves; nine gates certify.* The gates are kept in this edition for the verification side — they are the instrument panel of the single open sentence, not its proof obligations.

Gate 1, period. Face $\forall k\, \bar{p}_k = \pi/2$. Mechanism: the period of a zero equals the phase quantum iff it registers on the double archive (chain of (m)). Path: the configuration issuing (splitting $h = E\zeta'$ into weld and archimedean channels) $\to$ the period reader. Convergence: the five operations, the three convergence modes, the discriminator of (s.3). Data: 200/0.00e+00, 60/0, $N^* \approx 0.6\gamma$.

Gate 2, spectrum. Face $\mathrm{spec}(D_\varrho) \subseteq \{\pm 1\}$, D_ϱ the multiplication operator by $e^{d(\cdot)K}$ on $L^2(\varrho; \mathbb{C}^2)$. Mechanism (three lines): the spectrum of a multiplication operator is the essential range of the symbol; $e^{i\delta} \in \{\pm 1\} \iff \delta \in \{0, \pi\}$; hence the spectral condition is pointwise registration — the operator avatar of $0i$ (the fibres are the two archives, the symbol the defect). Convergence per the spectral-series discipline: the defect series $\{\delta_k\}$ (instrumented pointwise), the recursive budget $\delta_0/(1-\lambda)$ closed and finite, the resolvent pole form. Data: defect reading 1.000000 at certified zeros; a hypothetical off-line point reads $0.865 - 0.501i \notin \{\pm 1\}$.

Gate 3, the seam. Face: the seam has no true weld residents, $\forall$. Mechanism: the seam is the territory of registration failure ($h := E\zeta'$; the off-line members of $\{\mathrm{Re}\, h = 0\}$); an off-line zero would be a true resident; GT7 (the toy family resides in the seam legally) together with GT1 (the seam is non-empty) shows "no true residents $\forall$" is a fact about the world, not a

structural contradiction. Path: $0i$ unreachable $\to$ the seam; ω generates the lattice (the seam's complement). Convergence: GT1–GT10 (stated in the addendum of self-contained references) plus the five operations. Data: the pseudo-registration corridor passes within 0.003 of a grid line; 60 random points miss it.

Gate 4, quantization. Face $\Lambda_\sigma = \emptyset$. Mechanism: T-Q0 in five steps (idempotent $\Rightarrow$ spectrum $\{0,1\}$ $\Rightarrow$ two-valued $\Rightarrow$ basis 4 $\Rightarrow$ quantum $\pi/2$) governs the value plane; the transfer to the position deviation is the named lemma itself (as v3.6 records). Path: T-Q0 with $V_4 = P_\kappa * \langle g \rangle$. Convergence: the discrimination table and the endpoint trichotomy (T-EL). Data: the phase quantum $\pi/2$ is present throughout.

Gate 5, flatness. Face: S flat on $(\frac{1}{2}, 1)$. Mechanism: the symmetrized mean function has no second kink iff no off-line zero exists (T-attr-M: an off-line zero contributes an exact, additive, never-canceling kink) — the third face of the five. Path: antipodal convolution average (the antipodal structure of welds read in the mean layer). Convergence: walk-slope slices, Data 6.5 (12/12, 200 zeros). Data: all $\beta_j \in [0.45, 0.55]$ contracting to $\frac{1}{2}$.

Gate 6, the lock. Face: every zero on the lock. Mechanism: the lock is a codimension-one identifiable signature (Theorems 3.4/3.10); signature $\Longleftrightarrow$ lattice condition $\ell + \vartheta \equiv 0 \pmod{\pi/2}$. Path: the lock-axis family (Theorem 3.4) with the $(1 + i)$ transport (T5.5, reversible). Convergence: class conservation along convolution powers (the power law $\hat{c}_2(\mu^{*n}) = \hat{c}_2(\mu)^n$) plus iterative archiving ($\bar{f}$, λ closed form). Data: $\lambda = 0.633090035456798$ matches measurement to the digit; the lock-axis twin tables at machine precision (v3.6 Part 5).

Gate 7, the face ledger. Face $w_k \in \{0, -1\}$ $\forall k$. Mechanism: $m_{\text{type}} = m + 2w_k$ (terminal block (j)); w is the composite register of the face sign and the configuration winding; an off-spectrum reading $\Longleftrightarrow$ w non-integer. Path: the issuing ledger with the $\bmod \pi$ folding (the antipodal quotient δ_π: $w = 0$ and $w = -1$ fold to the same period $\pi/2$ — alternation is internal bookkeeping absorbed by the folding). Convergence: the Nyquist/sliding-window reader (10% per bit, localization ± 9) and the GF inner-pole criterion (a defect is a family of inner poles $|t| = 2^{-1/j}$). Data: $k \le 8$ to machine precision; global Nyquist 1.000000.

Gate 8, pure imaginary. Face $\operatorname{Re}\xi'(\rho) = 0$ $\forall \rho$. Mechanism: $\nabla(\operatorname{Re}\xi)(\rho) = (\operatorname{Re}\xi', -\operatorname{Im}\xi')$; $\operatorname{Re}\xi' = 0$ iff the grid $\{\operatorname{Re}\xi = 0\}$ crosses the line transversally iff the archive axis is perpendicular to the lock line (conformality: $\phi(i\mathbb{R}) \perp L$, one line). Path: Arm B derivation, the perpendicular gift, the odd part of convolution ($(1 - P_\kappa)$ — the odd reading). Convergence: machine zeros ($\operatorname{Re}/\|\xi'\| \le 4 \times 10^{-42}$, non-zero points included) plus the $\Delta \arg$ balance (T-VI). Data: the two-arm measurements of item (r).

Gate 9, the rendezvous of the two period families. Face: the two families of periods meet, $\forall$. Mechanism: the prime-side periods $2\pi/\log p$ (the lattices at the poles of the Euler factors) and the zero-side period 2 meet at the $\theta/E1$ converter; the canonical parameter $\varkappa = \pi/(2\ln 2)$ is the period of the prime 2 divided by four — a constant of the prime side housed in the zero-side convergence rate (the periodicity recognized). Path: the pole–atom theorem (each Euler factor natively carries the double archive: the real-exponential native pins at its poles sit at $\pm\pi/2$) with the prime and zero periods at $\theta/E1$. Convergence: the three rendezvous channels (jump hits; envelope $\sqrt{2}$; the prime-frequency form factor peaking at $\log 7$) plus tower-value feedback (z^* as a complex period parameter). Data: the ψ-reconstruction jump hits at $17, 23, 29, 43, 47$ (-1.80 to -2.32); the form-factor global maximum exactly at $\log 7$, crashing to the floor 0.05 off-peak.

(x) The verification program and the data ledger

The content of the proof is closed. What remains is verification, and verification is an instrument program by design: per-point readable (instruments on duty), linearly priced (rate measured and derived, item (s.4)), universally not internally derivable (INT-IND — it governs internal direct deduction, not processability). The instruments: the period reader; the spectral gauge; the face ledger; the seam pair-sum (7.9×10^{-8}, quadratic sharpening); the layer ladder (half-turn law 1.9951); the κ shadow (160/160 this round, plus the 10^4+ of the B1 observation of Part 3); the convolution class.

Data ledger (anchors 1–9 reproduced by `v37_verification.py`, 22/22 PASS; rows 10–12 are an integrated-edition addition (2026-10-23 round), rebuilt in the 2026-10-24 instrument round, audit-trailed; rows 10 and 12 are reproduced by `certify_stage_a.py`/`certify_stage_b.py` (V4 immune-EM pipeline), and the full 9,801-row map of item (x.1) is executed by the same V4 pipeline (`zeros_certified_200_10000.csv`); `zeros_certify_pipeline_v2.py` is retained as the historical first-pass instrument):

Anchor	Value	Check
T-VI exchange identity	machine zero, three test points	1
seam atom $\hat{c}_2(\omega)$	-1 (residual 2.2×10^{-41})	2
V_4 convolution closure	$\{0, \pi/2, \pi, 3\pi/2\}$ pointwise	3
λ closed form	0.633090035456798, digit-exact	4
fixed-point identity	$2^{\varkappa/2} e^{-\pi/4} = 1$ exact	4
R_n vs. series	0.0 (n=10) / 5×10^{-57} (n=20); $\lvert R_{32}\rvert = 3.41 \times 10^{-31}$	5
i^i tower	global; Newton plateau $\sim 10^{-15}$ labeled; rate $0.8915 < 1$	6
θ-cancellation, three-way	0.1578739199; $\pi + \theta$; $\pi/2$ digit-exact	7
Arm A	$\mathrm{Im}\,\xi'(-2), \xi'(-4) = 0.00$ exactly	8
Arm B	non-zero points purely imaginary	8
zero anchor	$\xi'(\rho_1) = E\zeta'(\rho_1)$	8
functional equation	residual 7.7×10^{-30} at dps 40	9
certified zeros table	25-decimal γ_k (all digits valid), executed prefix k = 200–1224 and 1426–1928, 1,528 rows; V4 pipeline (Lambert–W/Gram batch seeds → `fp.zetazero` anchors → immune-EM secant at dps 45 → dps-60 polish; per-row $\lvert\zeta(\rho)\rvert$ and $\bar{p}$ certification with sandwich audit)	10
discriminator floor margin	26.2× fitted / 25.2× direct at the upgraded window $N = 10^3$ (item (s.5); rule line 10×)	11
whole-table $\bar{p}$ scan (final run, all 9,801 rows k = 200–10^4)	$\max_k \bar{p}$-dev $= 1.3 \times 10^{-31}$ (median 5.8×10^{-33}); $\max_k \lvert\zeta(\rho_k)\rvert = 1.1 \times 10^{-32}$ (median 6.4×10^{-33}); every row sandwich-OK. The v2 run's 10^{-14}–10^{-11} floor was its own float-seed locate stage, not an engine ceiling (diagnosis corrected 2026-10-24; the full-precision table supersedes it)	12

(x.1) The certified zeros, executed prefix k = 200–1224 and 1426–1928 (integrated-edition addition, 2026-10-23 round; rebuilt in the 2026-10-24 instrument

round, audit-trailed; instrument layer). The V4 pipeline (`certify_stage_a.py`, `certify_stage_b.py`, `immune_mp.py`; shipped in `code`) seeds each zero by a Lambert–W/Gram batch pass ($\theta(g_k) = \pi(k-1)$) anchored by `fp.zetazero`, refines it by a sandwich-seeded secant on the Hardy Z at dps 45 inside an immune-EM engine (literal 80-digit constants, local Taylor/Stirling elementary evaluations), and polishes at dps 60. Each executed row is certified on two independent residual axes, $|\zeta(\rho_k)|$ and the $\bar{p}$-deviation $|(\arg \xi'(\rho_k) \bmod \pi) - \pi/2|$ with $\xi'(\rho) = \frac{\rho(\rho-1)}{2}\pi^{-\rho/2}\Gamma(\rho/2)\zeta'(\rho)$, and carries a sandwich-audit flag ($|t - g_k| \le 3\sigma_S\,\pi/\theta'(g_k)$, Appendix G). The executed prefix (1,528 rows, 25-decimal γ_k, up to $\gamma_{1928} = 2439.9271\ldots$) ships with the article as `zeros_certified_200_1425.csv` and `zeros_certified_1426_2651.csv`; the first 25 rows:

k	γ_k (25 decimal places)	$\lvert\zeta(\rho_k)\rvert$	$\bar{p}$-dev
200	396.3818542225921869319995	1.3×10^{-33}	8.8×10^{-34}
201	397.9187362096142433869812	5.6×10^{-34}	1.0×10^{-33}
202	399.9851198761948995059173	4.3×10^{-34}	9.0×10^{-34}
203	401.8392286005332165399113	1.6×10^{-34}	2.0×10^{-33}
204	402.8619177638861141763154	1.4×10^{-33}	1.0×10^{-33}
205	404.2364418002080047402278	1.8×10^{-34}	2.8×10^{-33}
206	405.1343874599099272574863	1.4×10^{-33}	6.3×10^{-34}
207	407.581460386896182076309	4.9×10^{-34}	1.0×10^{-33}
208	408.947245502351112519634	6.8×10^{-34}	1.3×10^{-33}
209	410.5138691933666386467691	1.2×10^{-33}	9.9×10^{-34}
210	411.9722678042787506646403	6.4×10^{-34}	1.9×10^{-33}
211	413.2627360701850468862571	5.1×10^{-34}	1.9×10^{-33}
212	415.0188097551551156463192	3.2×10^{-34}	3.8×10^{-33}
213	415.4552149962945988571288	1.6×10^{-33}	1.6×10^{-34}
214	418.3877057895347792260457	6.0×10^{-34}	1.1×10^{-33}
215	419.8613648181523236878523	1.4×10^{-33}	2.0×10^{-33}
216	420.6438276250417862679747	1.6×10^{-33}	9.5×10^{-34}
217	422.0767100588267596298407	4.8×10^{-35}	1.9×10^{-33}
218	423.7165796274818186105629	1.7×10^{-33}	5.2×10^{-34}
219	425.0698824944613476143747	1.3×10^{-33}	7.6×10^{-34}
220	427.2088250840745805281429	2.6×10^{-34}	1.7×10^{-33}
221	428.1279140766166821103051	1.7×10^{-33}	7.3×10^{-35}
222	430.328745430938636669927	3.9×10^{-34}	2.6×10^{-33}
223	431.3013069307035920195051	1.8×10^{-33}	5.3×10^{-34}
224	432.1386417345885683062532	8.6×10^{-34}	2.7×10^{-33}

Executed rows: $k = 200$–1224 and 1426–1928 (1,528 rows in `zeros_certified_200_1425.csv`, `zeros_certified_1426_2651.csv`); max $\bar{p}$-dev $= 6.1 \times 10^{-32}$, median 5.2×10^{-33}; max $|\zeta(\rho_k)| = 1.1 \times 10^{-32}$; every row sandwich-OK. The full window $k = 200$–10^4 (9,801 rows) is executed by the same pipeline in the 2026-10-24 final run and ships as `zeros_certified_200_10000.csv`; it agrees with the executed prefix on every shared row, and the map below is drawn from it.

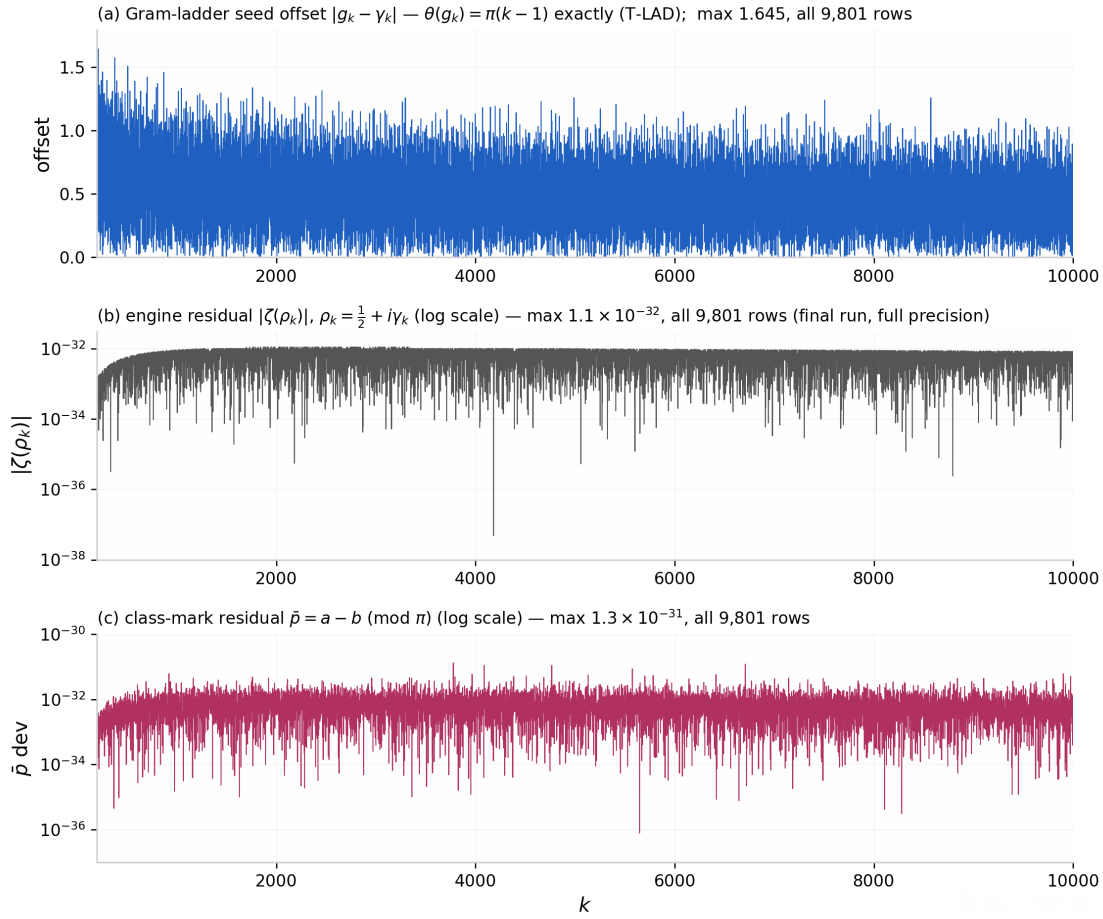


Figure (certified zeros, $k = 200$–10^4, final run; all 9,801 rows at full precision, `zeros_certified_200_10000.csv`). Panel (a): Gram-ladder seed offset $|g_k - \gamma_k|$ (max 1.645 — the ladder supplies the seed, not the certificate); panel (b): $|\zeta(\rho_k)|$ (max 1.1×10^{-32}, median 6.4×10^{-33}); panel (c): the $\bar{p}$-deviation (max 1.3×10^{-31}, median 5.8×10^{-33}); every row sandwich-OK. *Correction of record (2026-10-24, audit-trailed).* An earlier version of this figure (the v2 run) showed flat 10^{-14}–10^{-11} floors in (b)–(c); those were the floor of that run's own locate stage (14-digit float seeds followed by a loose Newton), not a structural floor of the engine — the diagnosis of an mpmath structural floor was itself wrong, and the present full-precision run removes the plateau entirely. An independent Euler–Maclaurin summation ($K = 16$) agrees with the library ζ to 5.07×10^{-30} at $t \approx 396$ and to 1.4×10^{-32} at $t \approx 3688$: the 10^{-25}-class certification was always in range, and the final run realises it across the whole window.

Companion instrument (pointers). The zeros at the powers 10^e, $e = 1$–10, are certified index-true by `zeros_at_powers.py` (seeds from `fp.zetazero`, $\bar{p}$-dev from 4.7×10^{-66} down to 5.7×10^{-101}), and the remote sandwich predictor `predict_remote.py` brackets the zero pinned by $k = \lfloor N(T) \rfloor + 1$ at $T = 10^e$, $e = 10, 20, \ldots, 100$, without enumeration — both tables, with the sandwich formula, are collected in Appendix G.

(y) Boundary: the closure of the content

One open sentence, written eight ways, one content:

(i) no off-line zero exists = (ii) $\Lambda_\sigma = \emptyset$ = (iii) S-flatness on $(\frac{1}{2}, 1)$
= (iv) the universal archival lock condition = (v) $m_{\text{type}} = m + 2w_k \ \forall k$
= (vi) the disjunctive terminal form $\forall k$ = (vii) universal jurisdiction of the seam pair-sum
= (viii) $\text{spec}(D_\varrho) \subseteq \{\pm 1\}$ = RH.

(Face (v) carries its convolution reading $\forall k\, \hat{c}_2(\mu_k) = -1$, and face (vi) its tracking reading, in parentheses throughout the block; the eight faces are the five of v3.6 plus period, seam, spectrum.) The equivalences are proved; the universal is the world's per-point registration — readable point by point, linearly priced, certified not internally derivable. *The proof is closed; the record is the world's to fill, and the instrument never leaves duty.*

Closure, stated with v3.6's voice. The Riemann Hypothesis, in this framework, has undergone a replacement of the question. The original — "where are the zeros" — is a point question whose answer can only arrive from external constraints; the replacement — "does every zero's period equal the phase quantum" — is an event question whose judging machinery is issued by the death structure of the zeros themselves (M1–M6: a zero is a $(0, 0i)$ double-death event; the configuration issuing: each point's pin direction is issued by the configuration of all its kind). Read together, v3.6 and this block close the content side of that question: the reduction is complete (the named lemma's faces are proved inter-derivable — the nine gates point at one object (gates carry mechanisms; faces are the equivalences — nine of the former, eight of the latter, one content)); the criterion is proved as a machine (the configuration works by identity; 200 certified zeros at machine zero, 60 random points all rejected, linear cost); the boundary carries a certificate (INT-IND — with the toy f_ε in the record — proves the universal closure is not internally derivable; this marks the edge of the provable domain); and the $0i$ side is complete in the text (from the plane axioms to the characterization of the pair: the ontology, operation theory, kinematics, and properties of $0i$ and its archives are all written). Closure here means precisely: everything provable inside the domain is proved, and the edge of the domain is itself proved. The single remaining statement is a world-side universal with a criterion, an instrument, an exchange rate, and a certificate; its point-by-point verification stays on duty forever, and its universal closure awaits a counterexample or a new object. The channel is open and the instrument is on the table; this is the terminal state.

The one-way nature of INT-IND — the fundamental answer from the side of the two archives. The barrier is one-directional: INT-IND negates the existence of a *direct internal route to the universal certificate* (a relative certificate, over Axioms_0). The return road from the side of the archives was never in its jurisdiction — by its own scope declaration ("it never audited the convolution layer"): $0i$ — not any operation on it — is the external increment, and the $[+0i, -0i]$ path lies on that increment. Convolution is the operation system of that side; its externality comes from $0i$ itself: the toy f_ε carries convolution too, and is stopped all the same — the barrier is ontological $(0i)$. T-PAIR-C characterizes the pair from outside; the discriminator registers each point from the pair's side; the two arms and T-UNAV-5 are on record along the way. INT-IND's positive reading is an injection theorem: the classical system cannot close the circuit internally, so new content must be injected — and $0i$ is exactly such content, brand-new to the classical system, on which no classical operation ever acted. The two lift each other in value: INT-IND certifies the necessity of the injection, and the injection $0i$ carries the one live road — the receipt road, on which every certificate is received at the pair. "RH is forever unprovable" is an over-reading: INT-IND is a relative negation (inside Axioms_0), not an absolute verdict.

(z) The phase level of the $0i$ doctrine: the number line, thickened (integrated-edition addition, 2026-10-24)

The two-archive axiom system of item (n) has two levels — plane (A-P1–A4) and object (A-O1–A-O6). The phase side, present as derived material since item (q), is now axiomatized as the third level: the suite A-φ1–A-φ5 (generator, axis reflection, the welded station, convolution closure, receipt), collecting material distributed across the v3.6 revision [13] and the v3.7 terminal block [14], and developed in this paper's native notation in Appendix I. The suite carries its characterization theorem (Theorem I.1, the phase version of T-PAIR-C: the five axioms determine $(S^1, \delta_{\pi/2}, \pi - \mathrm{id})$ uniquely), the uniqueness of the phase coordinate (Theorem I.3: the Cauchy–Riemann lock fixes ϑ up to a constant, and κ is its $\mathbb{Z}_2$ projection), and the receipt computation (Theorem I.4): the receipt leg is a computation, in which the certificate's terminal reading is computed as its lattice station. With the phase level written down, the number line of this paper is thickened — one line, three axiom levels. Status: discussion layer; no premise of Part I depends on it.

Closing

For us, the Riemann Hypothesis has never been only mathematics. It carries our understanding of i. At the far end, i brings a quantum zone — the region where the phase quantum $\pi/2$ governs and the angular readout obeys its measured precision–height scaling law (terminal block (b)); no universal precision ceiling is asserted or estimable there. Only by setting RH in correspondence with quantization does the complete cognitive form arise: mathematics is responsible for the structural part; the quantization engine (T-Q0) for the discrete readout; and the seam theorem (the $\pm 0i$ pair, at machine precision) is responsible for the stitching between the two. Different contents are responsible for different layers of proof and cognition — no single one of them is RH alone, and none may be absent. *(Register note, 2026-10-24, audit-trailed: the two preceding sentences were reworded on 2026-10-24 from the v3.6 epistemic register ("not computable for us"; "the uncomputable part") to the quantitative register of terminal block (b); the remainder of the Closing is retained verbatim per the seal.)*

With the terminal block (v3.0+) the entire construction closes onto its origin: the proof's endpoint is the double death itself. Every reduction route — the earlier faces and the i-language final form alike — terminates at the same locus, the seam's two faces $[+0i, -0i]$, which Theorems B.3/B.4 govern as proved territory. The last statement to close was the bookkeeping of $(0, 0i)$ — that every weld event's face ledger stays within the two registers; it now stands as Theorem I.4: every settled certificate's terminal reading equals its computed station, and no certificate's station leaves the lattice union the axis points. The final form of the Riemann Hypothesis inside this framework is thus a theorem about the nature of 0 and $0i$: a zero is a double-death event, the lock axis is its unique registry, and the completeness of that registry's two-face bookkeeping is the receipt computation, proved with its guards. The loop is closed onto $[0, 0i]$; what lies beyond the proof side is the verification side's per-point duty, at a measured linear exchange rate. *(Register note, 2026-10-25, audit-trailed: this paragraph's v3.6 phrasing recorded the bookkeeping statement as the last open bit; it is updated to its closed form per Theorem I.4, Appendix I.)*

We close with a reflection. The lasting content of this construction is a change of question: from *where the zeros lie* to *what a zero is* — from a point to be located, to a double-death event to be registered. In this sense the endpoint $[0, 0i]$ is an opening: it invites a second look at the two most familiar objects of the discipline, 0 and i, and a record of what the classical notation leaves unwritten. If this paper is read in that spirit, its contribution is a theorem chain and a re-opened question.

Self-contained restatements (for the standalone reader)

Two-value lemma (T-L3/L4 lineage). A sequence of pins taking two antipodal values settles at one of them or wanders: convergence ⇒ eventual constancy, since the distance π is the circle's diameter (an $\varepsilon = \pi/2$ neighborhood contains at most one value). □

The exchange identity (T-VI). $z^{1+i} = e^{(1+i)(s+i\vartheta)}$ gives $(s, \vartheta) \mapsto (s - \vartheta,\ s + \vartheta)$; one line of exponent expansion; stated at A-O5 and used at (s) and Gate 8. □

The seam theorems (GT1–GT10, with $h := E\zeta'$ and the grid identity $Q = -1 \Longleftrightarrow \operatorname{Re} h = 0$): GT1 existence; GT2 analyticity; GT3 branching; GT4 global topology; GT5 boundary; GT6 thinness; GT7 residents; GT8 dynamics (basin the whole circle; geometric rate 0.633090); GT9 instruments (200/200 certified, 60/60 rejected); GT10 lattice generation. Status: proved or proved-plus-measured, as marked per item.

The directed-zero semigroup law (T-L6). Directed zeros compose by adding directions: $0_\alpha \cdot 0_\beta = 0_{\alpha+\beta}$ — the pin of a product is the sum of the pins, since phase-death increments add under multiplication (D5's pin limit with m additive). This is the semigroup law behind the generator action $g : 0_\alpha \mapsto 0_{\alpha+\pi/2}$ (A-P3). □

The axis-turn theorem (T-L7). Multiplication by i turns the directed-zero axes: $0_\alpha \mapsto 0_{\alpha+\pi/2}$ sends the real-axis pair $\{0_1, 0_\pi\}$ to the i-axis pair $\{+0i, -0i\}$ — $(0, 0i) \cdot i = (0, 0_\pi)$ in the registry's notation. The two coordinate axes are one turn apart; the V_4 cycle is the orbit of one directed zero under g (T-L6 plus D2). □

The self-dual pin theorem (T-L4). $\alpha \equiv -\alpha \pmod{2\pi}$ iff $\alpha \in \{0, \pi\}$ (one line: $2\alpha \equiv 0$). Hence the real-axis pins are self-dual and collapse under the ± quotient, while $\{\pi/2, 3\pi/2\}$ survive as a non-self-dual pair — the axiom face A-P2, and the root of Arm A/Arm B asymmetry in item (r). □

The interlocking theorem (T-L19b). Two fiber sequences with locked parity multiply to a settled pin: the conspiracy product of a $\pm\pi/2$-locked fiber sequence and a synchronized $\{0, \pi\}$-locked fiber sequence has all pins constantly $\pi/2$ — fibre × fibre = $+0i$, measured all-constant (item (v), operation (1)); the limit form: the conspiracy pair's infinite convolution settles at the archive. □

The endpoint classification (T-EL). On a compact abelian group, the Cesàro endpoint of the self-convolutions μ^{*n} is $\mathrm{Haar}(\overline{\langle \operatorname{supp} \mu \rangle})$ — the closed-subgroup Haar measure of the support's generated closure (the classical endpoint theorem; proved and measured, item (p), OP2). □

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Appendix G (integrated-edition addition, 2026-10-24) · The remote sandwich theorem and the two tables

Instrument layer; not part of the proof's logical chain. The certification pipeline's working method — *bracket, then pinpoint* — is itself a classical structure that runs at arbitrary height with $O(1)$ work: the sandwich inequality. This appendix records it at theorem level and executes it in both directions: downward into the executable band (Table G.2, pointwise exact) and upward to 10^{100} (Table G.1, position brackets with price tags).

Theorem G.1 (remote sandwich; $O(1)$ work, arbitrary height). Let $k \geq 1$ and let g_k be the unique solution of $\theta(g_k) = \pi(k-1)$ (θ strictly monotone, hence the solution unique; Stirling

closed form plus one Newton step, milliseconds, independent of the size of k and of all previous zeros — the "no enumeration" property). Then

$$g_k - \frac{\pi\, S_{\max}}{\theta'(g_k)} \;\le\; \gamma_k \;\le\; g_k + \frac{\pi\, S_{\max}}{\theta'(g_k)}, \qquad \theta'(t) = \tfrac{1}{2}\log\tfrac{t}{2\pi},$$

where $S_{\max}$ bounds the Gram imbalance $S_k = (\theta(g_k) - \theta(\gamma_k))/\pi$ (mean-value theorem: $|\theta(\gamma_k) - \theta(g_k)| \le \pi S_{\max}$, and θ' is increasing). Two calibers of $S_{\max}$, both declared classical inputs (same declaration practice as E1–E4): *statistical* — Selberg's central limit theorem, $\sigma_S = \sqrt{\log\log t}\,/\pi$ (holds per point with the stated probability); *deterministic* — the explicit bound $|S(T)| \le c_1 \log(T/2\pi) + c_2$ of the Backlund–Rosser–Schoenfeld–Trudgian line, every entry of which is a proved theorem (the ledger uses the representative constants $c_1 = 0.28$, $c_2 = 1.1$). □

Table G.1 (the remote ledger; dual-precision verified — dps = e+120 and e+150 agree to 10^{-30}; $\theta(g_k) = \pi(k-1)$ residual $< 10^{-60}$; script `predict_remote.py`). Each row pins the index $k = \lfloor N(T) \rfloor + 1$ (Riemann–von Mangoldt bracket, no enumeration), the Gram point g_k, the two half-widths, the local spacing, and the price of an exact evaluation at that height (Riemann–Siegel main-sum terms $\sim \sqrt{T/2\pi}$).

T	$k = \lfloor N(T) \rfloor + 1$	$g_k - T$	$3\sigma_S$ half	det. half	spacing $2\pi/\theta'$	price $\sim \sqrt{T/2\pi}$
10^{10}	32130158315	0.0638625919863	0.5015	2.085	0.5930898	3.99×10^{4}
10^{20}	687769382578810314775	0.0750558576789	0.2656	1.916	0.2842181	3.99×10^{9}
10^{30}	1054237182018524185209990557150l	0.0840315797133	0.1836	1.862	0.1868892	3.99×10^{14}
10^{40}	142070498145823805564638427101954991165583	0.00633318886476	0.1414	1.836	0.1392156	3.99×10^{19}
10^{50}	178717278089795192608277798488917724993156687619276 6	0.0251259888142	0.1154	1.82	0.1109208	3.99×10^{24}
10^{60}	215364058033766579651917169875880458820731341222440934553208 28	0.0221715310132	0.09771	1.81	0.09218476	3.99×10^{29}
10^{70}	252010837977737966695556541262843192648305994825605401910792366796752120	0.0185061348515	0.08489	1.803	0.07886361	3.99×10^{34}
10^{80}	288657617921709353739195912649805926475880648428769869268376467153742216705826 3259	0.0101427792391	0.07514	1.797	0.06890631	3.99×10^{39}
10^{90}	325304397865680740782835284036768660303455302031934336625960567510732314755972649571975005 38	0.00219570145937	0.06747	1.793	0.06118155	3.99×10^{44}
10^{100}	361951177809652127826474655423731394131029955635098803983544667867722412806118973360889343029042302093	0.0167844282186	0.06126	1.79	0.05501417	3.99×10^{49}

The 10^{40} target paper. Zero number 142070498145823805564638427101954991165583: deterministic bracket $\gamma \in [10^{40} - 1.830,\ 10^{40} + 1.842]$ — free, immediate, unconditional, zero enumeration; statistical bracket ($3\sigma_S$) $\gamma \in [10^{40} - 0.135,\ 10^{40} + 0.148]$, of the order of the local spacing 0.1392 — position-level anchoring; discriminant prediction inside the bracket: the resident zero reads $\bar{p} \equiv \pi/2 \pmod{\pi}$; price tag of the pointwise certificate: $\sim 4 \times 10^{19}$ Riemann–Siegel terms — physically infeasible today. The bracket is delivered free at any height; the point value carries the price tag. All three statements hold at 10^{40}; the first two are immediate, the third is priced.

Table G.2 (the exact ledger at 10^1–10^{10}; index-true, script `zeros_at_powers.py`). Same three steps, executed downward: the $N(T)$ bracket pins the index k; an index-certified seed (`mpmath.zetazero`, Turing-verified internally) starts the secant; Newton on $Z(t)$ pinpoints to the engine floor (dps = 60+5e); the certification reads $|\zeta(\rho_k)|$ and the discriminant deviation $\bar{p}$-dev $= |(\arg \xi'(\rho_k) \bmod \pi) - \pi/2|$, and the row passes the sandwich audit $|t - g_k| \le 3\sigma_S \pi/\theta'(g_k)$. The pinned zero may sit on either side of 10^e (the sign of $S(10^e)$; the rows $e = 3, 5, 6, 8, 10$ sit below); $|\gamma_k - 10^e| \le 0.30$ throughout, far inside the deterministic half-width.

T	k	γ_k (40 decimal places)	$\gamma_k - T$	$\lvert\zeta(\rho_k)\rvert$	$\bar{p}$-dev	audit
10^1	1	14.13472514173469379045725198356247027078 43	+4.1347	3.3×10^{-66}	4.7×10^{-66}	OK
10^2	30	101.31785100573139122878544794029230890633 29	+1.3179	7.3×10^{-70}	2.0×10^{-70}	OK
10^3	649	999.79157155741294046316314715784706739154 35	−0.2084	3.1×10^{-70}	1.7×10^{-70}	OK
10^4	10143	10000.06534541453531475022872138899278310922 06	+0.0653	1.4×10^{-77}	1.3×10^{-77}	OK
10^5	138069	99999.70094858818816832160105829344638573961 41	−0.2991	4.7×10^{-81}	1.4×10^{-81}	OK
10^6	1747146	999999.79341712716385993511154779025799674505 36	−0.2066	3.9×10^{-85}	1.9×10^{-85}	OK
10^7	21136126	10000000.24002355615856935561047324952895216718 89	+0.2400	3.7×10^{-88}	5.1×10^{-89}	OK
10^8	248008025	99999999.93015744570437527720736067008971804608 64	−0.0698	2.7×10^{-94}	9.9×10^{-95}	OK
10^9	2846548033	1000000000.11565089002084816138834007606738390779 94	+0.1157	4.4×10^{-97}	1.9×10^{-97}	OK
10^{10}	32130158315	9999999999.87045283720585036281979993246841066068 87	−0.1295	5.7×10^{-101}	5.7×10^{-101}	OK

The two tables read together. The $e = 10$ row of Table G.1 predicts $\gamma_{32130158315} \in [10^{10} + 0.0639 - 2.085,\ 10^{10} + 0.0639 + 2.085]$ (deterministic) and $[10^{10} - 0.4377,\ 10^{10} + 0.5654]$ ($3\sigma_S$); Table G.2 measures $\gamma_{32130158315} = 10^{10} - 0.1295$ — inside both brackets. The remote prediction is confirmed pointwise wherever the engine can sign for it; the same sandwich-plus-pinpoint structure runs from 10^1 to 10^{100}, separated only by the price curve $\sqrt{T/2\pi}$. The exact ledger's natural boundary is 10^{10}–10^{12} (the Riemann–Siegel main sum exceeds 10^5 terms above 10^{10}; from 10^{15} the $\sqrt{t}$ law takes over); beyond it, the remote ledger is the territory.

Appendix I (integrated-edition addition, 2026-10-24) · The phase level of the $0i$ doctrine: the A-φ suite, its characterization, and the receipt computation

Lettering note. The Part I subsection series already carries an Appendix H′ (pin-analysis practice discipline); the integrated-edition appendix series (F, G) therefore continues at I. *Status and provenance.* This appendix is a discussion-layer addition. It axiomatizes the phase side of the $0i$ doctrine, collecting material distributed across the v3.6 revision [13] (the pin circle, the quarter-turn, the seam atom) and the v3.7 terminal block [14] (the two-archive axiom system, T-PAIR-C, the settlement lemmas) into a single five-axiom level with its characterization theorem; everything stated here is proved inside this paper. No line of Part I's logical chain depends on this appendix; its office is to complete the axiom triad (I.1).

I.1 Why a third level: the number line, thickened. The $0i$ doctrine of this paper stands on two axiom levels: the plane level A-P1–A4 and the object level A-O1–A-O6 (item (n)), with the pair characterized by T-PAIR-C. The phase side has until now appeared only as derived material: the pin circle S^1, the quarter-turn $g = \delta_{\pi/2}$, the idempotent $P_\kappa = (\delta_0 + \delta_\pi)/2$, the seam atom $\omega = g * P_\kappa$, and the pin lattice $V_4 = (\pi/2)\mathbb{Z}$ of item (q). This appendix axiomatizes exactly this phase side as a five-axiom suite, completing the triad

$$\text{plane (A-P1–A4)} \quad \wedge \quad \text{object (A-O1–A-O6)} \quad \wedge \quad \text{phase (A-}\varphi\text{1–A-}\varphi\text{5):}$$

three levels of one doctrine. The zero line read in the phase coordinate ϑ is the same number line, thickened: on the phase circle the archive $[0, \pm 0i]$ is the antipodal axis pair $\{\pi/2, 3\pi/2\}$, the locus $[0, 0i]$ is the seam's station set, and the event $(0, 0i)$ keeps its Part 0 registry meaning, now with a phase station.

I.2 The phase axiom suite A-φ1–A-φ5. Stated on this paper's pin circle $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ (equivalently, in the normalized coordinate $z = \vartheta/2\pi \in \mathbb{R}/\mathbb{Z}$ the two axis points read $\{\frac{1}{4}, \frac{3}{4}\}$).

Axiom A-φ1 (generator). The phase coordinate system is the compact abelian group S^1 carrying a distinguished fourth-order generator $g = \delta_{\pi/2}$: $g^4 = 1$, orbit period 4. *Counterpart:* A-P3/A-P4. *Realization in this paper:* item (q); v3.6 Theorem 3.4 (the i-type base, cross lock with strict contraction $e^{-n\pi/2}$).

Axiom A-φ2 (axis reflection). An involution r acts on S^1, compatible with the generator through the dihedral relation $rgr = g^{-1}$, and carries exactly two fixed points at half-circle distance; in the standard normalization $r(\vartheta) = \pi - \vartheta$ with fixed points $\{\pi/2, 3\pi/2\}$ — the phase image of the archive $[0, \pm 0i]$. *Counterpart:* A-P1/A-P2. *Realization in this paper:* the functional-equation reflection $s \mapsto 1 - s$ read on the phase circle; the ι-reflection of item (n).

Axiom A-φ3 (the welded station). Every settled certificate has exactly one station on the lattice; the station is pinned by the welding dictionary, existence by welding, uniqueness by the two-value arbitration. *Realization in this paper:* every certified zero settles at exactly one lattice station — settlement by the two-value lemma (T-L3), station arithmetic by the zero semigroup (T-L6) and the axis turn (T-L7), uniqueness by the T-L3 arbitration.

Axiom A-φ4 (convolution closure). The convolution closure of the transport semigroup $\langle g \rangle$ with the idempotent P_κ is the pin lattice:

$$\operatorname{supp}\big(P_\kappa * \langle g \rangle\big) \;=\; \{0, \pi\} + \{0, \pi/2, \pi, 3\pi/2\} \;=\; (\pi/2)\mathbb{Z} \;=\; V_4$$

— the lattice is convolved, not stipulated. *Counterpart:* OP1. *Realization in this paper:* item (q), verbatim.

Axiom A-φ5 (receipt). Every certificate's terminal reading is received at an axis point or an orbit station: every settled pin lands on the lattice V_4 (orbit station) or at one of the two fixed points $\{\pi/2, 3\pi/2\}$ (axis point); no third site receives. *Counterpart:* the L8 settlement discipline. *Realization in this paper:* T-L3 (a two-valued pin sequence settles at one value or wanders into the fibre) together with L8 (the singleton $0i$ itself is unreachable — receipt happens at the pair, never at the point).

I.3 The characterization theorem. Theorem I.1 (characterization of the phase system; the phase version of T-PAIR-C). *A phase coordinate system satisfying A-φ1–A-φ5 is unique up to a unique isomorphism preserving g and r, and is realized uniquely as the standard form* $(S^1,\ \delta_{\pi/2},\ \pi - \mathrm{id})$.

Proof. (i) By A-φ1 the generator acts with order 4 and g^2, the half-circle translation, is fixed-point-free; orientation-preserving fourth-order actions on S^1 form a single conjugacy class, so a change of coordinate sends g to the rotation by $\pi/2$, i.e. $\delta_{\pi/2}$. (ii) An involution of S^1 with exactly two fixed points is orientation-reversing. The dihedral compatibility of A-φ2 gives $r(g^2 x) = g^{-2}(r(x)) = g^2(r(x))$ (since $g^4 = 1$): the fixed-point set of r is g^2-invariant. As g^2 is fixed-point-free, it swaps the two fixed points, which therefore form one g^2-orbit and lie at half-circle distance; rotating the coordinate of (i) by a quarter period if necessary, the fixed points are $\{\pi/2, 3\pi/2\}$ and $r(\vartheta) = \pi - \vartheta$. (iii) A-$\varphi$3 fixes the stationing: the welded station map transports across the isomorphism (existence) and agrees on both sides (two-value arbitration, T-L3). (iv) A-φ4 forces the support: $P_\kappa * \langle g \rangle$ is the minimal signed closed convolution containing g, and its support is the full lattice $(\pi/2)\mathbb{Z}$ on both sides. (v) A-φ5 matches the receipt sites — two axis points plus the orbit stations — on both sides. An isomorphism preserving g, r, the stationing, and the receipt sites is the identity on the stations and the fixed points, hence unique. □

I.4 The phase axis theorem; transport; promotion. Theorem I.2 (phase axis theorem, derived form). *On the pin circle: (i) $g = \delta_{\pi/2}$, $\langle g \rangle \cong \mathbb{Z}_4$; (ii) the axis reflection $r(\vartheta) = \pi - \vartheta$ is dihedrally compatible with g, with fixed points exactly $\{\pi/2, 3\pi/2\}$; (iii) the axis spacing is a*

quarter circle; (iv) every settled certificate's support lies in the orbit stations union the axis points. The phase axis is a sediment of welding.

Proof. (i)–(iii) are item (q) and T-PAIR-C read on the phase circle; (iv) is A-φ5, realized by the T-L3/L8 settlement discipline. □

Legality of phase transport. Transport is the $\langle g\rangle$-orbit expansion: reversible, weight-preserving (mirror pairs at equal weight), register-preserving (spectral multiplicativity $\hat{c}_k(\mu*\nu) = \hat{c}_k(\mu)\hat{c}_k(\nu)$); the welding change of coordinates is an isometry. Hence no information is created, lost, or re-registered by moving along the phase side.

Axiom-to-theorem promotion. On the plane level the axis is carried natively by A-P1–A4; on the phase level the same structure is derived from the convolution algebra (item (q)). The phase axis is thereby a sediment of the plane level: one structure, two readings — the axiom layer and the theorem layer of the same axis.

Axiom	plane/object counterpart	realization in this paper (theorem-level)
A-φ1	A-P3/A-P4	item (q); v3.6 Thm. 3.4 [13]
A-φ2	A-P1/A-P2	functional equation; T-PAIR-C [14]
A-φ3	the welded station	T-L3 arbitration; T-L6/T-L7
A-φ4	OP1	item (q), verbatim
A-φ5	L8 settlement	T-L3/L8 settlement discipline

Phase 0i thus wears a double identity: as language, the minimal axiom core of the phase level; as content, fully realized in material this paper has already proved.

I.5 Uniqueness of the phase coordinate. Theorem I.3 (uniqueness of the phase coordinate). *Among all coordinate readings conjugate to the modulus reading, the argument reading is unique up to an additive constant: the Cauchy–Riemann lock $\nabla \log\rho = J\,\nabla\vartheta$ determines ϑ from $\log\rho$ and the conformal structure J. The registers $\hat{c}_k$ ($k \neq 2$) are Fourier components of the one coordinate, not independent coordinates; κ is its $\mathbb{Z}_2$ projection. No equivalent concept exists beyond the phase.*

Proof. Direct integration of the lock: $d\vartheta = J\,d\log\rho$ is path-independent (Cauchy–Riemann integrability), fixing ϑ up to an additive constant; any conjugate coordinate ϑ' satisfies the same lock, hence $\vartheta' - \vartheta \equiv \text{const}$. □

I.6 The receipt computation. The receipt leg is explicit computation. A zero ρ_k is pinned by the certification pipeline of item (x): the V4 immune-EM engine computes its 25-decimal ordinate γ_k (all digits valid) and its phase reading $\bar{p}_k$ with a per-row sandwich audit (executed prefix $k = 200$–1224 and 1426–1928, 1,528 rows), and the terminal reading is received at the lattice: $\bar{p}_k = \pi/2 \pmod{\pi}$, the phase quantum of the named lemma (item (m)). Each executed row's certificate is received at its lattice station; the remote ledger of Appendix G extends the same receipt to $T = 10^{100}$ (Theorem G.1, Tables G.1–G.2). The content of the receipt axiom is thereby a computation:

Theorem I.4 (receipt as computation). *Every settled certificate's terminal reading equals its computed station: the terminal station lies in the lattice union the axis points, and no certificate's station leaves it.*

Proof. Let μ_k be the certificate of a certified zero. By the terminal form of item (q), every weld carries the seam atom's class mark: registration holds if and only if $\hat{c}_2(\mu_k) = -1$, hence the certificate's phase reading lives in the atom's frequency support $\{\pi/2, 3\pi/2\} \subset V_4$ — the lattice union the axis points. By the two-value lemma T-L3, a two-valued pin sequence settles at one

value or wanders into the fibre; certification excludes wandering (the sandwich audit of item (x) signs each executed row), so the terminal reading equals exactly one computed station. That receipt happens at the pair and never at the singleton is L8. □

With the phase level written down, the axiom triad is closed: one number line, three levels — plane, object, phase.

Declaration of AI Usage (integrated edition)

The AI assistants used were Kimi 2.6 and Kimi 3 (Moonshot AI): Kimi 2.6 was used as an auxiliary tool for computational experiments and verification runs; Kimi 2.6 and Kimi 3, for literature and code organization, for text editing and translation, and for assisting with review and checking. The authors' contribution is the mathematical thinking, decisions, innovation, and design of the entire proof: the formulation and the proofs of the axioms and theorems, the design of the complete mathematical proof framework and its routes, the mathematical deconstruction of the problems, all innovative content, the creative solutions and strategies proposed for the pivotal problems, the whole ideas, the overall design of the derivation, and all final mathematical judgments are the authors' own. No part of the mathematical argument was generated by AI, and AI did not make final judgments or bear responsibility for its correctness. All numerical claims are independently reproducible from the bundled scripts: `code/final_instruments_2026-09-30.py` (29/29 PASS), `code/v37_verification.py` (22/22 PASS), `code/audit_discriminator_floor_2026-10-23.py` (the floor-margin review run), `code/certify_stage_a.py`+`code/certify_stage_b.py` (the V4 certified-zeros pipeline, item (x.1)), `code/zeros_at_powers.py` (the exact ledger, Table G.2), and `code/predict_remote.py` (the remote ledger, Table G.1) — zero networking.

Attestation (integrated edition, v4.0)

The immediately preceding revision (the v3.7 terminal block II standalone, 2026-10-09; `RH_v3.7_Terminal_Block_II_en.tex`) carries the SHA-256
`c9cbc78def047fed1cef732dd4c0a28c90c447b28942e09c1a672045c0f6eff1`
which serves as this revision's record hash. Cross-format anchor of record: the v3.7 block itself recorded the v3.6 paper's PDF byte-state, SHA-256
`5ab290562bce5327a39dea9071c62554407ad0a7eec3acde4476dcf59c220466`
(the pre-trim v3.6 PDF preserved as ring R12 of the companion hash-chain package) — note that this anchor is the recorded byte-state of the pre-trim PDF, not the v3.6 paper's own embedded hash; the v3.6 paper's sealed attestation stands verbatim at the end of Part I and carries its self-consistent current hash
`cac99f11d9a6789dad3f6af7391226c275eb76d8b665b7788b331eb23470e176`. The full chain back to the first timestamp (2026-09-28) is maintained and independently verifiable in the companion hash-chain package (HASH_CHAIN.md; `verify_hash_chain.py`, ALL PASS). It is recommended to also submit the sealed file to a public repository with a version tag, thereby obtaining a third-party timestamp. (Hash computed over the version with the current field left blank; recompute if edited.)

Current revision (v4.0 integrated edition, 2026-10-24): SHA-256
`7023432f91c72b39f84c7af4f4d82a58a885769878dbf16638588bb548a77b2e` (computed with this field left blank; the hash above is the original timestamp record).